Delivery Management

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Delivery Management

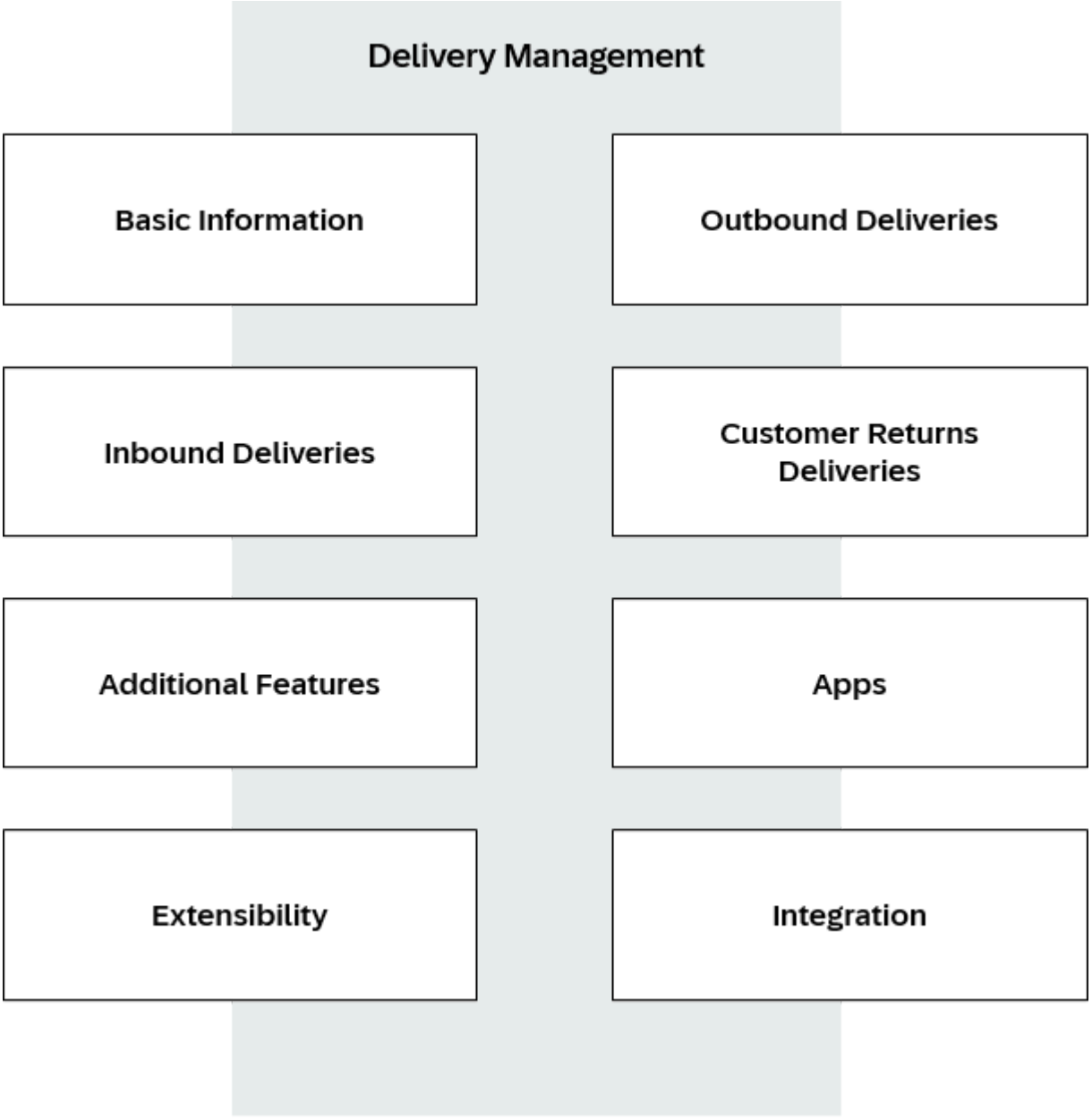
Delivery Management is an important part of the logistics chain in which guaranteed customer service and distribution planning support play major roles.

In Delivery Management, all delivery procedure decisions can be made at the start of the process by:

- Taking into account general business agreements with your customer
- Recording special material requests
- Defining shipping conditions in the sales order

The result is an efficient shipping process. To learn how the capabilities of Delivery Management in SAP S/4HANA Cloud Public Edition simplify your logistics processes, you can navigate through the graphic below.

This image is interactive. Hover over each area for a description. Click highlighted areas for more information.



Please note that image maps are not interactive in PDF outputs.

Basic Information

An overview of processes and configuration in Delivery Management.

Processes and Business Objects

Delivery Management is divided into three main processes. Each process corresponds to a specific business object and SD document category.

The table below outlines the processes and their corresponding SAP object types and SD document categories. A delivery needs the correct SD document category to function in a specific process. The system automatically assigns the SD document category when you create the delivery using the relevant apps or APIs.

Processes and Related Technical Objects in Delivery Management

Process	SAP Object Type	SD Document Category
Goods Issue with Outbound Delivery	Outbound Delivery (`OutboundDelivery`)	Delivery (`J`)
Goods Receipt with Inbound Delivery	Inbound Delivery (`InboundDelivery`)	Delivery/Shipping Notice (`7`)
Goods Issue with Customer Returns Delivery	Customer Returns Delivery (`CustomerReturnsDelivery`)	Returns Delivery for Order (`T`)

Delivery Documents

For each SAP object type in Delivery Management, there is a corresponding delivery document. These delivery documents contain all the details related to the delivery.

The table below gives an overview of these delivery documents and the apps you can use to view and change them:

Delivery Document	Apps for Displaying	Apps for Changing
Outbound Delivery	Display Outbound Delivery (`VL03N`)	Change Outbound Delivery (`VL02N`)
Returns Delivery		
Inbound Delivery	Display Inbound Delivery (`VL33N`)	Change Inbound Delivery (`VL32N`)

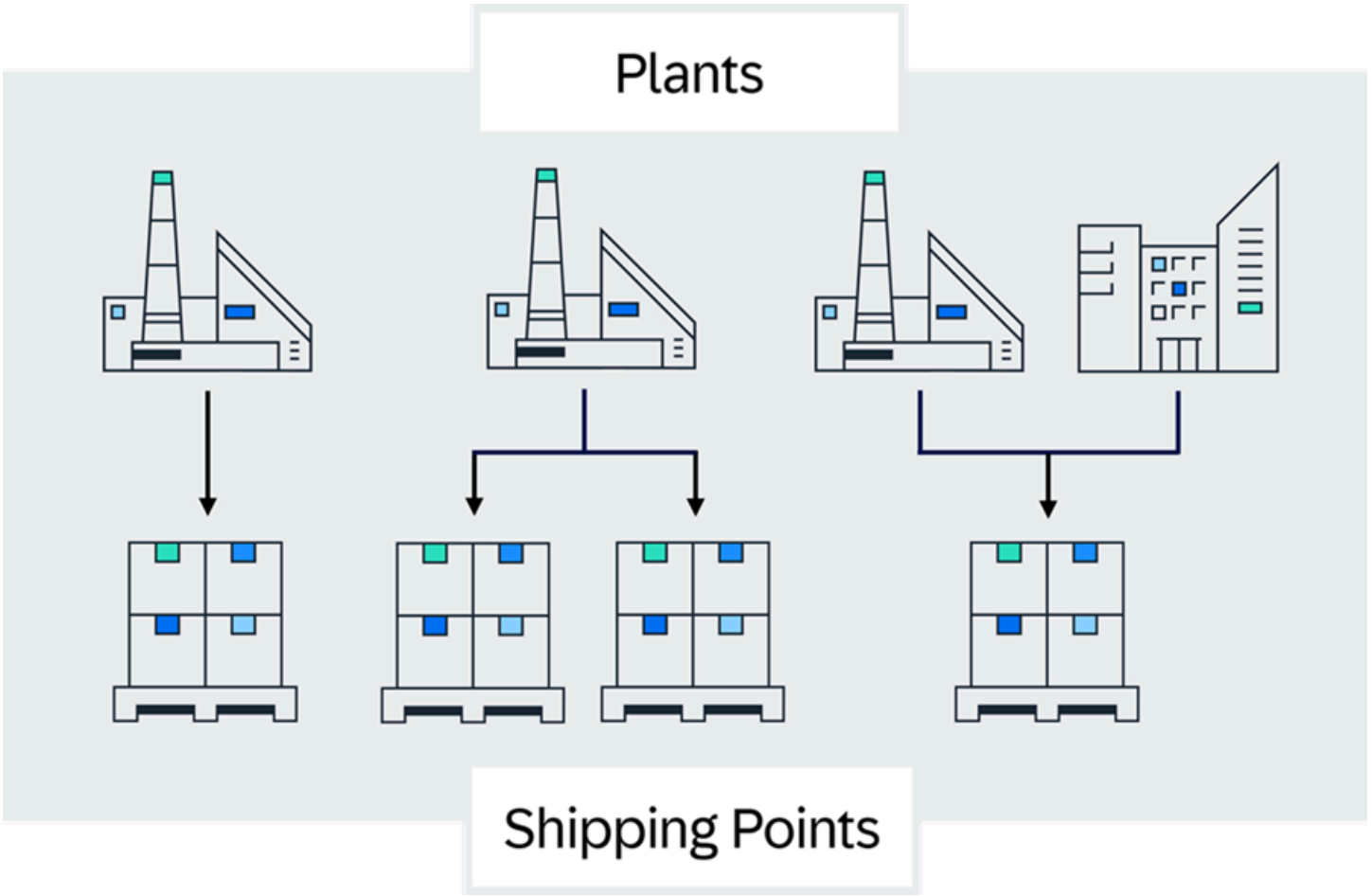
Shipping Points

Shipping points are the organizational units that handle shipping processing.

A plant can have multiple shipping points. Once you define the shipping points, you can configure your system to automatically determine the correct shipping point. Depending on your plant's complexity, you may have multiple shipping points in each plant. For instance, if you have multiple loading bays for trucks, you can create a shipping point for each. You can assign more than one

shipping point to a single plant to handle this. Similarly, two plants can share a single shipping point, and you can handle this scenario by assigning more than one plant to a single shipping point. This adaptability allows you to accurately model the network of plants and shipping points in your organization.

The diagram below illustrates the assignment of one plant to one shipping point, one plant to multiple shipping points, and multiple plants to one shipping point.



Note

You create shipping points in your configuration environment with the mandatory configuration activity **Organizational Structure** (ID: 100066). For more information about the available configuration environments, see Configuration Environment of SAP S/4HANA Cloud Public Edition.

Shipping Points and Deliveries

Shipping points are automatically determined based on the sales or stock transport order or shipping conditions and cannot be changed during the delivery creation process. A delivery can only have a single shipping point. To create a delivery from a sales document containing items from multiple shipping points, you need to create a separate delivery for each shipping point.

The table below explains how the system handles deliveries created from a sales document containing items with multiple shipping points:

Scenario	System Behavior
Single delivery for a specific shipping point based on a sales order containing items with multiple shipping points	You see an entry in the log that states the items with a different shipping point have not been included in the delivery.

Scenario	System Behavior
Multiple deliveries using **My Sales Orders – Due for Delivery** (ID: VL10A)	The system only informs you of errors that occur during processing.

! Restriction

Once a delivery is created, you cannot edit the shipping point. If you need to change the shipping point in a delivery that has been created, carry out the following steps:

1. Delete the delivery.
2. Make the necessary changes in the sales order.
3. Create the delivery again.

Alternative Shipping Points

When processing a sales document or stock transport order, the system determines a proposed shipping point for the line items based on the shipping point configuration. You can also configure the system to suggest alternatives, which you can then manually select in the order. Alternative shipping points allow you to choose a different location if the initially proposed shipping point is unavailable. If you enter a shipping point that is not maintained as a proposed or an alternative shipping point in the configuration, the system displays an error message.

¡ Note

You configure the proposed and alternative shipping points in your configuration environment with the mandatory configuration activity **Assign Shipping Points** (ID: 102126). For more information about the available configuration environments, see Configuration Environment of SAP S/4HANA Cloud Public Edition.

Example

Here's an example that shows how you can configure the shipping point determination:

Shipping Condition	Loading Group	Plant	Proposed Shipping Point	Alternative Shipping Point	Alternative Shipping Point
`01`	`CRANE`	`0001`	`0001`	`0002`	`0003`

The system selects shipping point `0001` based on the specific configuration settings for shipping condition, loading group, and plant. As an alternative to this shipping point, you can manually change the shipping point to `0002` or `0003` in the sales order or stock transport order. This can be done, for example, if it is known that shipping point `0001` is out of operation or doesn't have the capacity to handle the item.

Authorizations for Deliveries

In Delivery Management, you control access to delivery documents by maintaining the **Shipping Point / Receiving Point** restriction type using the **Maintain Business Roles** app.

The **Shipping Point / Receiving Point** (VSTEL) restriction type enables you to manage access to the following delivery documents:

- Outbound Delivery
- Inbound Delivery
- Customer Returns Delivery

You can assign the following access categories to users:

- Value Help
- Read
- Write

Example

You can use the **Maintain Business Roles** app to restrict access to shipping point 1010 by setting the permission in the user's role and specifying this shipping point for the **Shipping Point / Receiving Point** restriction field.

For more information, see How to Define Authorizations Based on Restrictions.

Related Information

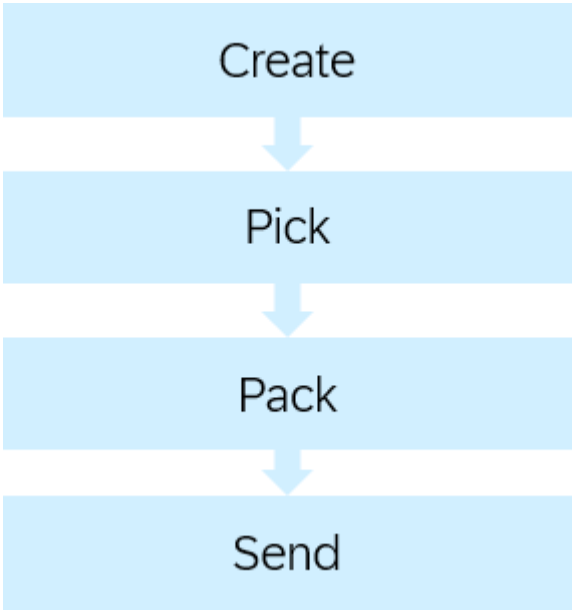
SAP Note 3227819

Business Catalogs for IAM Administration Apps

Outbound Delivery Process

The outbound delivery process includes all steps related to sending goods from a supplier to a ship-to party.

The outbound delivery process includes the following steps:



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Creating and Processing Outbound Deliveries

The outbound delivery process includes all shipping activities like picking, packing, transportation, and goods issue. After an outbound delivery has been created, picking or delivery scheduling are initiated, and data created during processing is added to the delivery.

You can create outbound deliveries as follows:

- With reference to a sales order or a stock transport order
- Without a reference

You can create outbound deliveries manually or automatically with worklists. You can make agreements with your customers for complete and partial deliveries and order combinations. Outbound deliveries can be combined to form a single group of deliveries. You can use overviews to monitor created outbound deliveries and remaining sales activities.

The following table outlines the options for creating outbound deliveries:

Manual creation (apps)	Create Outbound Delivery – with Order Reference
	Create Outbound Delivery – without Order Reference
	Create Outbound Deliveries – From Sales Orders
Automatic creation with batch processing (app)	Schedule Delivery Creation
Automatic creation with APIs	Outbound Delivery with Reference to a Sales Order – Create (A2X)
	Outbound Delivery (A2X)

Note

For scope item specific process descriptions, refer to **PRODUCT FEATURES** > **Scope Items** in the documentation for the relevant app in the SAP Fiori Apps Reference Library.

Related Information

Prerequisites for Creating Deliveries from Sales Orders

Working with Outbound Deliveries

For details about the apps you can use to work with outbound deliveries, refer to the following links:

Creating Outbound Deliveries

This topic provides details about creating and processing outbound deliveries.

Changing Outbound Deliveries

You can use the **Change Outbound Delivery** app to modify outbound delivery documents.

Displaying Outbound Deliveries

You can use the **Display Outbound Delivery** app to view and navigate outbound delivery details.

Deleting Outbound Deliveries

You can use the **Change Outbound Delivery** app to delete outbound deliveries.

Creating Outbound Deliveries

This topic provides details about creating and processing outbound deliveries.

When you create an outbound delivery, you initiate shipping activities such as picking and transportation scheduling. A delivery is processed through one shipping point.

Processing Deliveries

You can change a delivery after it has been created if the delivery situation changes in any way. You can also confirm picked quantities using the change function, or use the display function to access shipping information in a delivery.

Order items or schedule lines that have identical shipping criteria are combined into one delivery. If the data varies between items, for example, if a different ship-to party is defined for each item, two deliveries must be created for the order. Order items from different orders can be grouped together into one delivery if they have the same shipping criteria and sold-to party.

The system automatically updates the delivery status in the sales order at header and item level when the delivery is created. The delivery status of the entire order is based on the delivery status of all the items. The following delivery statuses are available:

- **Partially Delivered**: All items in the order have not yet been fully delivered.
- **Fully Delivered**: All items in the order have been fully delivered.

For an overview of the structure and data in a sales document, refer to Structure and Data in a Sales Document.

Related Information

Prerequisites for Creating Deliveries from Sales Orders
Shipping Points and Deliveries

Creating Outbound Deliveries with Order Reference

You can manually create an outbound delivery from a preceding sales document.

Context

Manually create an outbound delivery based on a single order. Only the items from this order that are due for delivery are included in the delivery. You can change the shipping data if necessary. For details about the prerequisites, refer to Prerequisites for Creating Deliveries from Sales Orders.

Procedure

1. From the SAP Fiori launchpad, open the Create Outbound Delivery – with Order Reference app.

   → **Tip**

   You can also create a single outbound delivery with order reference directly from the **Create Sales Orders** app by choosing **Sales Document** > **Deliver** in the menu of the overview screen.

2. Enter the appropriate shipping point in the **Shipping Point** field.
3. Specify the selection date in the **Selection Date** field. The current date is the default selection date.

> **! Restriction**
>
> You can only deliver schedule lines whose material availability date or transportation scheduling date is on or before the selection date.

4. Enter the sales order number in the **Order** field.
5. To define a range of items to be delivered, specify the appropriate item numbers in the **From Item** and **To Item** fields.
6. Press Enter or choose **Continue**.

   The outbound delivery is created according to your specifications and the outbound delivery overview screen appears. Data from the preceding sales document is copied into the outbound delivery. The system enters the delivery quantity in the delivery according to availability. On the header screens and item screens, you can enter additional specifications (for example, how the goods are to be transported). You can also change the specifications from the order.

   An error log entry is created if inconsistencies occur in a schedule line.

7. Save the delivery by choosing **Delivery** > **Save.**

   When the system has saved the document, it displays the document number of the outbound delivery.

Next Steps

The system creates an entry in a log if any errors have occurred. To view details about the error, you can navigate to the log from any of the overview screens in the delivery by choosing **Menu** > **Edit** > **Error log.** From there, you can change the shipping document directly.

Prerequisites for Creating Deliveries from Sales Orders

The table below outlines the prerequisites for creating deliveries from sales orders at the header, item, and schedule line level:

Level	Prerequisite
Header Level	No delivery block has been set. A delivery block can be set automatically if, for example, the customer has exceeded their credit limit. You can also enter a delivery block manually in the sales document. **i Note** A delivery block at the header level only works if it has been assigned to the appropriate delivery type in the configuration activity **Assign Blocking Reasons to Delivery Types** (ID: 102214). Ask your system administrator to perform this configuration as required.
	The sales document contains at least one item due for delivery.
Item and Schedule Line Level	The schedule line is due for shipping on the specified selection date. A schedule line is due for shipping when the material availability date or the transportation scheduling date has been reached.
	No delivery block has been set for the schedule line. You can manually set a delivery block for a schedule line in the sales document. No prior configuration is required to set a delivery block at the schedule line level.

Level	Prerequisite
	The delivery quantity is greater than zero.
	The items in the order are fully processed.
	The product status of the material is released for delivery.
	The required quantity of the material is available.

Creating Multiple Outbound Deliveries

You can create outbound delivery runs from a list of sales orders.

Context

You can manually create multiple outbound deliveries based on multiple sales orders using the **Create Outbound Deliveries – From Sales Orders** app.

→ Tip

As an alternative, you can use the Schedule Delivery Creation app to define a periodic job that automatically creates outbound deliveries in the background.

To manually create outbound delivery runs from a list of sales orders, carry out the following steps:

Procedure

1. From the SAP Fiori launchpad, open the Create Outbound Deliveries – From Sales Orders app.
2. Set the filters as needed to display the relevant sales orders.
3. Choose the **Go** button.
4. Select your sales orders and choose **Create Deliveries**.
5. Choose **Display Log**.

   With the **Analyze Delivery Logs** app, you can check system messages that have been logged during the collective creation run of deliveries. For more information, see Analyze Delivery Logs.

Creating Outbound Deliveries Without Order Reference

You can manually create outbound deliveries without order reference by manually entering ship-to party and materials. Specifying a delivery plant or storage location is optional.

Context

To create outbound deliveries without referring to an order, you need to manually enter the ship-to party and the materials. Data is taken over into the delivery from the appropriate master records. The system does not check the delivery for completeness. Therefore, you can create a delivery without specifying a delivering plant or a storage location, for example.

Procedure

1. From the SAP Fiori launchpad, open the Create Outbound Delivery – without Order Reference app.
2. Specify the following values according to your requirements:
    a. **Shipping Point**
    b. **Delivery Type**
    c. **Sales Organization**
    d. **Distribution Channel**
    e. **Division**
3. Choose ENTER. The delivery overview screen is displayed.
4. Specify the following values according to your requirements:
    a. **Ship-to Party**
    b. **Delivery Items** and **Deliv. Qty**
    c. Additional header and item data as required
5. Choose **Outbound delivery** > **Save** to save the outbound delivery.

Results

When the outbound delivery has been saved, the system displays a message that contains the outbound delivery document number.

Delivery and Transportation Scheduling

The system schedules relevant shipping activities to meet the requested delivery dates.

When entering an order, each line of an item can contain a requested delivery date, which is the date on which the goods should arrive at the customer. During order processing, the system can automatically schedule when shipping activities such as picking, loading, and transportation must begin to meet the requested delivery date.

Times and Dates in Scheduling

You must distinguish between the following terms used in scheduling:

- The times (the time duration) needed to carry out certain activities.
- The dates that are calculated based on these times.

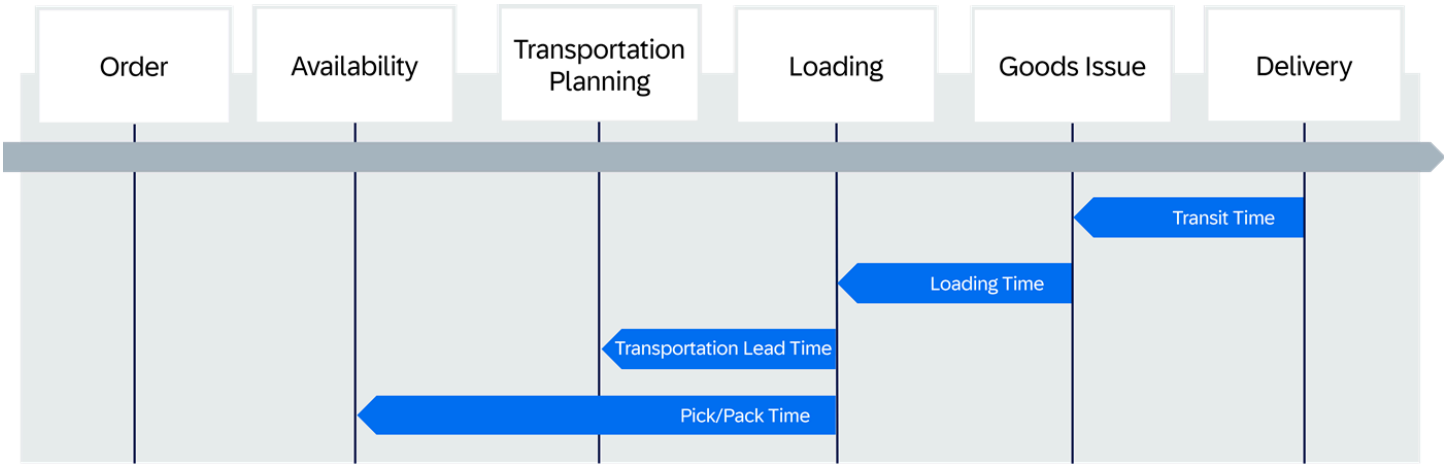
Times

The system uses the shipping department's experience to input transit time, loading time, pick/pack time, and transportation lead time:

Time	Description
Transit Time	The time in days that is required to deliver goods from your premises to the customer location. The system determines this time for a route.
Loading Time	The time in days that is required for loading a delivery item. The system determines this time from the shipping point, the route, and the loading group of the material.

Time	Description
Pick/Pack Time	The time required to pick and pack goods for a delivery. The system determines this time using the shipping point, the route, and the weight group of the order item.
Transportation Lead Time	The time in days needed to organize the shipping of the goods. This might include booking a ship and reserving a truck from a forwarding agent. The system determines this time for a route.

The following figure shows the different times considered during delivery scheduling:



ℹ Note

If you have **not** maintained any working times in the shipping point, the unit for the time specification is the day (workday = 24 hours) with two decimal places.

If you've maintained the working times in the shipping point, the unit for the time specification is hours and minutes.

You can also enter the factory calendar of the forwarding agent for the route when you define the transit time. This factory calendar can be different from your company's calendar. For example, the forwarding agent might operate a 6-day-week.

ℹ Note

You may only enter the transit time in days with two decimal places, even if you've maintained the working times in the shipping point.

Dates

The following dates are important for delivery processing:

Date	Description
Material Availability Date	The date when you must start picking and packing activities. This date must be selected early enough so that the goods are ready by the given loading date.
Transportation Planning Date	The date when you must start to organize the transportation of the goods. This date must be selected early enough to ensure that the means of transport is available by the loading date.
Loading Date	The date on which the goods must be available for loading and on which all vehicles required to ship these goods must be ready for

Date	Description
	loading. After the time needed to load the goods (loading time) has expired, the goods issue can be carried out.
Goods Issue Date	The date when the goods leave the company to arrive punctually at the customer's location.
Delivery Date	The date when the goods are to arrive at the customer's location. The difference between the goods issue date and the delivery date is calculated from the transit time required for the route between the delivering plant and the customer. **Note** The delivery date can be the customer's requested delivery date or the confirmed delivery date (that is, the earliest date on which you can deliver goods to the customer).
Delivery Creation Date / Planned Creation Date	The date on which the delivery creation is planned. This date is determined during the sales order item/schedule line scheduling by taking the earliest of the following dates: • The material availability date • The transportation planning date

Shipping Data in the Sales Order

In the sales order, the relevant data is displayed on the Shipping details screen. You can call up this screen as follows:

1. On one of the overview screens in the order, select the item whose shipping data you want to look at. Then, choose Goto > Item > Schedule Lines.
2. Place the cursor on the appropriate schedule line and choose the Shipping button.

Shipping Data in the Delivery

In the delivery, you can find important shipping data at header level. Choose Goto > Header > Shipment from a header or overview screen to display the relevant data.

You can also find important data at item level, such as the material availability date, which can be different for each item. Select the item for which you want to display the material availability date from the initial overview screen and then choose Goto > Item > Picking. You find the Mat.Avail.Date field in the Warehouse section.

In the sales order, the system can determine precisely when goods must become available, be loaded, and sent so that they arrive at the ship-to-party location on the requested delivery date. The two different types of scheduling that are carried out are delivery scheduling and transportation scheduling.

Delivery and Transportation Scheduling Details

During delivery scheduling, all activities that must be carried out before the goods can be delivered to the customer are considered. This includes loading, picking, and packing. Delivery scheduling also determines the material availability date and the loading date.

Transportation scheduling considers all dates used for preparing and carrying out the transportation of goods. This includes the transit time and the transportation lead time that you need to order a foreign forwarding agent or arrange a truck from your company's truck fleet. Transportation scheduling also determines the goods issue date and the transportation planning date.

Dependencies

For each sales document type, you can define whether delivery and/or transportation scheduling are carried out.

You can also specify whether the system proposes loading and pick/pack times for delivery scheduling. Use the following criteria to determine these times:

- Shipping point
- Shipping point, route, and weight and/or loading group
- Shipping point, weight and/or loading group

Backwards Scheduling

Starting from the requested delivery date, backwards scheduling calculates when the material must be in stock and when transportation planning must begin to ensure that delivery is on time. The calculation considers the factory calendars as follows:

1. The transportation time is subtracted from the requested delivery date to get the goods issue date (the date the transport has to start).
2. The loading time is subtracted from the goods issue date to determine the loading date (the date the goods must be ready for loading).
3. From the loading date, the material availability date is calculated by subtracting the picking and packaging time from the loading date.
4. The transportation planning date is calculated by subtracting the transportation lead time from the loading date.

Material Availability date

The earlier of these last two dates (material availability date and transportation scheduling date) determines when an outbound delivery should be created.

Forwards Scheduling

In the following cases, the system automatically carries out forwards scheduling to determine the earliest possible shipping date:

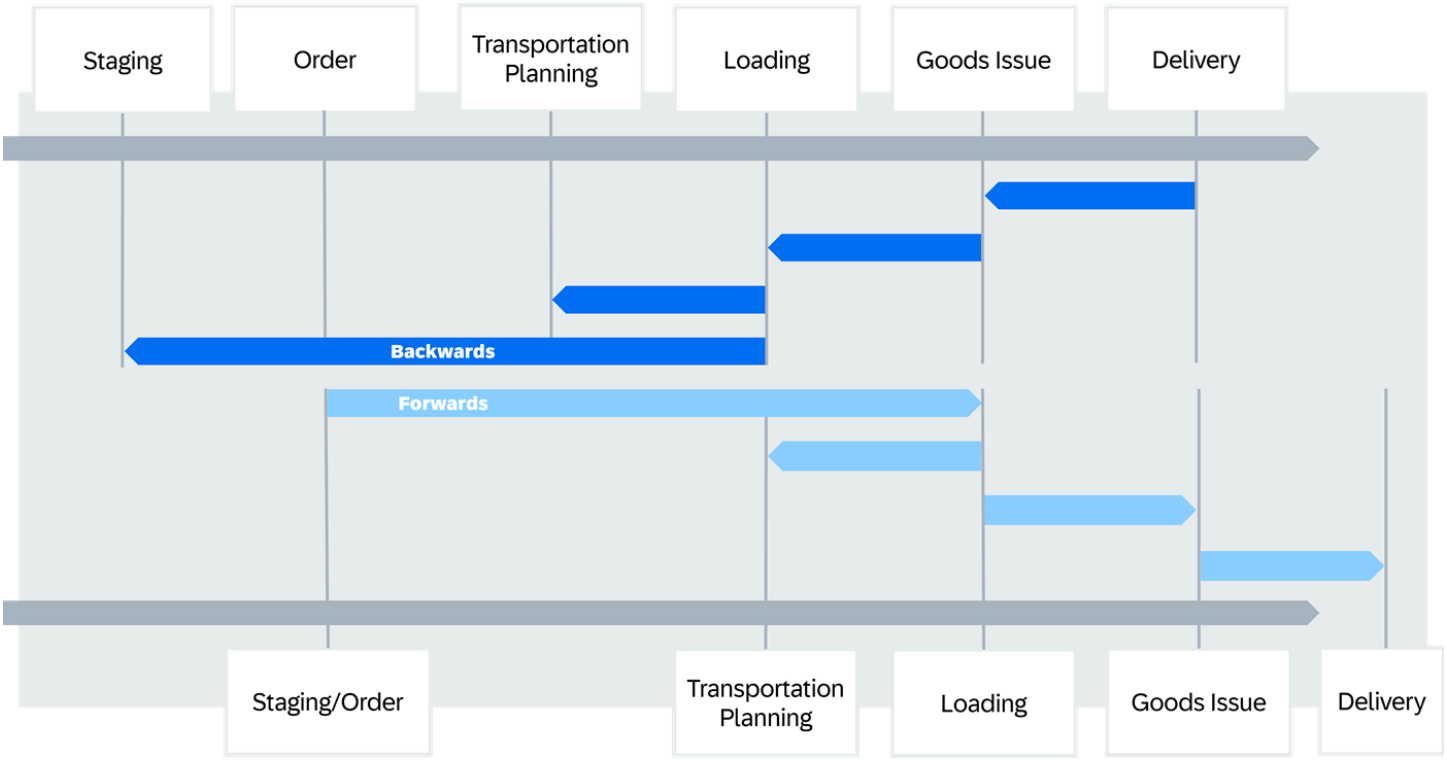
- If backwards scheduling determines a date in the past as the date on which a schedule line is due for shipping.
- If the material is unavailable on the dates determined.

For example, starting from the current date, the system calculates the loading date, the goods issue date, and then the confirmed delivery date. It does this by adding the shipping times mentioned earlier.

→ Remember

Backwards scheduling is always carried out for delivery and transportation scheduling. Suppose that backwards scheduling determines a material availability date or transportation scheduling date in the past. In that case, the system automatically carries out forwards scheduling to determine a delivery date that can be confirmed.

The following figure shows how backwards and forwards scheduling is calculated:



Rounding Rules for Data

You must maintain the working times for the scheduling to be calculated correctly. In addition, you can define different buffer times for each shipping point and add them to the loading time and the pick/pack time. This enables you to consider possible delays in the shipping activities in advance.

⚠ Caution

If you don't maintain the working times, the system always rounds the scheduling dates up or down according to standard business practice. Depending on the rounding rule, the system may determine that loading scheduled for the afternoon according to the pick/pack time won't start until the following day. This way, if there's a delay in picking and packing, the truck is still on time.

ℹ Note

You can make all relevant settings for **Delivery and Transportation Scheduling** with the following configuration activities:

- **Define Scheduling By Sales Document Type** (ID: 102129)
- **Define Shipping Point** (ID: 102125)
- **Maintain Working Hours** (ID: 102436)
- **Maintain Duration** (ID: 102523)

For more information, see Configuration Environment of SAP S/4HANA Cloud Public Edition.

Changing Outbound Deliveries

You can use the **Change Outbound Delivery** app to modify outbound delivery documents.

Context

To change an outbound delivery with the **Change Outbound Delivery** app, carry out the following procedure:

Procedure

1. In the SAP Fiori launchpad, open the Change Outbound Delivery app.
2. Enter the number of the outbound delivery document that you want to change, or use the search help to find the corresponding delivery.
3. Choose **Continue** to display the delivery overview screen.

   → Tip

   Alternatively, you can Choose **Menu** > **Goto** to navigate to specific views of the delivery that you want to modify.

4. Change the outbound delivery as required and choose **Save** (Ctrl + S).

Adding Items to Outbound Deliveries

Context

You can add the following items to existing outbound deliveries using the **Change Outbound Delivery** app:

- Order independent items
- Individual order items from other orders
- Entire orders

! Restriction

You can only add items to existing outbound deliveries if:

- Goods issue has not yet been posted
- The business transaction has not yet been invoiced

For a detailed description of creating order independent items in outbound deliveries, see Creating Order Independent Items in Outbound Deliveries

Creating Order Independent Items in Outbound Deliveries

Context

You can only add an item that is not dependent on an order to an outbound delivery if the delivery permits the item category for that item. The item category is determined based on the delivery type and item category group in the material master record. For this reason, items cannot be added to an outbound delivery that has no preceding order. Your system configuration determines which materials can be included in an outbound delivery. Carry out the following procedure:

Procedure

1. From the SAP Fiori launchpad, open the Change Outbound Delivery app.
2. Enter the number of the outbound delivery to which you want to add an item.

3. Choose **Continue**.
4. Enter the material numbers and delivery quantities as needed and choose **Save** (Ctrl + S).

Displaying Outbound Deliveries

You can use the **Display Outbound Delivery** app to view and navigate outbound delivery details.

Context

To display an outbound delivery using the **Display Outbound Delivery** app, carry out the following procedure:

Procedure

1. From the SAP Fiori launchpad, open the Display Outbound Delivery app.
2. Enter the document number of the outbound delivery that you want to display in the **Outbound Delivery** field or select the delivery using the search help.
3. Choose **Continue**.

Results

The overview screen of the outbound delivery is displayed. From here, you can navigate to the details related to the delivery. If you would like to change to the edit mode, choose the **Display ↔ Change** button.

Deleting Outbound Deliveries

You can use the **Change Outbound Delivery** app to delete outbound deliveries.

Context

To delete an entire outbound delivery, carry out the following procedure:

Procedure

1. From the SAP Fiori launchpad, open the Change Outbound Delivery app.
2. Enter the number of the outbound delivery you want to delete.
3. Press the Enter key.

   The outbound delivery screen appears.
4. Choose **Menu > Outbound Delivery > Delete**.
5. The **Delete Delivery Processing** dialog box appears. The system asks you to confirm that you really want to delete the delivery.
6. Choose **Yes** and press the Enter key.

Results

You receive a message informing you that the delivery has been deleted.

Deleting Delivery Items

Context

To delete individual items from an outbound delivery, carry out the following procedure:

Procedure

1. Open the Change Outbound Delivery app.
2. Select the item or items in the outbound delivery that you want to delete.
3. Choose **Delete item(s)** at the bottom of the screen.
4. Choose **Yes** in the dialog box that appears to confirm that you want to delete the items.
5. Choose **Save** (Ctrl + S) to save the changes.

Picking

Picking is part of the delivery and shipping process. In the picking process, items are taken from storage, placed in a staging area, and prepared for loading and shipping.

You can configure picking relevance for each delivery item category. Based on this configuration, the system behaves differently.

Overview of Picking Relevance Scenarios

The following table lists the configuration-dependent system behavior for picking relevance:

Picking Relevance	Result	Use Case
Not Relevant	No picking quantity can be maintained.	Items that do not require physical handling before shipment.
Relevant	A picked quantity must be entered.	Items that are physically picked, labeled, and packaged before shipping.

Picking Items

You indicate picked materials in the outbound delivery by entering a picked quantity. You can manually record the picked quantity to confirm the delivery data and reflect the items that have been physically picked. Entering a picked quantity for all items is a prerequisite for some follow-up activities on the outbound delivery.

⚠ Caution

The process of picking items differs depending on whether it is managed in delivery documents or with Warehouse Management. The following topics highlight the main points of each method:

- Picking in Delivery Documents
- Picking with Warehouse Management

Delivery Management Apps for Picking

For more information on Delivery Management apps you can use for picking, see the Editing section in the Feature Comparison for Outbound Deliveries topic.

Picked Quantity and Delivery Quantity

You can post goods issue without changes if the picked quantity matches the delivery quantity. If the picked quantity is less than the delivery quantity, adjust the delivery quantity to match the picked quantity before posting goods issue.

For more information on maintaining the delivery and picked quantity, see:

- Maintaining the Delivery Quantity
- Maintaining the Picked Quantity

⚠ Caution

For warehouse-relevant deliveries, the picked quantity is not updated in the SAP S/4HANA Cloud Public Edition. For more information, see the **System Behavior and Delivery Document Updates** subsection in Picking with Warehouse Management.

Picking Status and Goods Issue

After you enter the picked quantity, the system updates the picking status for the delivery item. The status is maintained per item, allowing you to see which items have been picked to adjust quantities if needed. If Warehouse Management is integrated, manual adjustments are handled through the Warehouse Management process.

You can view the picking status on the delivery's **General Information** tab when you open a delivery using the Outbound Delivery app.

The following table outlines the statuses available at the item level for each item and at the header level for the entire delivery:

Status	Description
"Blank"	Not relevant
A	Not yet processed
B	Partially processed
C	Fully processed

ℹ Note

A delivery item is partially processed if the picked quantity is less than the delivery quantity.

Once you have entered an item picked quantity and the system has set the picking status to **Fully processed**, you can post goods issue.

Determining a Picking Storage Location

Picking always occurs from a particular storage location. If a delivery item is relevant for picking, you must enter a storage location in the delivery. Picking and storage location requirements are handled as follows:

- Goods are issued from the stock at the picking storage location specified in the delivery.

- If you use Warehouse Management for picking, the storage location in the delivery determines which storage location is used for picking.

The item category of the delivery item determines the picking relevance for the item and whether or not you must specify a storage location. For example, text items are not relevant for picking, so you do not need to specify a storage location.

ℹ Note

You determine the rule for your storage location for picking with the configuration activity **Define Delivery Item Categories** (ID: 102974). For more information, see Configuration Environment of SAP S/4HANA Cloud Public Edition.

Since the information necessary for specifying a picking location is unavailable when entering the order, the person processing sales orders usually cannot decide about the picking location. This decision is made when the delivery is created at the shipping point.

The picking location can be determined in two ways:

- Manually in the delivery item. For further information, see Determining the Picking Storage Location Manually.
- Automatically, when the delivery is created and the picking storage location is entered into the delivery item. For more information on this, see Determining the Picking Storage Location Automatically.

Related Information

Picking List

Maintaining the Picked Quantity

Maintain item-level picked quantities for non-warehouse-relevant deliveries in the Change Outbound Delivery app.

Procedure

1. From the SAP Fiori launchpad, open the Change Outbound Delivery app.
2. Enter the relevant delivery number in the **Outbound Delivery** field and choose **Continue**.
3. On the **Picking** tab, enter the picked quantity in the **Picked Qty** field and choose **Save**.

   After saving, the **Picked Qty** field is updated. You can post goods issue once the system sets the picking status to **Fully processed**.

You can also maintain item-level picked quantities for non-warehouse-relevant deliveries in the Pick Outbound Delivery app.

Procedure

1. From the SAP Fiori launchpad, open the Pick Outbound Delivery app.
2. Enter the relevant delivery number in the **Delivery** field and press Enter.
3. On the **Delivery Items** section, enter the picked quantity in the **Picking Quantity** field and choose **Save**.

   After saving, the system updates the **Picking Quantity** field. You can post goods issue only when the delivery quantity matches the picked quantity. The system then sets the picking status to **Fully processed**.

Maintaining the Delivery Quantity

If the picked and delivery quantities differ, copy the picked quantity into the **Delivery Quantity** field; if they match, no changes are required before posting the goods issue.

Procedure

1. From the SAP Fiori launchpad, open the Pick Outbound Delivery app.
2. Select at least one delivery item.
3. Choose **Copy Picking Quantity**.

   The system copies the picking quantity over into the delivery quantity.

Picking in Delivery Documents

You record the results of the physical picking process directly in delivery documents, such as outbound deliveries. This basic option is suitable for physical warehouses or storage rooms that do not require complex handling or high workload management.

System Behavior

This section outlines how the system updates delivery statuses and quantities throughout picking and packing in delivery documents:

- When you create the delivery, you see the picking status as **Not Yet Picked**.
- You start picking for the delivery and enter the picked quantities per item. The system updates the item's picked quantity, the remaining open quantity, and the picking status. The header status shows the overall status of all items.
- If you pick fewer items than planned, you need to resolve the difference between the picked quantity and the delivery quantity. For example, you can reduce the delivery quantity to match the actual picked quantity.
- The system recalculates totals such as weight and volume, and refreshes dates and capacities as needed.

Picking with Warehouse Management

You can perform picking in more detail at the warehouse level. Here are the options for picking with Warehouse Management:

Using Warehouse Management in SAP S/4HANA Cloud Public Edition

This requires mapping your warehouse structure in the system and managing material movements in Warehouse Management in SAP S/4HANA Cloud Public Edition.

For more information, see Warehouse Management.

Integration with External Warehouse Management Systems

You can also manage warehouse operations using external warehouse management systems integrated with SAP S/4HANA Cloud Public Edition:

- **SAP Extended Warehouse Management (SAP EWM)**

SAP S/4HANA Cloud Public Edition supports integration with SAP Extended Warehouse Management (SAP EWM).

For more information, see Basic Integration with SAP Extended Warehouse Management (SAP EWM).

- **Third-Party Logistics (3PL) Warehouse Management Systems**

  You can use SOAP interfaces to integrate third-party warehouse management systems.

  This enables connectivity with external software solutions that provide Warehouse Management functionality, including those operated by your logistics service provider.

  For more information, see Third-Party Warehouse Management.

  ! Restriction

  There are functionality restrictions related to changing deliveries after an initial distribution to a 3PL provider Warehouse Management system. For more information, see KBA 3260042 .

System Behavior and Delivery Document Updates

If you use Warehouse Management in SAP S/4HANA Cloud Public Edition, SAP Extended Warehouse Management (EWM), or a third-party (3PL) warehouse management system, the following applies:

Picked quantities	Picked quantities are not maintained directly in delivery documents. Instead, deviating picked quantities are reflected automatically as changes to the delivery quantity.
Picking status	All warehouse-relevant items (identified by a warehouse number assigned to the **Plant** and **Storage Location** fields) have their **Picking Status** set automatically to **Not Relevant**. You cannot manually maintain picked quantities in delivery documents. This prevents inconsistencies with actual picking results in the warehouse system.
Batches and serial numbers	Batches and serial numbers assigned at warehouse level are displayed in delivery documents based on updates from the warehouse system.

Determining the Picking Storage Location Automatically

The system automatically determines the picking storage location based on the following criteria:

- Shipping point
- Plant
- Storage condition

The system displays the determined picking storage location in both the delivery item and the **Picking** overview screen of your delivery.

i Note

In the configuration activity **Define Rules for Picking Location Determination** (ID: 101107), you assign a suitable picking rule to each delivery type to automatically determine the picking location. For more information, see Configuration Environment of SAP S/4HANA Cloud Public Edition.

- The configuration activity offers you standard picking rules that you cannot change.
- Define the picking rules only for delivery types that require automatic picking location determination.

For more details on storage conditions in delivery processing, see Storage Condition.

Storage Condition

A storage condition is a predefined set of requirements that specify how goods should be stored to ensure proper handling and organization.

Maintain the storage condition in the material master record so it is available during delivery processing. This makes the storage condition information available during delivery. The system can then use it to determine the picking location automatically.

To set this up, review the storage conditions used in your company and configure how they should be applied for automatic picking location determination. Ensure the storage condition is accurately maintained in all relevant material master records.

ℹ Note

You define your storage conditions with the configuration activity **Define Storage Conditions** (ID: 102128). For more information, see Configuration Environment of SAP S/4HANA Cloud Public Edition.

To learn how to display the storage condition in your delivery, see Displaying the Storage Condition.

Prerequisites for Automatic Determination of the Picking Storage Location

The system determines the picking storage location based on the delivery type and the item category of the delivery item as follows:

- Each delivery type must have rules configured for automatic storage location determination.
- The system determines a storage location for a delivery item only if storage location determination is defined for that item's item category.

ℹ Note

You define attributes for delivery types and item categories in your configuration environment using the configuration activities **Define Delivery Types** (ID: 102123) and **Define Delivery Item Categories**. For more information, see Configuration Environment of SAP S/4HANA Cloud Public Edition.

! Restriction

You can only change the system-determined picking storage location if picking has not started. For more information on how to make these changes, see Determining the Picking Storage Location Manually.

Displaying the Storage Condition

Display the storage condition of your deliveries using the Display Material app.

Procedure

1. First, open the Manage Outbound Deliveries app from the SAP Fiori launchpad.
2. Enter the relevant search criteria, choose **Go**, and select a delivery from the list.

   The system opens the delivery in the Outbound Delivery app.

3. In the **Items** overview, select a material and choose the Display Material app from the popover.
4. On the **Select View(s)** page, choose **General Plant Data /Storage 1**, then choose **Continue**.
5. Select a plant and choose **Continue**.

   The system displays the **Plant data/stor. 1** tab. Locate the **Storage** conditions field in the **General data** section.

You can also display the storage condition using the Display Outbound Delivery app.

Procedure

1. From the SAP Fiori launchpad, open the Display Outbound Delivery app.
2. Enter the document number in the **Outbound Delivery** field or use the search help to select the delivery.

   Choose **Continue** to open the overview.
3. Select an item and navigate to **Menu** > **Environment** > **Material Master Data**.

   This opens the **Select View(s)** page.
4. Choose **General Plant Data /Storage 1** and select **Continue**.
5. Select a plant and choose **Continue**.

   The **Storage Conditions** field is located in the **General data** section of the **Plant data / stor. 1** tab.The **Storage Conditions** field is located in the **General data** section of the **Plant data / stor. 1** tab.

Determining the Picking Storage Location Manually

Manually determine the picking storage location in the delivery item overview for **one item**.

Procedure

1. From the SAP Fiori launchpad, open the Change Outbound Delivery app.
2. Enter the relevant delivery number in the **Outbound Delivery** field and choose **Continue**.
3. Select a delivery item on the delivery's **Item Overview**. Then choose **Menu** > **Goto** > **Item** > **Picking**.
4. In the **Warehouse** section, enter the picking storage location in the **Location** field.

Manually determine the picking storage location for **multiple items** from the overview screen.

Procedure

1. From the SAP Fiori launchpad, open the Change Outbound Delivery app.
2. Enter the relevant delivery number in the **Outbound Delivery** field and choose **Continue**.
3. On the delivery's **Picking** tab, enter the picking storage location in the **SLoc** field of each delivery item and choose **Save**.

   You can change the location specified in this field if picking has not started.

Packing

You specify the packing of delivery items directly in the delivery itself. You can manually select available handling units or use previously defined packing instructions.

Note

This is a function shared by more than one delivery type and its processes. For more information about this function, see the related links.

Related Information

Automatic Packing – Packing with Packing Instructions
Packing in Deliveries

Goods Issue

Goods issue confirms that the goods have left the company, leading to a reduction in warehouse stock. Once goods issue is posted, the shipping activity is finished.

You have the following options for posting goods issue:

- Directly in the Change Outbound Delivery app for a single outbound delivery
- Collective processing to manually post goods issue for multiple outbound deliveries using the Outbound Deliveries for Goods Issue app
- Collective processing to automatically post goods issue for multiple outbound deliveries in the background using the Schedule Goods Issue for Deliveries app

Remember

The data from the preceding delivery document is copied to the goods issue document and can't be manually changed in the goods issue document. If you want to change the data in the goods issue document, you need to reverse the goods issue, change the outbound delivery, and post goods issue again.

Proof of Delivery

After receiving an outbound delivery, the ship-to party can send you proof of delivery (POD). After receiving proof of delivery, you update the POD quantities. You can update them manually, or the system updates them automatically depending on your configuration.

When you confirm proof of delivery, you provide the following information:

- When you sent the confirmation
- Which order you're confirming proof of delivery for
- Received items
- Received quantities
- Differences between the received and ordered quantities and the reasons for these differences

The proof of delivery status informs you whether the ship-to party reported a proof of delivery. The table below outlines the statuses available on the item level for each item and the header level for the entire delivery:

Status	Description
[empty]	Not relevant for the POD process
A	Relevant for the POD process - not yet processed
B	Relevant for the POD process - differences reported (The POD needs to be manually confirmed.)
C	Relevant for the POD process - quantities were verified and confirmed (The POD is automatically confirmed.)

Configuring Proof of Delivery

Context

For SAP S/4HANA Cloud Public Edition to accept proof of delivery, you need to specify the following:

- Reasons for quantity differences
- POD-relevance depending on delivery item category
- POD relevance for sales areas of all corresponding ship-to-parties

ℹ Note

This is a setting that requires you to make changes in your configuration environment. For more information about the available configuration environments, see Configuration Environment of SAP S/4HANA Cloud Public Edition.

Procedure

1. Configure the reasons for quantity differences.
    a. In your configuration environment, use the search function to open the **Define Reasons for Quantity Differences** activity. (ID: 102359)
    b. Create any reasons that you need.

    ### ℹ Note

    For the generation of POD difference quantities and reasons to be carried out fully or partially automatically by the system, you need to create one reason for overdelivery and one reason for underdelivery and mark them as default reasons.

2. Configure POD-relevance depending on delivery item category.
    a. In your configuration environment, use the search function to open the **Set POD-Relevance Depending on Delivery Item Category** activity. (ID:101904)
    b. Use the input help to set the field **POD-relevant** as required.
    c. Use the input help to set the field **POD automatic** as required.

    The following **POD automatic** settings are available:

    - V = fully automatic processing: If you have defined default reasons for overdelivery or underdelivery, the system displays the default reason, and you cannot change it.

- T = partially automatic processing: If you have defined default reasons for overdelivery or underdelivery, the system displays the default reason, and you can change it as required.
- M = manual processing: The system does not display reasons for overdelivery or underdelivery and you need to enter the reason manually.

3. Configure the ship-to party.

   You need to set the sales area of every ship-to party that you want to receive proof of delivery from as POD-relevant.

   a. In the **Manage Customer Master Data** app, select a ship-to party and open or create a sales area.

   > **Note**
   >
   > The business role Master Data Specialist - Business Partner Specialist `SAP_BR_BUPA_MASTER_SPECIALIST` is required to use the **Manage Customer Master Data** app.

   b. Go to the sales area details and find the shipping section.

   c. Set the sales area as relevant for proof of delivery.

Receiving Proof of Delivery (API)

You can use the API `Proof of Delivery - Update (B2B)` to automatically receive proof of delivery (POD) for an outbound delivery from an external system. If there are no differences in the received POD, the service automatically confirms the POD.

Prerequisites

- Activation of the scope item **Supply Chain Automation with SAP Business Network** (5I2)
- Carrying out the steps described in the section *Configuration in SAP S/4HANA Cloud Public Edition* in the setup instructions for the scope item 5I2 in the SAP Signavio Process Navigator. Choose the **Accelerators** tab and then select the **Set-up instructions** tile.

Context

The Proof of Delivery – Update (B2B) SOAP API allows your ship-to parties to send you proof of delivery.

For each item in an outbound delivery, you can update the following information:

- Actual received quantities
- Differences between delivery and received quantities

If you update the received quantities only, the service compares the received quantities with the delivery quantities and calculates any differences. Alternatively, if you update the differences between the received quantities and the delivery quantities, the service calculates the received quantities. SAP recommends sending only one of these values and allowing the service to calculate the remaining values.

You can also enter a reason for any differences. SAP recommends sending a specific reason for quantity difference. If a reason is not provided and you defined a default reason in your configuration environment, the system automatically sets the default reason.

After posting, the service updates the proof of delivery quantities in the outbound delivery that you specify.

Procedure

1. Your ship-to party receives a delivery and compares the received quantities with the quantities in the outbound delivery.
2. Your ship-to party sends proof of delivery using the inbound `Proof of Delivery - Update (B2B)` API.
    - If there are no differences, the service updates the values in the outbound delivery and sets the proof of delivery status to **Confirmed**.
    - If there are differences, the service updates the values in the outbound delivery and sets the proof of delivery status to **Differences reported**. In this case, you must confirm the proof of delivery quantities manually.

Manually Confirming Proof of Delivery

If you received a proof of delivery (POD) using the API and there are quantity differences, you need to manually confirm the POD.

You can use the following apps to manually confirm POD:

- Deliveries Worklist – Proof of Delivery
- Change Outbound Delivery – Proof of Delivery

Related Information

Proof of Delivery – Update (B2B)

(Optional) Creating a Worklist

If you want to confirm proof of delivery (POD) for multiple deliveries, you can create a worklist that allows you to see all of the deliveries relevant for processing.

Procedure

1. In the SAP Fiori Launchpad, open the **Deliveries Worklist - Proof of Delivery** app.
2. Enter a date and time for the proof of delivery confirmation.
3. Enter the search criteria that you want to use to generate the worklist and choose **Run** (F8).

   The app displays outbound deliveries that match your search criteria.
4. Select the deliveries that you would like to confirm proof of delivery for and choose **Confirm.**
5. (Optional) To add differences for individual deliveries or to confirm proof of delivery individually, select a delivery and choose **Menu** > **Goto** > **Process Delivery**.

   The **Proof of Delivery** screen opens.

   You can now continue as described in step 3 of Confirming Proof of Delivery (App).

Confirming Proof of Delivery (App)

Confirm the quantities for single deliveries using the **Change Outbound Delivery – Proof of Delivery** app.

Procedure

1. In the SAP Fiori launchpad, open the **Change Outbound Delivery – Proof of Delivery** app.
2. Enter the ID of the outbound delivery that you want to update proof of delivery for and choose **Continue**.

The **Proof of Delivery** screen opens.

3. On the **Overview POD** tab, for each item, enter the received quantities in the **POD quantity** field.
4. Use the input help to enter the sales unit for the material in the **SU** field.
5. Use the input help to specify a reason for overdelivery or underdelivery in the **Reason** field.
6. (Optional) To be able to add multiple reasons and quantities for an item, choose *POD Detail: Reasons and Qnties* in the **Details: Proof of Delivery** field and enter the details as required.
7. To confirm the POD choose  (Confirm Proof of Delivery) above the item list, to the right of the **POD Date** field.

Canceling Proof of Delivery

After confirming proof of delivery, you can cancel it to discard any entered quantities.

Context

You can only cancel the POD confirmation if:

- Billing documents have not yet been created
- No subsequent corrections have been made for value or quantity

After canceling, you can change the difference quantities and reasons and reconfirm if necessary.

Procedure

1. In the SAP Fiori launchpad, open the **Change Outbound Delivery – Proof of Delivery** app.
2. Enter the ID of the outbound delivery that you want to cancel proof-of-delivery confirmation for and choose **Continue**.

   The **Proof of Delivery** screen opens.
3. Next to the **POD Date**, choose the **Cancel Proof of Delivery** icon.
4. Choose **Save** to save the changes to the POD status.

Output Management

The output for delivery documents can be in various formats that are visible outside of SAP S/4HANA Cloud Public Edition. For instance, you can create PDFs for printing or e-mails for sending. It is possible to generate output for multiple document types, such as delivery notes and quality certificates.

The output for outbound delivery documents and delivery items uses the **Output Control** in SAP S/4HANA Cloud Public Edition. It is based on SAP Interactive Forms, form data providers, and BRFplus based configurations. To configure the output of delivery documents, you can use the **Output Parameter Determination** app.

Related Information

Manage Output Items for Deliveries

Schedule Output for Deliveries

Quality Certificate

Delivery Note

Picking List

Packing List

Output Management for Outbound Delivery Items

The output for outbound delivery items is based on the **Output Control** in SAP S/4HANA Cloud Public Edition. It is based on SAP Interactive Forms, form data providers, and BRFplus based configurations.

Key Features

- You can display a list of output items issued for a delivery document item.
- You can view the status of output items (for example, in preparation, to be output, completed, error).
- You can assign the output type (for example, quality certificate).
- You can choose the recipient and channel, the printer settings, the email settings of the sender and receiver, the output queue settings, and the form (PDF) template used for your output items.
- You can create your own form template consisting of a custom layout of the output PDF and the fields in the standard form data provider.
- You can preview output items (for example, the PDF of a quality certificate).
- You can send output and resend output.

Example

Under **Menu** > **Extras** > **Delivery Output** > **Item** in the **Change Outbound Delivery** app, you can preview the PDF of an outbound delivery document by choosing **Display Document** or make additional settings as needed.

Business Rules

As an administrator, you define when and how to issue an output in the **Output Parameter Determination** app. With this app, you set output rules for the object type **Outbound Delivery Item**.

Note

If you change a business rule, this change is not automatically applied to existing deliveries (except in the case of the output relevance rule). You must manually update the outputs for deliveries that already exist. For example, if you change the form template rule, you have to manually change the form template for existing outputs.

Output Type

In the **Output Type** determination step, you define the business rules that control which output type is issued for each delivery.

Supported output types:

- `QUALITY_CERTIFICATE` (Quality Certificate)

Application Condition Parameters

You can use the following application condition parameters to define your business rules:

- Item Category
- Sales Organization
- Shipping Point
- Distribution Channel
- Ship-To Party
- Sold-To Party
- Material Group
- Material
- Material by Customer
- Material Type
- Plant
- Delivery Type
- Division
- SD Document Category
- Overall Picking Status
- Total Goods Movement Status

ℹ Note

To avoid performance issues, only include the application condition parameters that you really need in your output type determination step.

Receiver

The receiver is required to determine the recipient address.

Channel

In the **Channel** determination step, you define how to issue the output type (for example, print, email, EDI, `EXTOM` (External Output Management)).

ℹ Note

Prerequisites for the channel `EXTOM` (External Output Management):

- Activation of the scope item 7DJ (Integration of External Output Management Solution)
- Creation of a technical communication user and a communication system as described in the setup instructions for the scope item 7DJ
- Setting up the communication arrangement as described in the setup instructions for the scope item 7DJ

The setup instructions for the scope item 7DJ can be found in the **Accelerators** tab of Integration of External Output Management Solution (7DJ) in the SAP Signavio Process Navigator. For more information about this channel, see External

Output Management.

Printer Settings

If you want to print the output, you can configure the print queue.

Email Recipient

If you want to print the output, you can configure the email type codes (TO, CC, BCC) and the email address. This step is optional. In case no values are maintained in this step, the email address is defaulted based on the receiver for which role is maintained in the “Receiver” step.

Output Queue Settings

If you want to send the output in XML format to an external system using the channel `EXTOM`, an output queue for the external system needs to be available for each output type in the **Manage Output Queues** app. In this step, you assign the output queue to the output type. The output queue owner is the technical user you have defined for the communication scenario `SAP_COM_0881`. For more information, see Manage Output Queues.

Form Template

The form template defines the layout and content of the PDF document that is to be created for printouts and email attachments. You can create your own form template, or you can copy an existing form template and adapt it as required. After creating the form template, you assign it to the output type.

Output Relevance

Output relevance defines the conditions that must be met for the output to be issued.

An output can be issued only once. To resend an output, you must manually create a new output.

ℹ Note

Note

Outputs are not issued automatically when a delivery is saved if:

- A delivery block is set where printing is blocked
- The product marketability status is set to either **Blocked** or **Partially Blocked**
- The dangerous goods status is set to **Blocked**

You can, however, manually issue the output.

Quality Certificate

A quality certificate confirms that the quality of goods is acceptable. For example, it contains inspection results, characteristics for the batch, and delivery data.

Prerequisites

You’ve configured the following:

- Quality certificate profile
- Certificate profile assignment

If you haven't met these prerequisites, output items that are created have the status **Error**. All output items remain in the status **In Preparation** until you have saved the relevant delivery. If you create the output manually, saving the related delivery doesn't change the status of the quality certificate.

Supported Output Channels

You can create the following types of output for quality certificates for outbound deliveries:

- Print
- Email
- EDI
- EXTOM (External Output Management)

Configuration

The quality certificate should be created only after picking is complete. Therefore, the **Overall Picking Status** in **Output Type** and **Output Relevance** must always be set to **C** (complete).

Supported dispatch time is **1** (Immediately).

Supported role is **WE** (Ship to party).

Extensibility

To adapt the layout of the quality certificate to your needs, copy one of the following form templates for the corresponding output channel:

- `QM_QUALITY_CERTIFICATE_FORM` for a printout
- `QM_QUALITY_CERT_SERIAL_FORM` for a printout that takes serial numbers into account with the inspection results sorted by serial number
- `QM_QUALITY_CERTIFICATE_EMAIL` for an e-mail

Related Information

Form Templates

Output Management for Outbound Delivery Documents

The output for outbound delivery documents is based on the **Output Control** in SAP S/4HANA Cloud Public Edition. It is based on SAP Interactive Forms, form data providers, and BRFplus based configurations.

Key Features

- You can display a list of output items issued for a delivery document.

- You can view the status of output items (for example, in preparation for sending, to be output or processed, pending processing, successfully processed, error while processing).
- You can assign the output type (for example, delivery note, delivery picking list).
- You can choose the recipient and channel (for example, by email, printing), the printer settings, the email settings of the sender and receiver, and the form (PDF) template used for your output items.
- You can create your own form template consisting of a custom layout of the output PDF and the fields in the standard form data provider.
- You can preview output items (for example, the PDF of a delivery note).
- You can send output and resend output.

Example

Under Menu > Extras > Delivery Output > Output Control in the Change Outbound Delivery app, you can preview the PDF of an outbound delivery document by choosing Display Document or make additional settings as needed.

Business Rules

As an administrator, you define when and how to issue an output in the Output Parameter Determination app. With this app, you set output rules for the object type Outbound Delivery.

ℹ Note

If you change a business rule, this change isn't automatically applied to existing deliveries (except in the case of the output relevance rule). You must manually update the outputs for deliveries that already exist. For example, if you change the form template rule, you have to manually change the form template for existing outputs.

Output Type

In the Output Type determination step, you define the business rules that control which output type is issued for each delivery. When the delivery is saved for the first time, the outputs for the delivery are determined.

ℹ Note

When a goods issue posting is reversed, the output type determination for delivery note is executed again and a new output can be added.

Supported output types:

- `DELIVERY_NOTE` (Delivery Note)
- `DELIVERY_PICK_LIST` (Delivery Picking List)
- `DELIVERY_PACKING_LIST` (Delivery Packing List)

Supported dispatch times:

- 1 (Immediately)

Application Condition Parameters

You can use the following application condition parameters to define your business rules:

- Delivery Number
- Document Category
- Delivery Type
- Sales Organization
- Shipping Point
- Route
- Ship-To Party
- Sold-To Party
- Total Goods Movement Status
- Overall Picking Status
- Status Decentralized Warehouse
- Warehouse Execution Status

ℹ Note

To avoid performance issues, only include the application condition parameters that you really need in your output type determination step.

Channel

In the **Channel** determination step, you define how to issue the output by choosing one of the following supported channels:

- `PRINT` (Printout)
- `EMAIL` (Email) – only relevant for delivery notes
- `EDI` (Electronic Data Interchange) – only relevant for delivery notes
- `EXTOM` (External Output Management) – only relevant for packing list

  For more information about setting up external output management for handling unit related outputs, see Output Management for Handling Unit Management.

Form Template

The form template defines the layout and content of the PDF document that is to be created for printouts and email attachments. You create your own form template and assign it to the output type.

Receiver

The receiver is required to determine the recipient address.

Output Relevance

Output relevance defines the conditions that must be met for the output to be issued.

An output can be issued only once. To resend an output, you must manually create a new output.

ℹ Note

Outputs aren't issued automatically when a delivery is saved if:

- A delivery block is set where printing is blocked
- The product marketability status is set to either **Blocked** or **Partially Blocked**
- The dangerous goods status is set to **Blocked**

You can, however, manually issue the output.

Related Information

Output Parameter Determination

Delivery Note

Delivery notes describe the contents of a delivery. They include details such as addresses, business partners, products, dates, quantities and weights, dangerous goods, and handling units information. The system creates a delivery note at different times depending on the type of delivery. For outbound deliveries, it creates the note when you post goods issue.

Note

If a delivery block prevents a goods movement, the system doesn't create a delivery note when you save a delivery. You can manually create a delivery note if necessary.

After creating a delivery note, you can re-create the output PDF document at any time. If you re-create the document, it's issued as a changed delivery note. You use the output-management reuse service to re-create a delivery note.

Supported Output Channels

You can create the following types of output for delivery notes:

- Printout
- E-mail
- EDI

Supported Delivery Types

You can create this output document for outbound deliveries.

Extensibility

To adapt the layout of the delivery note to your needs, copy form template `FDP_V2_DELIVERY_NOTE_V1`.

You can extend the delivery note using the **Custom Fields** app by adding fields to the **Shipping: Delivery** or **Shipping: Delivery Item** business context.

Picking List

Picking lists show all of the items in a delivery that need to be picked. They include details such as product information, business partners, storage locations, quantities and weights, and dangerous goods information. The system automatically creates the picking list when you create a delivery.

ℹ Note

If a delivery block prevents a goods movement, the system doesn't automatically create a picking list when you save a delivery. You can manually create a picking list if necessary.

After creating a picking list, you can re-create the output PDF document at any time. If you re-create the document, it's issued as a changed picking list. You use the output-management reuse service to re-create a picking list.

Supported Output Channels

You can create the following types of output for picking lists:

- Printout

Supported Delivery Types

You can create this output document for outbound deliveries.

Extensibility

To adapt the layout of the picking list to your needs, copy form template `FDP_V2_PICKING_LIST`.

You can extend the picking list using data source extensions with the **Print Delivery Pick List** data source.

Packing List

Packing lists show the handling units with the products they contain, including the serial numbers, quantities, and batch data.

ℹ Note

To output a packing list with handling units in a **WM-managed** storage location, first set the WM outbound delivery to the status **Ready for Shipping**. You can do this in the **Manage Outbound Delivery Orders** app (see Manage Outbound Delivery Orders). Once the status is set, the output form of the packing list is updated with the handling unit information when you trigger the packing list output for the Delivery Management outbound delivery.

Configuration

To create output for a packing list, a key user for configuration must maintain an entry for the form template in the **Assign Form Templates** configuration activity (ID: 102313) in your configuration environment.

Supported Output Channels

You can create the following types of output for packing lists:

- `PRINT` (Printout)
- `EXTOM` (External Output Management)

Supported Delivery Types

You can create this output document for outbound deliveries.

Extensibility

To adapt the layout of the packing list to your needs, copy form template `FDP_PACKING_LIST`.

You can extend the packing list using the **Custom Fields** app by adding fields to the **Handling Unit**, **Shipping: Delivery**, or **Shipping: Delivery Item** business context.

Specific extensibility options:

- You can add export control statements on your custom form template. For more information, see Print Statements.
- You can add information about configurable materials in Advanced Variant Configuration (AVC) to your custom form template. For more information, see Output Management.

For more information, see Output Extensibility.

How to Add Additional Business Partner Roles to a Custom Form Template

Add additional partner role information to your document output.

Prerequisites

A business role that contains the business catalog `SAP_CORE_BC_OM` (**Output Management**) is assigned to your user.

Standard Fields

The form templates that SAP delivers include several standard business partner roles:

- Sold-to party (`SP`)
- Ship-to party (`SH`)

You can add additional business partner information, such as the name of responsible employee and contact person information, to your custom form templates.

Adapt the Configuration

1. In your configuration environment, use the search function to open the **Set Up Partner Determination for Delivery** (ID: 101287) activity. For more information on configuration environments, see Configuration Environment of SAP S/4HANA Cloud Public Edition.
2. Select the relevant partner determination procedure (for example, for outbound delivery) and open the partner functions in this procedure.
3. To add a specific partner function to your custom form templates, mark it as relevant for printing. You can also enter a text for the print label.

   After saving your changes, you can download the form template which will now include the print-relevant partners.

> **Note**
>
> This configuration applies to the picking list and the delivery note. If a partner function is not used in a document, the field doesn't contain a value in the output.

Related Information

[How to Create and Use a Custom Form Template]

How to Create and Use a Custom Form Template

Create your own form template and use it for your document output.

Prerequisites

A business role that contains the following business catalogs is assigned to your user:

- `SAP_CORE_BC_OM` (**Output Management**)
- `SAP_CA_BC_IC_LND` (**End to End Implementation Experience - Implementation Cockpit**)
- `SAP_CA_BC_OC_PC` (**Output Management - Output Control**)

Download a Predelivered Template

1. In the **Maintain Form Templates** app, open the predelivered templates.

   The following templates are available:

Form Template	Outbound Delivery Document or Outbound Delivery Item
`FDP_V2_PICKING_LIST`	Picking list
`FDP_V2_DELIVERY_NOTE_V1`	Delivery note
`FDP_PACKING_LIST`	Packing list
`QM_QUALITY_CERTIFICATE_FORM`	Form for quality certificate
`QM_QUALITY_CERT_SERIAL_FORM`	Form for quality certificate that takes serial numbers into account with the inspection results sorted by serial number

2. Choose the form template you need (for example, `FDP_V2_PICKING_LIST`), and download the language-dependent form templates you need.

Adapt the Template to Your Needs

1. Open the form template you downloaded in Adobe LiveCycle Designer.

   For more information about the Adobe LiveCycle Designer, see [Install Additional Software].

2. Change the form template according to your business needs. Pay attention to the data view tab.

→ **Tip**

To view a clearly arranged, searchable overview of all fields in the template, you can also open the template .XSD file in a spreadsheet application.

3. Save your adapted form template under a new name.

Upload Your Custom Form Template

1. In the **Maintain Form Templates** app, create a new custom template.
2. For the form template file, choose the file that you adapted earlier.
3. Choose the correct data source (for example, `FDP_V2_PICKING_LIST_SRV` for outbound delivery).

   The following data sources are available:

Data Source	Outbound Delivery Document or Outbound Delivery Item
`FDP_V2_PICKING_LIST_SRV`	Picking List
`FDP_V2_DELIVERY_NOTE_V1_SRV`	Delivery Note
`FDP_LOHUM_HU_PACKINGLIST_SRV`	Packing List
FDP_QC_QUALITY_CERTIFICATE_SRV	Form for Quality Certificate

4. Choose the content form type.

Assign Your Custom Form Template

1. In your configuration environment, use the search function to open the configuration activity **Assign Form Templates**. For more information, see Configuration Environment of SAP S/4HANA Cloud Public Edition.
2. Create a new entry. Use, for example, `OUTBOUND_DELIVERY` as the application object type and `DELIVERY_NOTE` as the output type. For the form template ID, use the one of the custom form template that you created earlier.

Define Business Rules for Your Custom Form Template

1. In the **Output Parameter Determination** app, show the rules for the outbound delivery or outbound delivery item and the determination step for form templates.
2. Select the relevant rule (for example, for outbound delivery), and edit it.

   Note that alternatively, you can also create a new rule.
3. In the form template column to the very right, choose the entry (for example, `FDP_V2_PICKING_LIST`).
4. Change the form template name to the one of the custom form template that you created earlier.

Flexible Grouping of Delivery Items on the Delivery Note

Your business may require that the items on a delivery note are grouped in a specific way — for example, by handling unit header attributes, by customer purchase order, or by a combination of both. The rules by which the items are grouped may vary from one

delivery to the next, depending on the attributes of the delivery being output. You can achieve this goal by using flexible delivery note grouping to define custom grouping logic and apply it dynamically via a Cloud BAdI.

Standard Output of the Delivery Note

SAP predelivers a standard form template for the delivery note (`FDP_V2_DELIVERY_NOTE_V1`). The standard output renders all delivery items in a flat list without any grouping.

You can create copies of the standard form template and modify them to create custom form templates that render grouped output.

→ Tip

For more information about form templates, see How to Create and Use Custom Form Templates.

Grouped Output Using the Cloud BAdI

If you implement the Cloud BAdI `LE_SHP_DELIVERY_NOTE_GROUP`, the Delivery Note Form Data Provider assembles a two-level group structure and includes it in the output payload alongside the existing flat item list. The grouped structure is additive. The flat `DeliveryItemNode` list is always populated and is not affected. Standard form templates that do not use the group structure continue to work without modification.

The group structure supports the following two levels:

- **Level 1 (`GroupNode`):** The top-level grouping criterion.
- **Level 2 (`SubGroupNode`):** An optional sub-grouping criterion within each Level 1 group.

Each level is identified by an `AggregationID` (a key value, up to 20 characters) and an `AggregationName` (a descriptive label, up to 120 characters). The BAdI implementation decides what values to assign to these fields. The system imposes no constraints on which delivery document fields are used as grouping criteria.

Items are assigned to groups via item nodes at each level:

Level	Item Node	Supported Item Keys
Level 1	`GroupItemNode`	Delivery item (`DeliveryItem`) or handling unit (`HandlingUnitInternalId`)
Level 2	`SubGroupItemNode`	Delivery item (`DeliveryItem`) or handling unit (`HandlingUnitInternalId`)

You can use Level 1 item nodes alone (without any Level 2 groups), or combine both levels. An item can appear in multiple groups if your business logic requires it.

How the Group Structure Affects the Form Template

The standard predelivered form template for the delivery note does not render the group structure. To output grouped delivery notes, you must use a custom form template that binds to the `GroupNode/GroupItemNode/SubGroupNode/SubGroupItemNode` elements of the payload.

Within the form template, each grouping level is represented as a repeating subform:

- `GroupNode[*]`: One repetition per Level 1 group. Bind to `AggregationID` and `AggregationName` to display the group header.
- `GroupItemNode[*]`: One repetition per item assigned directly at Level 1 (used when no Level 2 sub-grouping is needed). Navigate to the delivery item data via the `DeliveryItemNode` navigation node, or to HU data via the `HandlingUnitHeaderNode` navigation node.
- `SubGroupNode[*]`: One repetition per Level 2 group within a Level 1 group. Bind to `SubAggregationID` and `SubAggregationName`.
- `SubGroupItemNode[*]`: One repetition per item within a Level 2 group. Navigate to the delivery item data via the `DeliveryItemNode` navigation node, or to HU data via the `HandlingUnitHeaderNode` navigation node.

ℹ Note

Navigation nodes inside `GroupItemNode` and `SubGroupItemNode` are assembled automatically by the Form Data Provider based on the item keys returned by the BAdI. You do not need to duplicate item data in the BAdI implementation.

Implementing the BAdI

Create a BAdI implementation for `LE_SHP_DELIVERY_NOTE_GROUP`.

The BAdI is called once per delivery note during payload assembly. It receives the full delivery header, all delivery items, and all handling unit data as input. The BAdI implementation returns flat output tables that the Form Data Provider uses to assemble the nested group XML.

The following output tables must be populated:

`groups_out` - Level 1 groups

One entry per Level 1 group. Required fields:

Field	Description
`DeliveryDocument`	The delivery document number (copy from `delivery_document_in`).
`AggregationID`	The group key (max. 20 characters).
`AggregationName`	The group label (max. 120 characters), for example a descriptive name.

`subgroups_out` - Level 2 groups (optional)

One entry per Level 2 group. Required fields:

Field	Description
`DeliveryDocument`	The delivery document number.
`AggregationID`	Foreign key to the parent Level 1 group entry.
`SubAggregationID`	The sub-group key (max. 20 characters), for example the customer purchase order number.
`SubAggregationName`	The sub-group label (max. 120 characters).

`group_items_out` - items at Level 1 (optional)

One entry per item assigned directly to a Level 1 group. Required fields:

Field	Description
`DeliveryDocument`	The delivery document number.
`AggregationID`	Foreign key to the parent Level 1 group entry.
`DeliveryItem`	The delivery item number. Populate this field for delivery item references; leave blank for HU references.
`HandlingUnitInternalId`	The handling unit internal ID. Populate this field for HU references; leave blank for delivery item references.

`subgroup_items_out` - items at Level 2

One entry per item assigned to a Level 2 group. Required fields:

Field	Description
`DeliveryDocument`	The delivery document number.
`AggregationID`	Foreign key to the parent Level 1 group entry.
`SubAggregationID`	Foreign key to the parent Level 2 group entry.
`DeliveryItem`	The delivery item number (for delivery item references).
`HandlingUnitInternalId`	The handling unit internal ID (for HU references).

Implementation Procedure

Perform the following steps to implement flexible delivery note grouping.

1. Create a BAdI implementation for `LE_SHP_DELIVERY_NOTE_GROUP`.
2. Create a custom form template (XDP) based on the standard delivery note template. Add repeating subforms bound to `GroupNode[*]`, `SubGroupNode[*]`, and `SubGroupItemNode[*]` (or `GroupItemNode[*]` for Level 1 item grouping) to render the grouped output.
3. Assign your custom form template to the relevant output type in the output configuration so that it is used when delivery notes are generated.
4. Test the output using a delivery document that your BAdI implementation will group. Verify that the group structure in the rendered output matches the expected grouping logic.

Output Extensibility

You can extend the output documents by adding additional standard fields and your own custom fields.

Adapting Output Forms

You can add fields to and customize the layout of output forms using Adobe LiveCycle Designer. You can add fields to output forms if they are available in the form data provider. You can add fields to the form data provider by adding them to the business context, extending the data sources, and using the output parameter determination. For more information about adapting output forms, see **Developing Custom Form Templates**.

Customizable Objects

To adapt the layout of the delivery note to your needs, copy form template `FDP_V2_DELIVERY_NOTE_V1`.

To adapt the layout of the packing list to your needs, copy form template `FDP_PACKING_LIST`.

To adapt the layout of the picking list to your needs, copy form template `FDP_V2_PICKING_LIST`.

To adapt the layout of the quality certificate to your needs, copy one of the following form templates for the corresponding output channel:

- `QM_QUALITY_CERTIFICATE_FORM` for a printout
- `QM_QUALITY_CERT_SERIAL_FORM` for a printout that takes serial numbers into account with the inspection results sorted by serial number
- `QM_QUALITY_CERTIFICATE_EMAIL` for an e-mail

Adding Standard Fields (Business Context)

Context

You add additional standard fields to document templates using the **Custom Fields** app. Carry out the following procedure:

Procedure

1. Go to the **Custom Fields** app.
2. Select a business context that you want to add fields to.

   > **Note**
   >
   > You can extend the packing list using the **Custom Fields** app by adding fields to the **Handling Unit**, **Shipping: Delivery**, or **Shipping: Delivery Item** business context.
   >
   > You can extend the delivery note using the **Custom Fields** app by adding fields to the **Shipping: Delivery** or **Shipping: Delivery Item** business context.

3. Add the new field to your output form.

   For more information, see Adapting Output Forms.

Adding Standard Fields (Data Source Extensions)

Context

Add additional standard fields to outbound delivery document templates by creating a data source extension in the **Custom Fields** app.

Procedure

1. Create a data source extension.
    a. In the **Custom Fields** app, open the data source extensions tab.
    b. Create a new data source extension for a valid data source and enter a description.

    > **Note**
    > You can extend the picking list using data source extensions with the **Print Delivery Pick List** data source.

    The system automatically creates the extension ID based on your description.
2. Add fields to the data source extension.
    a. Expand the list of available standard fields to navigate to the field you want to add.

    Choose, for example, the outbound delivery header level and select the planned goods issue date.

    b. If needed, change the name of the selected field.
    c. Save and publish your data source extension.
3. Add the new field to your form template.

    For more information, see Adapting Output Forms.

Results

When you add fields to custom form templates using data source extensions, these fields are not displayed in the PDF preview, only in the final output.

Adding Custom Fields (Output Parameter Determination)

Context

You add custom fields to your deliveries in the **Custom Fields** app. Carry out the following procedure:

Procedure

1. In the **Custom Fields** app add the custom field to an outbound delivery document.
2. Go to the **Output Parameter Determination** app and refresh the condition parameters of the application.

    See the section *Refresh Condition Parameters of Application* in https://support.sap.com/content/dam/SAAP/Sol_Pack/S4C/Library/Setup/1LQ_Set-Up_EN_XX.pdf for detailed instructions.
3. Add the new field to your form template.

    For more information, see Adapting Output Forms.

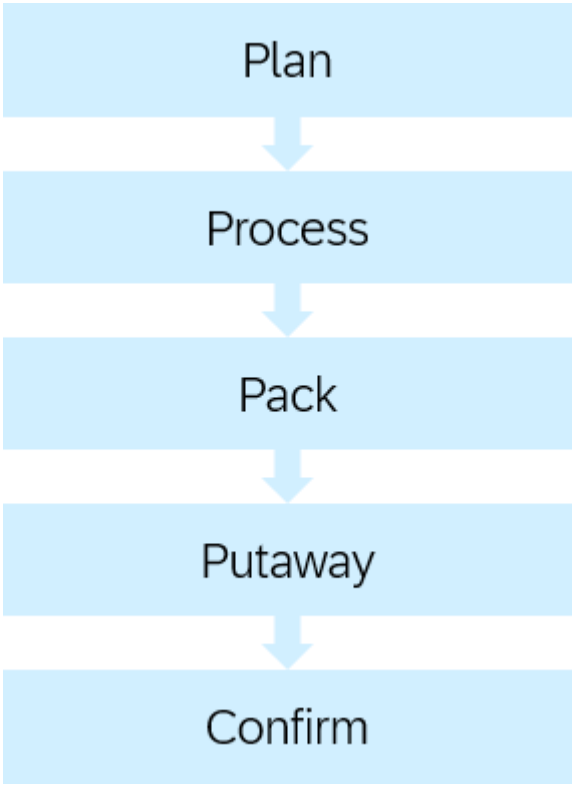
Related Information

SAP S/4HANA Output Control
Planning of Outgoing Quality Certificates
Custom Fields App and Custom Logic App
Key User Extensibility
Custom Fields

Inbound Delivery Process

The inbound delivery process includes all steps related to the receipt of goods from the supplier.

The inbound delivery process includes the following steps:



Please note that image maps are not interactive in PDF outputs.

Creating and Processing Inbound Deliveries

When an outbound delivery is underway, a corresponding inbound delivery can be created to record the incoming goods. This document contains information such as the items in the delivery, weight, and delivery data. An inbound delivery can be manually or automatically created based on an order document. The goods receiving point is determined by the system based on configuration settings.

You can create inbound deliveries with reference to a:

- Purchase order
- Shipping notification
- Customer returns order

Overviews help you monitor the created inbound deliveries and related activities. After goods receipt, you can update the putaway quantity or any other data that has been changed in the delivery. You can also use the display function to view information about the inbound delivery.

The following table outlines the options for creating inbound deliveries:

Manual creation (apps)	Create Inbound Delivery

	Inbound Deliveries for Purchase Orders
Automatic creation with batch processing (app)	Schedule Inbound Delivery Creation
Automatic creation with APIs	Inbound Delivery (A2X)
	Inbound Delivery – Create (B2B)

Note

For scope item specific process descriptions, refer to **PRODUCT FEATURES** > **Scope Items** in the documentation for the relevant app in the SAP Fiori Apps Reference Library.

Working with Inbound Deliveries

You can manually process outbound deliveries using the corresponding app. The following links provide details about working with inbound deliveries:

Creating Inbound Deliveries

You can create a single inbound delivery based on a purchase order and change the delivery as required.

Changing Inbound Deliveries

You can change an inbound delivery using the **Change Inbound Delivery** app.

Displaying Inbound Deliveries

You use the display mode to view specific information about an inbound delivery. You cannot make changes to the inbound delivery in display mode.

Deleting Inbound Deliveries

You can use the **Change Inbound Delivery** app to delete entire inbound deliveries and their corresponding delivery items.

Batches in Inbound Deliveries

Batch functionality can be used with inbound deliveries.

Creating Inbound Deliveries

You can create a single inbound delivery based on a purchase order and change the delivery as required.

Context

The system automatically generates the inbound delivery with the items and information from the purchase order.

Procedure

1. From the SAP Fiori launchpad, open the Create Inbound Delivery app.
2. Select a supplier and enter the PO number of the purchase order you want to create the delivery for.
3. Select the delivery date.

   The delivery date is the date that goods are expected to arrive based on information from your supplier.
4. Press the Enter key.
   - The system creates the inbound delivery according to your specifications.

- The inbound delivery overview screen is displayed.
- Data from the purchase order is copied to the inbound delivery.

You can enter additional details on the header and item screens and change the information in the purchase order.

Inconsistencies that occur in a schedule line are recorded in the error log.

> **Note**
> If you select a purchase order whose items are to be put away in different warehouses with warehouse numbers, the system automatically creates several inbound deliveries in the background.

5. Choose **Menu** > **Inbound delivery** > **Save** to save the delivery.

Results

When the system has saved the document, it displays the number of the inbound delivery.

The system determines the goods receiving point according to the receiving plant and receiving storage location that you specified in the inbound delivery. You can define several entries for the goods receiving point and assign a priority to each one.

Next Steps

The system records errors that occurred during delivery creation in a log. The system displays a message in the case of errors.

You can navigate to the log from any of the overview screens in the delivery by choosing **Menu** > **Edit** > **Error log**. From there, you can make changes directly to the document.

Creating Multiple Inbound Deliveries

Context

You can manually create multiple inbound deliveries with reference to multiple purchase orders using the **Inbound Deliveries for Purchase Orders** app. The system copies the data from the purchase orders to the inbound deliveries.

> **Tip**
> As an alternative, you can use a job scheduling app to define a periodic job that automatically creates inbound deliveries in the background. For more details, see Schedule Inbound Delivery Creation.

To manually create multiple inbound deliveries with reference to multiple purchase orders, carry out the following steps:

Procedure

1. From the SAP Fiori launchpad, open the Inbound Deliveries for Purchase Orders app.
2. Enter the criteria you want to use to select purchase orders.
3. (Optional) Select the **Test Run** option to test the selection before creating inbound deliveries.
4. Choose **Run** (F8).

Changing Inbound Deliveries

You can change an inbound delivery using the **Change Inbound Delivery** app.

Context

In addition to changing header and item data, you can change the structure of inbound deliveries. For example, you can delete individual items from the inbound delivery or add new items to it that exist in the purchase order.

Procedure

1. From the SAP Fiori launchpad, open the Change Inbound Delivery app.
2. Enter the number of the inbound delivery you want to change or choose the input help to open a dropdown for defining search criteria for selecting deliveries.
3. Choose **Continue**.
4. Choose **Menu** > **Goto** to navigate to a corresponding view of the inbound delivery and make changes as required.
5. Choose **Menu** > **Inbound delivery** > **Save** to save the changes.

Results

The system displays a message confirming that the inbound delivery has been saved.

View Inbound Deliveries Available for Changes

You can view open inbound deliveries before deciding which to change.

Procedure

1. From the SAP Fiori launchpad, open the General Inbound Delivery List app.
2. Enter your selection criteria.
3. Choose **Execute**.

   The system displays a list of inbound deliveries that meet your selection criteria.
4. Select the inbound deliveries to be changed and choose **Change Inbound Deliveries** on the bottom right of the screen.

   The overview screen of the delivery you selected appears.
5. Make the required changes.
6. Save the changes. If you have selected multiple deliveries, the system displays a message asking if you would like to process the next delivery from your selection.

Displaying Inbound Deliveries

You use the display mode to view specific information about an inbound delivery. You cannot make changes to the inbound delivery in display mode.

Procedure

1. From the SAP Fiori launchpad, open the Display Inbound Delivery app.
2. Enter the number of the inbound delivery that you want to display or choose the input help to display a list of deliveries based on your defined search criteria.

3. Press the Enter key.

Searching for Inbound Deliveries

To find a specific inbound delivery, you use the Enterprise Search in the header bar of the SAP Fiori launchpad. Before you enter any attributes in the search input field, select the category **Inbound Delivery**.

Context

In the search input field, you can search for an inbound delivery. For example, you could search for a material number. To reduce the number of search results, you can also combine several attributes, such as a material number and a sold-to party. The search takes into account the following fields:

- Delivery document type
- Created on
- Created by
- Shipping point
- Material number
- Supplier (ID and name)
- External delivery ID
- Material description
- Receiving point

Results

A list of inbound deliveries appears as a result. For the first inbound delivery, the display area includes the header, the summary, and details. For all other inbound deliveries, you only see a header and a summary. To display the details of other inbound deliveries, you can expand the display area.

Filter your search results to the ones that are relevant to you. You can also sort the search results according to your needs.

Related Information

Searching from the Fiori Launchpad

Deleting Inbound Deliveries

You can use the **Change Inbound Delivery** app to delete entire inbound deliveries and their corresponding delivery items.

Procedure

1. From the SAP Fiori launchpad, open the Change Inbound Delivery app.
2. Enter the number of the inbound delivery you want to delete.
3. Press the Enter key.

The inbound delivery overview screen appears.

4. Choose the **Delete** button at the top of the overview screen.
5. The **Delete Delivery Processing** dialog box appears. The system displays a message asking you to confirm that you want to delete the delivery.
6. Choose **Yes**.

Results

The system displays a message confirming that the delivery has been deleted.

Deleting Delivery Items

Delete individual items from an inbound delivery.

Procedure

1. Open the Change Inbound Delivery app.
2. Select the item or items in the outbound delivery that you want to delete.
3. Choose the **Delete item(s)** button at the bottom of the screen.

   A dialog box appears in which you must confirm that you want to delete the items.
4. Choose **Yes**.

   The selected items are deleted and no longer appear on the overview screen.
5. Choose **Menu** > **Inbound delivery** > **Save** to save the inbound delivery.

Batches in Inbound Deliveries

Batch functionality can be used with inbound deliveries.

Note

This is a function shared by more than one delivery type and its processes. For more information about this function, see the related links.

Related Information

Batches in Delivery Management
Automatic Batch Creation for Inbound Deliveries

Packing

You specify the packing of delivery items directly in the delivery itself. You can manually select available handling units or use previously defined packing instructions.

Note

This is a function shared by more than one delivery type and its processes. For more information about this function, see the related links.

Related Information

Automatic Packing – Packing with Packing Instructions
Packing in Deliveries

Putaway

The putaway functions in Delivery Management allow you record how many items are put into storage in a warehouse.

You can use Delivery Management for putaway in the following scenarios:

Scenario	Details
Inventory Management	With Inventory Management, you can report putaway quantities in the returns delivery directly.
Warehouse Management	Warehouse Management (WM) refers to the warehouse management capabilities in SAP S/4HANA Cloud Public Edition. For more information, see Warehouse Management.
Third-Party Warehouse Management	You can use the Third-Party Warehouse Management APIs to send and receive data regarding the putaway status of a delivery to a third-party solution. For more information, see: • Third-Party Warehouse Management • Scope item Logistics with Third-Party Warehouse Management (1ZQ) in SAP Signavio Process Navigator
SAP Extended Warehouse Management (SAP EWM)	You can integrate your decentralized on-premise instances of SAP EWM into your SAP S/4HANA Cloud Public Edition. For more information, see Basic Integration with SAP Extended Warehouse Management (SAP EWM).

Putaway Status

The **OvrllPickStatus** field shows the following putaway statuses for an inbound delivery:

Indicator	Meaning
Empty	The item isn't relevant for putaway.
A	Putaway of the item has not been started yet.
B	The item has been partially put away. A delivery item is partially put away if the quantity that was put away is smaller than the delivery quantity.
C	The item has been completely put away.

Putaway Status and Goods Receipt

You can only post goods receipt for a delivery once all items relevant for putaway have been completely put away.

The delivery cannot be put away in the system if the putaway quantity that was reported by the warehouse is smaller than the delivery quantity. In this case, you must decide whether to reduce the delivery quantity and post goods receipt immediately to determine if the quantity put away in the warehouse is in fact lower than the delivery quantity.

Viewing the Putaway Status for a Whole Delivery

You can see the overall putaway status for a delivery and an overview of the putaway status for each of the items in a delivery on the **Status Overview** tab.

Procedure

1. Display a delivery.
2. Choose the **Status Overview** tab.

Results

You can see the overall status of the putaway activities for the delivery in the **Overall Putaway Status** column. You can seethe status of individual items in the **Delivery Item Status** screen area.

Viewing the Putaway Status of Delivery Items

You see the putaway status on the **Putaway** item details screen.

Procedure

1. Display a delivery and select the item that you want to see putaway information about.
2. Choose **More** > **Goto** > **Item** > **Putaway**.

Results

You see an overview of all putaway activities for the item. The status indicator is in the **Warehouse** screen area. You can also see the status of the Warehouse Management (WM) transfer order,if relevant.

Putting Away Delivered Goods

Inbound deliveries include information about the delivery time and storage location of materials in a delivery. When the delivery arrives at the warehouse, you can pass this information on to warehouse management to ensure that goods are put away in the correct storage location.

Record Putaway Quantities

You record how many items have been put away in the inbound delivery.

Procedure

1. From the SAP Fiori launchpad, open the Inbound Deliveries for Putaway app.
2. Enter the delivery date and inbound delivery number and choose **Execute**.
3. Select the inbound delivery that you want to report putaway quantities for and choose **Change Inbound Deliveries**.
4. On the **Item Overview** tab, enter a putaway quantity for each item.
5. Choose **Save**.

Goods Receipt

Posting a goods receipt for an inbound delivery means that the inbound delivery process is completed. You can also use the goods receipt process to provide proof of delivery and monitor differences between the ordered and delivered quantities.

You have the following options for posting goods receipt:

- You can manually post goods receipt for one or more inbound deliveries.
- Goods receipt for multiple inbound deliveries can be automatically posted in the background.

The data needed for the goods receipt is copied from the inbound delivery and can't be changed in the goods receipt document. If you need to update the goods receipt, you must change the inbound delivery, and this changed data is copied to the goods receipt document.

When you post a goods receipt for an inbound delivery, the system performs the following tasks based on the goods receipt document:

- The warehouse stock of the material is increased by the delivery quantity.
- Value changes are posted to the balance sheet account in inventory accounting.
- Goods receipt posting is automatically recorded in the document flow.

After posting goods receipt, the possibilities for changing the delivery document are restricted. This restricted editing helps prevent any differences between the goods receipt document and the inbound delivery.

Confirming Goods Receipt

After receiving items in a delivery and putting them into storage, you can post goods receipt to confirm that you've received all goods in the delivery.

Prerequisites

The data in the inbound delivery is complete.

The items are put away and confirmed.

Posting Goods Receipt

You post goods receipt to confirm the receipt of goods for an inbound delivery.

Procedure

1. In the SAP Fiori Launchpad, open the Post Goods Receipt for Inbound Delivery app.
2. Enter the number of the inbound delivery that you want to confirm the goods receipt for.
3. Press the Enter key.
4. If necessary, you can add any missing information to the inbound delivery.
5. Choose **Post**.

Results

The app displays a link to the material document that was created. If you have the correct authorization, you can choose the link to view the document.

Posting Goods Receipt for Multiple Deliveries

Context

You can use the **Post Goods Receipt for Inbound Delivery** app to confirm the receipt of goods for multiple deliveries.

→ Tip

You can also use a job scheduling app to define a periodic job that automatically posts goods receipt in the background. For more details, see Schedule Goods Receipt for Inbound Deliveries.

To manually confirm the receipt of goods in multiple deliveries, carry out the following steps:

Procedure

1. In the SAP Fiori Launchpad, open the Inbound Deliveries for Goods Receipt app.
2. To select the inbound deliveries, enter the selection criteria.
3. Choose **Execute**.
4. Select the deliveries for which you want to post a goods receipt. To select more than one delivery, use the Ctrl key.
5. To post the goods receipt for the selected inbound deliveries in the overview, choose **Subsequent Functions** > **Post Goods Receipt**. Alternatively, you can choose the **Post Goods Receipt** button at the bottom right of the screen.

Canceling Goods Receipt

You can reverse the goods receipt to make a delivery editable again.

Procedure

1. In the SAP Fiori Launchpad, open the Reverse GI - Outbound Delivery, Reverse Goods Receipt for Inbound Delivery app.
2. Enter the number of the inbound delivery that you want to reverse the goods receipt for.
3. Choose **Execute**.
4. Select the inbound delivery and choose **Reverse**. You can now edit the inbound delivery and post an updated goods receipt for it.

Proof of Delivery

As a ship-to party, after receiving an inbound delivery, you can record proof of delivery (POD) of the goods in the delivery. You do this by updating the delivery values in the inbound delivery.

When you update the values in the inbound delivery, if the item is relevant for proof of delivery, you can report overdeliveries or underdeliveries and provide a reason for them. Depending on your configuration, the system prompts you to select a reason or automatically provides one whenever you enter a quantity that is different from the quantity in the preceding purchase order.

Status

The proof of delivery status indicates whether the ship-to party has sent proof of delivery or not. The following statuses are available:

Status	Description
[empty]	Not relevant for the POD process
A	Relevant for the POD process - not yet processed
B	Relevant for the POD process - differences reported (The POD needs to be manually confirmed.)
C	Relevant for the POD process - quantities were verified and confirmed (The POD is automatically confirmed.)

Configuring Proof of Delivery

Before you can create proofs of delivery, you must configure reasons, item categories, and sales orders and mark them as relevant for proof of delivery.

Procedure

1. Configure the reasons for quantity differences.
    a. In your configuration environment, use the search function to open the **Define Reasons for Quantity Differences** activity. (ID: 102359)
    b. Create any reasons that you need.

    Note

    For the generation of POD difference quantities and reasons to be carried out fully or partially automatically by the system, you need to create one reason for overdelivery and one reason for underdelivery and mark them as default reasons.

2. Configure POD-relevance depending on delivery item category.
    a. In your configuration environment, use the search function to open the **Set POD-Relevance Depending on Delivery Item Category** activity. (ID:101904)
    b. Use the input help to set the field **POD-relevant** as required.
    c. Use the input help to set the field **POD automatic** as required.

    The following **POD automatic** settings are available:

- V = fully automatic processing: If you have defined default reasons for overdelivery or underdelivery, the system displays the default reason, and you cannot change it.
- T = partially automatic processing: If you have defined default reasons for overdelivery or underdelivery, the system displays the default reason, and you can change it as required.
- M = manual processing: The system does not display reasons for overdelivery or underdelivery and you need to enter the reason manually.

3. Configure the ship-to party.

You need to set the sales area of every ship-to party that you want to receive proof of delivery from as POD-relevant.

a. In the **Manage Customer Master Data** app, select a ship-to party and open or create a sales area.

> **Note**
> The business role Master Data Specialist - Business Partner Specialist `SAP_BR_BUPA_MASTER_SPECIALIST` is required to use the **Manage Customer Master Data** app.

b. Go to the sales area details and find the shipping section.

c. Set the sales area as relevant for proof of delivery.

Automation Level

Configure the level of automation for the POD of inbound deliveries.

You set the automation level of the POD for inbound deliveries with the configuration activity **Set POD-Relevance Depending on Delivery Item Category** (ID: 101904).

> **Note**
> This is a setting that requires you to make changes in your configuration environment. For more information about the available configuration environments, see Configuration Environment of SAP S/4HANA Cloud Public Edition.

You set the automation level by entering the corresponding value in the **POD automatic** field. The following automation levels are available:

Automation Level	Description
V: Fully-Automated POD Processing	If you have defined default reasons for overdelivery or underdelivery, the system displays the default reason, and you cannot change it.
T: Partly-Automated POD Processing	If you have defined default reasons for overdelivery or underdelivery, the system displays the default reason, and you can change it as required.
M: Manual POD Processing	The system does not display reasons for overdelivery or underdelivery, and you need to enter the reason manually.

> **Tip**
> These details can also be displayed directly using the F1 field help for the **POD automatic** field in the configuration step **Set POD-Relevance Depending on Delivery Item Category** (ID: 101904).

Creating Proofs of Delivery (PoD)

You can manually send proof of delivery upon receipt of goods.

Prerequisites

You have set the automation level for inbound delivery PoD to **Manual**. For more information, see Automation Level.

Procedure

1. In the SAP Fiori launchpad, open the Change Inbound Delivery.
2. Enter the number of the inbound delivery you want to change.
3. Choose **Continue**.
4. Change the delivery quantity to the quantity received.

   > **Note**
   > See Adjusting Quantities if you create quantity differences that you do not want to report to the vendor.

   A screen on which you can make entries for the proof of delivery appears.
5. Enter a reason for deviation and the corresponding difference in quantity.
6. Choose **Copy**.

   You return to the inbound delivery overview screen.
7. Save the changes by choosing **Inbound delivery** > **Save**.

   A message appears informing you that the inbound delivery has been saved.

Results

The system uses the reasons for deviation that you entered to send an automatic message about the quantity differences to the supplier.

Creating Proofs of Delivery (PoD) Automatically

You can set your system so that if there are differences between the quantity that was actually delivered and the quantity in the shipping notification, the system automatically sends a message with preset standard reasons to the vendor at goods receipt.

Prerequisites

You have set the automation level for inbound delivery PoD to **Full**. For more information, see Automation Level.

You have define one default reason for underdelivery and one for overdelivery.

Procedure

1. In the SAP Fiori launchpad, open the Change Inbound Delivery.
2. Enter the number of the inbound delivery you want to change.
3. Choose **Continue**.

4. Change the delivery quantity to the quantity received.

> **Note**
>
> See Adjusting Quantities if you want to create quantity differences that you do not want to report to the vendor.

5. Save the changes by choosing **Inbound delivery** > **Save**.

   A message appears informing you that the inbound delivery has been saved.

Results

The system uses the standard reasons and reasons for deviation that you have set up for overdelivery or underdelivery to send an automatic message about the quantity differences to the supplier.

Adjusting Quantities

You can create a quantity difference for an inbound delivery without notifying the supplier about it. You might choose to do so, for example, if you want to generate an inbound delivery from a scheduling agreement, but are still awaiting arrival of other inbound deliveries in this scheduling agreement.

Context

Based on a purchase order, your supplier has notified you of and delivered a certain amount of a certain material.

Procedure

1. Create an inbound delivery with reference to a purchase order.

   The system suggests the purchase order quantity as the delivery quantity.

2. On the overview screen of the inbound delivery, change the delivery quantity to the actual delivered quantity.

   A screen on which you can make entries for the proof of delivery appears. Here, what you can do depends on the automation level you selected for the delivery item type.

3. Adjust any proofs of delivery, if necessary and possible.
4. Choose **Adjust quantity**.
5. If a standard reason appeared in the previous step, it's now removed.

> **Note**
>
> Repeat this step for each quantity change.

6. Choose **Copy**.
7. Save the changes by choosing **Inbound delivery** > **Save**.

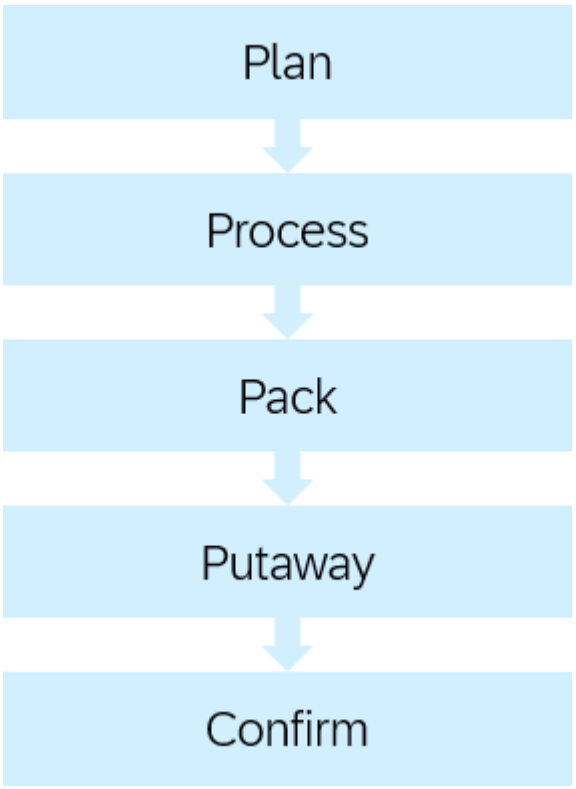
Results

The supplier doesn't receive a notification about the reduction in quantity that you've carried out using the **Adjust quantity** function.

Customer Returns Delivery Process

The customer returns delivery process includes all steps related to the return of goods from the customer to the supplier.

The customer returns delivery process includes the following steps:



Please note that image maps are not interactive in PDF outputs.

Creating and Processing Customer Returns Deliveries

When a customer returns goods to the supplier, a corresponding returns delivery can be created to record the incoming goods. This document contains information such as the items in the delivery, weight, and delivery data. A returns delivery can be manually or automatically created based on a customer returns order. The goods receiving point is determined by the system based on configuration settings.

You can update a delivery to report the putaway quantity or any other changes to the delivery. You can also use the display function to view information about the returns delivery.

The following table outlines the options for creating customer returns deliveries:

Manual creation (apps)	Create Outbound Delivery – with Order Reference
	Create Outbound Deliveries – From Sales Orders
Automatic creation with batch processing (app)	Schedule Delivery Creation
Automatic creation with APIs	Outbound Delivery with Reference to a Sales Order – Create (A2X)
	Outbound Delivery (A2X)

ℹ Note

For scope item specific process descriptions, refer to **PRODUCT FEATURES** > **Scope Items** in the documentation for the relevant app in the SAP Fiori Apps Reference Library.

Working with Customer Returns Deliveries

You can manually process returns deliveries using the corresponding app.

ℹ Note

Although the customer returns process is about the physical receipt of goods returned by a customer, it technically uses the apps and processes of the goods issue process. Therefore, some of the terminology used in the apps applies to the outbound process even though this is a goods receipt-related process. For more information, see Outbound Specific Terminology for Customer Returns Deliveries.

The following links provide details about working with customer returns deliveries:

Creating Customer Returns Deliveries

You can manually create single or multiple returns deliveries based on customer returns orders and subsequently change the deliveries as required.

Changing Customer Returns Deliveries

You use the **Change Outbound Delivery** app to change a returns delivery.

Displaying Customer Returns Deliveries

You use the **Display Outbound Delivery** app to display a returns delivery.

Deleting Customer Returns Deliveries

You use the **Change Outbound Delivery** app to delete entire returns deliveries and their corresponding delivery items.

Creating Customer Returns Deliveries

You can manually create single or multiple returns deliveries based on customer returns orders and subsequently change the deliveries as required.

Creating Single Customer Returns Deliveries

You can manually create a single returns delivery for a specific customer returns order. Only the order items from this order that are due for delivery are included in the delivery. You can make any necessary changes to the returns delivery after creating it.

Procedure

1. From the SAP Fiori launchpad, open the Create Outbound Delivery – with Order Reference app.
2. Enter a receiving point as required in the **Shipping Point** field.
3. Enter the ID of the specific returns order in the **Order** field.
4. Choose **Continue**. The system displays the overview of the returns delivery.
5. Edit the returns delivery as required and choose **Save**.

Results

When the system has saved the document, it displays the number of the returns delivery.

The system determines the goods receiving point according to the receiving plant and receiving storage location that you specified in the returns delivery.

Next Steps

The system records errors that occurred during delivery creation in a log. The system displays a message in the case of errors.

You can navigate to the log from any of the overview screens in the delivery by choosing **Edit** > **Error log** from the menu. From there, you can make changes directly to the document.

Creating Multiple Customer Returns Deliveries

Context

You can manually create multiple returns deliveries based on multiple customer returns orders using the **Create Outbound Deliveries – From Sales Orders** app. The system copies the data from the customer returns orders to the returns deliveries.

→ Tip

As an alternative, you can use a job scheduling app to define a periodic job that automatically creates returns deliveries in the background. For more details, see Schedule Delivery Creation.

To manually create multiple returns deliveries with reference to multiple customer returns orders, carry out the following steps:

Procedure

1. From the SAP Fiori launchpad, open the Create Outbound Deliveries – From Sales Orders app.

   The UI shows a list of sales orders.

2. To display a list of your customer returns orders, set the following filter criteria:

   a. In the **Adapt Filters** dialog, choose **SD Document Category**.

   b. In the **SD Document Category** select dialog, choose **Returns (H)**.

   c. Choose the **Go** button.

3. If no data is found, try adjusting the search or filter criteria.
4. Choose the **Go** button.
5. Select your returns orders and choose **Create Deliveries**.
6. Choose **Display Log**.

   With the **Analyze Delivery Logs** app, you can check system messages that have been logged during the collective creation run of deliveries and navigate to any deliveries that have been created. For more information, see Analyze Delivery Logs.

Changing Customer Returns Deliveries

You use the **Change Outbound Delivery** app to change a returns delivery.

Procedure

1. In the SAP Fiori launchpad, open the Change Outbound Delivery app.
2. Enter the number of the returns delivery document that you want to change, or use the input help to find the corresponding delivery.
3. To make changes in a specific view of the delivery, choose **Menu** > **Goto** and the corresponding view.

4. Choose **Continue**. The overview screen (or the specific view you chose) appears.
5. Make your changes in the returns delivery as required.
6. Choose **Save**.

Displaying Customer Returns Deliveries Available for Changes

You can display a list of open returns deliveries available for changes and then select specific returns deliveries from this list that you would like to change.

Procedure

1. From the SAP Fiori launchpad, open the General Outbound Delivery List app.
2. Enter your selection criteria.
3. Choose **Execute**.

   The system displays a list of returns deliveries that meet your selection criteria.
4. Select the returns deliveries to be changed and choose **Change Outbound Deliveries** on the bottom right of the screen.

   The overview screen of the delivery you selected appears.
5. Make the required changes.
6. Save the changes. If you have selected multiple deliveries, the system displays a message asking if you would like to process the next delivery from your selection.

Displaying Customer Returns Deliveries

You use the **Display Outbound Delivery** app to display a returns delivery.

Procedure

1. From the SAP Fiori launchpad, open the Display Outbound Delivery app.
2. Enter the number of the returns delivery that you want to display or choose the input help to display a list of deliveries based on your defined search criteria.
3. Press the Enter key.

Searching for Customer Returns Deliveries

To find a specific returns delivery, you use the Enterprise Search in the header bar of the SAP Fiori launchpad. Before you enter any attributes in the search input field, select the category **Returns Deliveries**.

Context

In the search input field, you can search for a returns delivery. For example, you could search for a material number. To reduce the number of search results, you can also combine several attributes, such as a material number and a sold-to party. The search takes into account the following fields:

- Delivery document type description
- Created on

- Created by
- Material number
- Supplier (ID and name)
- External delivery ID
- Material description
- Receiving point

Results

A list of returns deliveries appears as a result. For the first returns delivery, the display area includes the header, the summary, and details. For all other returns deliveries, you only see a header and a summary. To display the details of other customer returns deliveries, you can expand the display area.

Filter your search results to the ones that are relevant to you. You can also sort the search results according to your needs.

Related Information

Searching from the Fiori Launchpad

Deleting Customer Returns Deliveries

You use the **Change Outbound Delivery** app to delete entire returns deliveries and their corresponding delivery items.

Procedure

1. From the SAP Fiori launchpad, open the Change Outbound Delivery app.
2. Enter the number of the returns delivery you want to delete.
3. Press the Enter key.

   The returns delivery screen appears.
4. Choose **Outbound Delivery** > **Delete**.
5. The **Delete Delivery Processing** dialog box appears. The system asks you to confirm that you really want to delete the delivery.
6. Choose **Yes** and press the Enter key.

Results

You receive a message informing you that the delivery has been deleted.

Deleting Delivery Items

Delete individual items from a returns delivery.

Procedure

1. Open the Change Outbound Delivery app.

2. Select the item or items in the returns delivery that you want to delete.
3. Choose **Delete item(s)**.

   A dialog box appears in which you must confirm that you want to delete the items.
4. Save the delivery by choosing **Outbound delivery** > **Save**.
5. Choose **Yes**.

   The items selected are deleted and no longer appear on the overview screen.

Packing

You specify the packing of delivery items directly in the delivery itself. You can manually select available handling units or use previously defined packing instructions.

> **Note**
> This is a function shared by more than one delivery type and its processes. For more information about this function, see the related links.

Related Information

Automatic Packing – Packing with Packing Instructions
Packing in Deliveries

Putaway

The putaway functions in Delivery Management allow you record how many items are put into storage in a warehouse.

You can use Delivery Management for putaway in the following scenarios:

Scenario	Details
Inventory Management	With Inventory Management, you can report putaway quantities in the returns delivery directly.
Warehouse Management	Warehouse Management (WM) refers to the warehouse management capabilities in SAP S/4HANA Cloud Public Edition. For more information, see Warehouse Management.
Third-Party Warehouse Management	You can use the Third-Party Warehouse Management APIs to send and receive data regarding the putaway status of a delivery to a third-party solution. For more information, see: • Third-Party Warehouse Management • Scope item Logistics with Third-Party Warehouse Management (1ZQ) in SAP Signavio Process Navigator
SAP Extended Warehouse Management (SAP EWM)	You can integrate your decentralized on-premise instances of SAP EWM into your SAP S/4HANA Cloud Public Edition. For more information, see Basic Integration with SAP Extended Warehouse Management (SAP EWM).

→ Remember

Although the customer returns process is about the physical receipt of goods returned by a customer, it technically uses the apps and processes of the goods issue process. Therefore, some of the terminology used in the apps applies to the outbound process even though this is a goods receipt-related process. For more information, see Outbound Specific Terminology for Customer Returns Deliveries

Putaway Status

The **OvrllPickStatus** field shows the following putaway statuses for a returns delivery:

Indicator	Description
Empty	Not relevant
A	Not yet processed
B	Partially processed
C	Completely processed

Putaway Status and Goods Receipt

You can only post goods receipt for a delivery once all items relevant for putaway have been completely put away.

The delivery cannot be put away in the system if the putaway quantity that was reported by the warehouse is smaller than the delivery quantity. In this case, you must decide whether to reduce the delivery quantity and post goods receipt immediately to determine if the quantity put away in the warehouse is in fact lower than the delivery quantity.

Putting Away Delivered Goods

Returns deliveries include information about the delivery time and storage location of materials in a delivery. When the delivery arrives at the warehouse, you can forward this information to warehouse management to ensure that goods are put away in the correct storage location.

Recording Putaway Quantities

You record how many items have been put away in the returns delivery.

Procedure

1. From the SAP Fiori launchpad, open the Outbound Deliveries for Picking app.
2. Enter the receiving point and the picking date.
3. Choose **Execute**.
4. Select the returns delivery that you want to report putaway quantities for and choose **Change Outbound Deliveries**
5. On the **Picking** tab, enter a putaway quantity for each item in the **Picked Qty** field.
6. Choose **Save**.

Goods Receipt

When you receive goods, posting a goods receipt means the delivery is completed. You can also use the goods receipt process to provide proof of delivery and keep track of any differences between the ordered and returned quantities.

You have the following options for posting goods receipt:

- You can manually post goods receipt for one or more returns deliveries.
- Goods receipt for multiple returns deliveries can be automatically posted in the background.

The goods receipt document is created based on the returns delivery. The information needed for the goods receipt is copied from the returns delivery and can't be changed in the goods receipt document. If you need to update the goods receipt, you must change the returns delivery, and this changed data is copied to the goods receipt document.

When you post a goods receipt for a returns delivery, the system performs the following tasks based on the goods receipt document:

- Warehouse stock of the material is increased by the delivery quantity.
- Value changes are posted to the balance sheet account in inventory accounting.
- Goods receipt posting is automatically recorded in the document flow.

After posting the goods receipt, the possibilities for changing the delivery document are restricted. This helps prevent any differences between the goods receipt document and the returns delivery.

→ Remember

Although the customer returns process is about the physical receipt of goods returned by a customer, it technically uses the apps and processes of the goods issue process. Therefore, some of the terminology used in the apps applies to the outbound process even though this is a goods receipt-related process. For more information, see Outbound Specific Terminology for Customer Returns Deliveries

Confirming Receipt of Goods

After receiving items in a delivery and putting them away, you can post a goods receipt to confirm that you've received all goods in the delivery.

Prerequisites

The data in the returns delivery is complete.

The items are put away and confirmed.

Confirming Receipt

You post a goods receipt for a single returns delivery to confirm that you have received the goods.

Procedure

1. In the SAP Fiori Launchpad, open the Manage Outbound Deliveries app.

2. In the **Adapt Filters** dialog, choose **Delivery Document**.
3. Enter the number of the returns delivery that you want to confirm goods receipt for.
4. Choose the **Go** button.
5. Select the returns delivery and choose **Post GI**.

Results

The app displays the material document that was created. If you have the correct authorization, you can view the document by choosing the link provided.

Confirming Receipt of Goods for Multiple Deliveries

Context

You can confirm the receipt of goods in multiple returns deliveries using the **Manage Outbound Deliveries** app.

→ Tip

As an alternative, you can use a job scheduling app to define a periodic job that automatically posts goods receipts in the background. For more details, see Schedule Goods Receipt for Inbound Deliveries.

To manually confirm the receipt of goods in multiple deliveries, carry out the following steps:

Procedure

1. In the SAP Fiori Launchpad, open the Manage Outbound Deliveries app.

   The app displays a list of outbound deliveries.

2. To display a list of your returns deliveries, set the following filter criteria:

   a. In the **Adapt Filters** dialog, choose **Delivery Type**.

   b. In the **Delivery Type** select dialog, sort the **SD Category Document** category in descending order to bundle the sales category documents with the value `T`. `T` stands for **Returns Delivery for Order** and is the sales document category of the returns deliveries. Then, proceed with selecting the delivery type according to your business needs.

   c. Choose the **Go** button.

3. Select the returns deliveries for which you want to post a goods receipt.
4. Choose **Post GI**.

Outbound Specific Terminology for Customer Returns Deliveries

The following tables list the apps used for returns deliveries and provide details about the UI elements that use outbound delivery terminology and the UI elements that have been adapted for returns deliveries:

SAP Fiori Apps

App	App ID	UI Texts Using Outbound Specific Terminology	UI Texts Adapted for Returns Deliveries
Create Outbound Deliveries – From Sales Orders	`F0869A`	**Planned GI Date** **Shipping Point**	

App	App ID	UI Texts Using Outbound Specific Terminology	UI Texts Adapted for Returns Deliveries
		Ship-To Party	
Analyze Delivery Logs	F0870A		You use the Display Returns Delivery link to display your returns deliveries.
Schedule Delivery Creation	F2228	Goods Issue Date	In Step 3: Shipping Point/Receiving Point Unloading Point
Pick Outbound Delivery	F0868	Picking Date Picking Status Picking Quantity Loading Date Shipping Point	Actual GR Date Planned GR Date GR Status GR Ready/GR Not Ready (status that is displayed)

SAP Web GUI Apps

App	App ID	UI Texts Using Outbound Specific Terminology		UI Texts Adapted for Returns Deliveries
		Tabs	Overview Screen	
Create Outbound Delivery – with Order Reference	VL01N	Picking Loading	Actual GI Date Loading Loading Date Loading Point OvrllPickStatus Planned GI Pick Date/Time Pick Confirmation	After entering the shipping point and the order number and choosing Continue, the overview screen displays the Post Goods Receipt button instead of the Post Goods Issue button that is usually displayed when creating a standard outbound delivery. Confirmation Picking / Putaway Confirmation column on the Status Overview tab
Change Outbound Delivery	VL02N			Post Goods Issue button When you choose this button, the system recognizes that it is a returns delivery and displays a Post Goods Receipt button.
Display Outbound Delivery	VL03N			

App	App ID	UI Texts Using Outbound Specific Terminology		UI Texts Adapted for Returns Deliveries
		Tabs	Overview Screen	
Outbound Deliveries for Picking	VL06P		Picking Date Picking Data Only Picking Without WM Only WM Picking Both Picking Types Picking Data	Shipping Point/Receiving Pt
Reverse Goods Receipt	VL09	Only outbound specific terminology available		

Additional Features

Some functions that Delivery Management offers are used in multiple processes. This section gives you an overview of these features.

The following additional features are available:

Batches in Delivery Management

Batch splitting describes specifying quantities from more than one batch for a particular delivery item. You activate automatic batch creation for a given delivery item category. When active, the system creates batches for the delivery item category whenever you create an inbound delivery from a purchase order.To configure your system to automatically create batches for inbound deliveries, you activate automatic batch creation in your configuration environment. For more information about the available configuration environments, see Configuration Environment of SAP S/4HANA Cloud Public Edition.

Flexible Delivery Document Numbering

When a new delivery document is created, the system can evaluate specific delivery document header fields to dynamically determine an appropriate number range interval from which to draw the new document number.

Batches in Delivery Management

Batches are subsets of materials in stock that are managed seprately from other subsets of the same material. Here are some examples of batches:

- Different production lots (such as paints, dyes, and pharmaceutical products)
- Delivery lots
- Quality grades of a material

You can specify a batch in the sales order item detail screen for a material requiring batch management. You find the indicator in the material master record in the Storage and Purchasing screens. When a delivery for a sales order or purchase order is created, the batch is copied to the delivery. If no batch was entered in the sales order, you can enter it in the **Picking Overview** screen in the delivery. In addition, you can automate batch creation, although we recommend checking the accuracy of the created batches to avoid conflicts or errors.

Related Information

Splitting Batches in Deliveries

Batch splitting describes specifying quantities from more than one batch for a particular delivery item.

Context

The batch split function in the delivery includes a batch selection screen and an availability check. In addition, batch processing is linked to the SAP classification system, enabling you to select batches according to classes and characteristics.

In Materials Management (MM), you can specify basic data for each batch, such as available from and expiration dates, and data about the origin of the batch.

Procedure

1. When editing or creating a delivery, select the item for which you want to split batches and choose **Goto** > **Item** > **Batch Split**.

   The overview screen for batch split items is displayed.

2. Use batch determination to select the required batches.

   The relevant quantity is proposed by the system and you assign it to the delivery item.

3. Enter the split quantity for the required batches until you have assigned the entire requirement quantity and choose **Copy**.

   The system creates batch split items for the selected batches.

   The open quantity is decreased by the assigned quantity.

4. Choose **Back** to return to the delivery overview screen.

> **Note**
> For delivery items with a batch split, the quantity that has been picked or put away is determined for each individual split item. If you want to maintain or check this quantity for split items, you need to switch to the batch structure view for the relevant main item by choosing the + button.

Automatic Batch Creation for Inbound Deliveries

You activate automatic batch creation for a given delivery item category. When active, the system creates batches for the delivery item category whenever you create an inbound delivery from a purchase order.

> **Restriction**
> You cannot automate the creation of batches for delivery items that contain a batch split.

The system transfers the following data from the preceding document to the delivery:

- Supplier Number
- Supplier Batch Number
- Date of Manufacture

- Best-Before Date (BBD) or Shelf Life Expiration Date (SLED)
- Country/Region of Origin of Material (Non-Preferential Origin)
- Region of Origin of Material (Non-Preferential Origin)

⚠ Caution

Because every SAP implementation is different, check whether automatically created batches are valid and correct as some errors can occur depending on your setup.

Activating Automatic Batch Creation

To configure your system to automatically create batches for inbound deliveries, you activate automatic batch creation in your configuration environment. For more information about the available configuration environments, see Configuration Environment of SAP S/4HANA Cloud Public Edition.

Context

! Restriction

Batches can only be automatically created for inbound deliveries.

Procedure

1. In your configuration environment, use the search function to open the **Activate Automatic Batch Determination for Sales Order Item Categories** activity. (ID: 102171)
2. For the Item Category TP0P (Inbound Delivery), activate the **Auto batch determin** checkbox and choose **Save**.

Next Steps

To ensure completeness, it is recommended to manually check the data for automatically created batches.

Product Hierarchies in Batch Determination for Deliveries

Product hierarchies are often necessary and helpful to organize materials or products on multiple levels. You can use product hierarchies for batch determination in delivery documents if batch determinations are set up with a dependency on product hierarchy data.

Product Hierarchy in Delivery Documents

To determine the product hierarchy in a delivery document, the system compares the data assigned in the **Manage Product Hierarchy Assignments** app with the data in the delivery document. It then selects the most precise product hierarchy from these two options. For this procedure, the system considers product hierarchy timeframes that are in the status **Active** only.

When the system copies items from preceding documents to subsequent documents (for example, from a sales order to a delivery item), it also copies the product hierarchy data from the preceding document (in this example, from the sales order).

You can view the product hierarchy data for a delivery document on the item level by displaying the **Materials** tab page of a delivery from the **Change/Display Outbound Delivery** app. This viewable information includes the product hierarchy ID, parent node, and the **Valid From/To** dates of the hierarchy timeframe.

You can redetermine the product hierarchy in delivery documents as follows:

- You can set up the system to determine the product hierarchy during automatic batch determination when creating the delivery document. In this case, automatic batch determination needs to be activated for the delivery item category used (**Activate Automatic Batch Determination for Delivery Item Categories** configuration activity).
- The system automatically redetermines the product hierarchy when you create a delivery document with the related **Delivery Item Category**.

Setting Up Product Hierarchies

If you want to use product hierarchies in delivery documents, a product hierarchy must exist in the system. In the **Manage Product Hierarchies** app, you must create a product hierarchy based on a specific timeframe and the materials or products that you want to use in delivery documents.

Create Assignments

You must create an assignment for a product hierarchy to enable its use in delivery documents.

When creating a new assignment in the **Manage Product Hierarchy Assignments** app, the system compares the validity period of the assignment with the timeframe of the product hierarchy. The validity period of an assignment can be partially or entirely within the timeframe of the product hierarchy.

To determine the product hierarchy for a delivery document, the system compares the dates in the **Valid From/To** field of the product hierarchy assignment with the date on which the batch determination is carried out. For batch determination in the delivery, this date cannot be configured.

! Restriction

For the product hierarchy determination in a delivery document, the validity period of the assignment must be within the timeframe of the product hierarchy.

The system prevents you from creating assignments with overlapping validity periods for the same combination of purpose, sales organization, and distribution channel. You can create assignments for product hierarchies with the status **Active** or **In Revision**.

For the batch determination to work properly for product hierarchies in delivery documents, you must assign the **Batch Determination** purpose for materials or products that are enabled for batch management.

You must complete the delivery-related configuration activities for batch determination in your configuration environment before you can use the product hierarchy determination for the **Batch Determination** purpose.

Related Information

Batch Determination
Manage Product Hierarchies
Manage Product Hierarchy Assignments
Product Hierarchies in Sales

Flexible Delivery Document Numbering

When a new delivery document is created, the system can evaluate specific delivery document header fields to dynamically determine an appropriate number range interval from which to draw the new document number.

Standard Delivery Document Numbering

When documents (for example, outbound deliveries or inbound deliveries) are created in Shipping, the system assigns unique document numbers to them by drawing consecutive numbers from predefined number range intervals of the number range object RV_BELEG. For newly created delivery documents of delivery type LF outbound delivery, the system by default draws the document numbers from number range interval 01, which is defined as ranging from 0000000001 to 0049999999. (The leading zeroes express that the maximum technical length of the number range is 10, but only the last 8 digits are in use.)

If you implement flexible delivery document numbering, you can assign additional number range intervals and, optionally, number range prefixes to delivery documents that fulfill specific criteria (for example, specific values for delivery type).

As a configuration expert, you can maintain internal number range intervals by performing the **Define Number Ranges for Deliveries** configuration activity. Find this activity by using the search function in your configuration environment. For more information, see Configuration Environment of SAP S/4HANA Cloud Public Edition.

ⓘ Note

In contrast to number range prefixes, which are transported automatically, number range intervals cannot be transported from the system in which they were created to the production (P) system. Therefore, while you can transport number range prefixes from their origin system to the P system, any number range intervals you have maintained for these prefixes in the origin system (as well as any other number range intervals) must be reworked manually in the P system.

→ Remember

Always release transports containing extensibility items (for example, custom fields to be used as parameters in flexible delivery document numbering routines) before releasing the standard configuration transports. This prevents errors and inconsistent transports.

Prefixes for Delivery Document Numbering

When you use flexible delivery document numbering, internal number range intervals can be combined with alphanumeric prefixes of up to five characters length. For example, use of the prefix ABC in conjunction with an interval ranging from 0000001 to 9999999 could result in a delivery document number of ABC1234567. The following are the two main reasons why you may want to work with number range prefixes:

- The space of possible delivery document numbers provided by purely numeric number ranges is insufficient for your business requirements

  Each prefix opens up a new subspace for number range intervals that enables the creation of entirely new, unique subranges of delivery document numbers. For example, if you create the prefix AB and define a number range interval 01 within this prefix, with the same upper and lower limit as the standard numbering interval 01 of delivery type LF outbound delivery, you have effectively doubled the quantity of numbers available for delivery document numbering without creating a new interval.

  You can do this for as many number range prefixes and internal number range intervals as required. Clearly, this mechanism greatly increases the flexibility of number range assignment.

- You want to categorize specific delivery documents in a way that helps fulfill legal requirements and makes them instantly recognizable.

  For example, an international corporation could choose to use the prefix DE for all delivery documents created by its German branch, and the prefix FR for all delivery documents created by its French branch.

ⓘ Note

Any prefixes that you define must conform to the following format:

- Alphanumeric characters only (A-Z, 0-9, no special characters)
- Maximum length of five characters
- The last character must be a letter (for example, 1A or 1234B)
- You cannot assign P, S, PBD and TMP as prefixes because they are reserved for system use

The technical maximum length of a delivery document number is 10 characters. When a prefix is used, this reduces the length of the possible number range by the length of the prefix. For example, a standard delivery document number could be 1234567890. If you apply the number range prefix ABC, the format would change to something resembling ABC1234567. This leaves only 7 characters for the purely numeric part of the document number, which is dictated solely by the number range interval.

For more information about defining number range prefixes, see the configuration help of the **Define Number Range Prefixes for SD Documents** configuration activity.

Applying Flexible Delivery Document Numbering

If the standard, static delivery document numbering described above cannot fulfill your business requirements, you can implement a flexible form of delivery document numbering. Flexible delivery document numbering assigns newly created delivery documents to internal number range intervals (and, optionally, number range prefixes) based on the evaluation of specific delivery document attributes (certain header fields of the newly created delivery document).

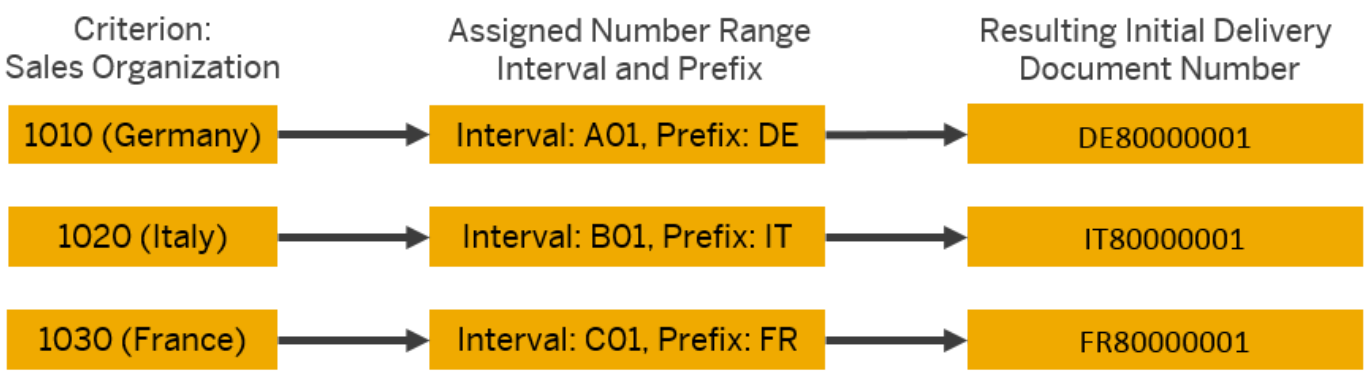
The following example illustrates a typical use case for flexible delivery document numbering.

Example

Consider a multinational corporation that consists of eight sales organizations, each based in a different country, including Germany and Italy.

When a new delivery document of type LF outbound delivery is created by the German company, custom logic implemented by the corporation tells the system to use the next available number in number range interval A01 within number range prefix DE. When, on the other hand, an LF outbound delivery is created by the Italian company, the custom logic tells the system to use the next available number in number range interval B01 with number range prefix IT. If a company based in France later joins the corporation, delivery documents originating from this company could be assigned to the interval C01 within prefix FR. In this way, for each delivery document, a different number range interval and number range prefix can be determined and used for each company.

The following figure shows how the sales organization of the newly created delivery document is used as a criterion to determine the number range interval and number range prefix from which the delivery document number is drawn.



The above example is a very simple use case for illustration purposes. When implementing flexible delivery document numbering, configuration experts can create as complex a set of conditions for determining the number range interval as required. Assignments can be made freely based on combinations of values of certain delivery document header fields.

In addition, **custom fields** that have been added to the delivery document header through the extensibility business context **Shipping: Delivery** can also be evaluated to determine a specific number range interval and prefix.

Restrictions

Note that flexible document numbering only functions with internal number range intervals that already exist in the system. The assignment of delivery documents to external number range intervals is not possible.

Also, flexible document numbering cannot be applied to any process that triggers early number assignment such as ARM (Advanced Returns Management), for example.

How to Implement Flexible Delivery Document Numbering

Configuration experts and developers can implement flexible delivery document numbering using the Configurable Parameters and Formulas (CPF) framework.

For more information about how to implement, see the link below.

Related Information

Using Configurable Parameters and Formulas (CPF) to Implement Flexible Delivery Document Numbering

Using Configurable Parameters and Formulas (CPF) to Implement Flexible Delivery Document Numbering

Using the Configurable Parameters and Formulas (CPF) framework, you as a configuration expert can define rules that the system uses to determine number range intervals for the numbering of newly created delivery documents. These number range intervals will then replace the default number range interval for internal numbering of the delivery document type. These rules enable a flexible allocation of delivery documents to number range intervals and, optionally, number range prefixes that is controlled by one or more header fields of the delivery document to be created.

Configuration experts and developers can implement flexible delivery document numbering using the Configurable Parameters and Formulas (CPF) framework.

This approach does not require knowledge of ABAP coding to create your custom numbering routines. Instead, you can set up decision tables to define the rules that the system processes to evaluate the delivery document header fields when a delivery document is created. In addition to defining your own number range intervals (and, optionally, number range prefixes), setup involves performing two configuration activities to create a CPF formula and mapping it to a routine number, which is in turn mapped to a specific delivery type.

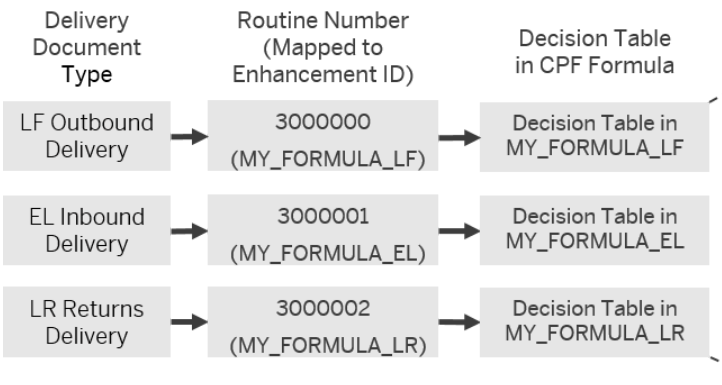
The CPF implementation doesn't require any programming effort. It evaluates rules specified in a decision table within a CPF formula. In this context, a CPF formula comprises the combination of relevant parameters and one decision table.

Configuration Overview

By mapping a formula ID to a routine number, then mapping the routine number to a delivery type, each delivery type that requires flexible numbering can be assigned to a different set of numbering rules. You maintain these rules within the decision table of the formula.

Example

The following figure shows an overview of how three different delivery types could be mapped to three different routine numbers, which in turn are mapped to three different enhancement IDs. In each case, as the enhancement ID you enter the formula ID of the formula containing the target decision table. Doing so facilitates the intended mapping between a routine number and a specific set of numbering rules within the decision table.



Rules in Decision Table

Input Parameters			Result Parameters	
Delivery Document Type	Sales Organization	Shipping Point	Number Range Interval	Number Range Prefix
LF	1000	1001	A01	DE
LF	1000	1002	A02	AT
LF	1000	1003	A03	CH
EL		1000	A01	IT
LR	2000		A01	FR

In the example above, the same sales organization `1000` is responsible for Germany, Austria and Switzerland, while a different shipping point is assigned to each of these countries. That means that country-specific prefixes can be used in the resulting delivery document number, for example.

For delivery document type EL, there is no sales organization assigned to Italy (‘blank/space’), which means that sales organizations are ignored during the determination of the number range interval.

As only the combination of all parameters needs to be unique, the same number range interval `A01` can be used each time, unless the same country-specific prefix is used more than once. This could be the case if there are two shipping points in Germany that should each receive a separate number range interval, for example.

Assigning a routine number to a delivery type causes the associated formula's decision table to override the delivery type's standard numbering procedure. In the standard numbering procedure, the number is derived from the number range provided for the delivery type or the group.

The designated formula is called whenever a delivery document of the delivery type that is mapped to the corresponding routine number is created. In effect, each routine number represents a specific decision table in a specific formula. You can make all required settings within the configuration activities **Define Formulas for Flexible Delivery Document Numbering** and **Assign Custom Routines to Delivery Document Types**. You can find them by using the search function in your configuration environment. For more information, see Configuration Environment of SAP S/4HANA Cloud Public Edition. For step-by-step instructions, see the section *Configuration Procedure* below.

Note

If necessary, the same routine number can be assigned to multiple delivery document types.

Input Parameters and Result Parameters

The rules in the decision table determine the correct number range interval and, optionally, the correct prefix for a given delivery type by evaluating whether specific delivery document header fields (represented in the formula as input parameters) satisfy conditional rules that you have specified within the rows of the decision table. The decision table can evaluate the following delivery document header fields:

- Delivery type
- Shipping point

- Sales organization
- Any custom field that has been added to the delivery document header through the extensibility business context Shipping: Delivery

> **Note**
>
> As a configuration expert, you can add custom fields as additional input parameters in the Define Parameter Catalog for Flexible Delivery Document Numbering configuration activity. For more information about how to add custom fields as additional input parameters, see the configuration help of this configuration activity.

Decision Table Logic

You create your decision table by creating one table row per rule that you want the system to evaluate based on the parameter values. Each row represents a numbering rule. Each rule is a conditional expression that specifies values for some or all of the input parameters, as well as for the result parameter `NUMBERRANGEINTERVAL`.

Through the input parameters, the system compares the header fields of the delivery document that is being created to the values you have maintained in the decision table. It processes the rules in the decision table sequentially from top to bottom. For each parameter, you can specify an operator as follows:

- Operator = (Equal to)

  If this operator is set, the system considers the corresponding parameter as a hit whenever the parameter value provided by the delivery document field is equal to the value you have specified in the corresponding column of the decision table. If every parameter in a row provides a hit, the rule represented by the row is applied and the number range interval (and, if applicable, the prefix) is assigned according to how they are specified in that row through result parameters `NUMBERRANGEINTERVAL` and `NUMBERRANGEPREFIX`.

- No Operator (Do not consider)

  If the operator cell is left empty (‘blank/space’), the corresponding parameter is ignored during the evaluation of that row, that is, the parameter cell is treated as a wildcard. This setting is useful when you want to treat a parameter for which ‘blank/space’ is a valid value as a wildcard.

After you have entered the rows representing the rules to be evaluated, the system automatically orders them based on the priority that you have assigned to each parameter in a previous step. To achieve the final order, the system sorts the rows as follows:

1. First, the rows are sorted according to the parameter with the highest priority (that is, the parameter in the left-most column). Rows in which a specific value (as opposed to a wildcard) is filled for this parameter rise to the top.
2. After this initial sort, the same principle is applied for the parameter with the second highest priority (second column from the left) to establish the relative order of the rows within the overarching order established by the previous sort.
3. This process is repeated for each subsequent parameter column to establish the final order of the rows, which ultimately dictates the sequence in which the system processes the rules that you have specified.

All cells are evaluated by checking the value you have entered into the cell against the corresponding parameter value provided by the delivery document. For each cell, a boolean result is returned. If all cells in a row are evaluated as true, the evaluation stops and the system returns the result parameter `NUMBERRANGEINTERVAL` (and, if applicable, `NUMBERRANGEPREFIX`) maintained in the current row. Otherwise, processing continues with the next table row until either a matching set of conditions is found or the end of the table is reached.

Example

You fill a row so that for any delivery document of type LF (using operator =) created for sales organization 1000 (using operator =), the system assigns number range interval A01 with prefix ABC.

The next row, however, assigns documents of delivery type LF (operator =) of any sales organization (no operator, that is, wildcard, and empty parameter cell) to number range interval B02 with no prefix.

The third row assigns delivery documents of any delivery document type (no operator, empty parameter cell) of sales organization 2000 (using operator =) to number range interval C03 with number range prefix XYZ.

If no rule is satisfied and the end of the table is reached, the standard number range interval for delivery type LF outbound delivery is used (without any prefix). In an actual decision table, these three rows would look as shown in the following table:

Input Parameters				Result Parameters	
Operator	DELIVERYTYPE	Operator	SALESORGANIZATION	NUMBERRANGEINTERVAL	NUMBERRANGEPREFIX
=	LF	=	1000	A01	ABC
=	LF			B02	
		=	2000	C03	XYZ

Configuration Procedure

Perform the following steps to get your CPF implementation of flexible delivery document numbering up and running.

→ Tip

You can find the configuration activities listed below by using the search function in your configuration environment. For more information, see Configuration Environment of SAP S/4HANA Cloud Public Edition. For more detailed information about each configuration activity, see the dedicated configuration help within each configuration activity.

1. (Optional) If you want to use prefixes, ensure that you have defined them in the **Define Number Range Prefixes for SD Documents** configuration activity. After the prefixes are defined, you can refer to them in your decision table.
2. To create custom number range intervals that the system can assign dynamically, configuration experts must perform the configuration activity **Define Number Ranges for Deliveries**.
3. (Optional) If you want to include custom fields as additional formula parameters, you can enable these custom fields as parameters in the **Define Parameter Catalog for Flexible Delivery Document Numbering** configuration activity.

   ### → Tip

   Developers can create custom fields in the **Custom Fields** app.

   ### → Remember

   Always release transports containing extensibility items, such as custom fields to be used as parameters in a CPF formula, before releasing the standard configuration transports. This prevents errors and inconsistent transports.

4. In the configuration activity **Define Formulas for Flexible Delivery Document Numbering**, create a CPF formula (including a decision table) that fulfills your business requirements. Save your formula and make a note of the formula ID.
5. Perform the configuration activity **Define Custom Routines for Flexible Delivery Document Numbering**. Here, register a routine number and assign it your formula by selecting the implementation type **Configurable Parameters and Formulas**

(CPF) and entering the formula ID as the enhancement ID. Your formula (including its decision table) is now mapped to the routine number.

6. Perform the configuration activity **Assign Custom Routines to Delivery Document Types**. Choose the delivery type (for example, LF outbound delivery) for which you want to trigger the formula that you have created. Then, enter the routine number that you registered in the previous step to assign the chosen delivery type to the target formula.

If everything is configured correctly, when a delivery document of the relevant delivery type is created, the system now calls the assigned CPF formula to determine the number range interval (and, if applicable, the prefix) from which the new delivery document number is drawn.

! Restriction

The formula can only function correctly if it refers to internal number range intervals that already exist in the system. Referencing external number range intervals is not possible.

Related Information

Custom Fields App and Custom Logic App
Configuration Environment of SAP S/4HANA Cloud Public Edition

Apps

Delivery Management provides apps that help you manage goods receipt and goods issue processes. The apps that you see may be different depending on the configuration of your system.

If you cannot see one of the apps in the launchpad but think you need access to it, contact your system administrator.

The following apps are available:

Apps in Delivery Management

App	App ID	Additional Information
Inbound Delivery	F0232A	Access this app using the Enterprise Search on the launchpad. Link to SAP Fiori Apps Reference Library
Outbound Delivery	F0233A	Access this app using the Enterprise Search on the launchpad. Link to SAP Fiori Apps Reference Library
Returns Delivery	F0234A	Access this app using the Enterprise Search on the launchpad. Link to SAP Fiori Apps Reference Library
Monitor Handling Units	F4557	Link to SAP Fiori Apps Reference Library
Reverse GI - Outbound Delivery, Reverse Goods Receipt for Inbound Delivery	VL09	Link to SAP Fiori Apps Reference Library
Create Inbound Delivery	VL31N	Link to SAP Fiori Apps Reference Library
Change Inbound Delivery	VL32N	Link to SAP Fiori Apps Reference Library

App	App ID	Additional Information
Display Inbound Delivery	VL33N	Link to SAP Fiori Apps Reference Library
General Inbound Delivery List	VL06IF	Link to SAP Fiori Apps Reference Library
Inbound Deliveries for Goods Receipt	VL06IG	Link to SAP Fiori Apps Reference Library
Inbound Deliveries for Putaway	VL06IP	Link to SAP Fiori Apps Reference Library
Inbound Deliveries for Purchase Orders	VL34	Link to SAP Fiori Apps Reference Library
Display Delivery Change Documents	VL22(N)	Link to SAP Fiori Apps Reference Library
Schedule Goods Receipt for Inbound Deliveries	F2776	Link to SAP Fiori Apps Reference Library
Expected Goods Receipt	F3298	Link to SAP Fiori Apps Reference Library
Manage Inbound Deliveries – VL06I	VL06I	Link to SAP Fiori Apps Reference Library
Schedule Inbound Delivery Creation	F2798	Link to SAP Fiori Apps Reference Library
Receive Replacement Materials – From Vendor	MSR_VRM_GR	Link to SAP Fiori Apps Reference Library
My Follow up Activities – For Replacement Materials	MSR_VRM_LFU	Link to SAP Fiori Apps Reference Library
Enter Inspection Results – For Warehouse	MSR_INSPWH	Link to SAP Fiori Apps Reference Library
Post Goods Receipt for Inbound Delivery	F2502	Link to SAP Fiori Apps Reference Library
Stock – Single Material	F1076	Link to SAP Fiori Apps Reference Library
Stock – Multiple Materials	F1595	Link to SAP Fiori Apps Reference Library
Overdue Materials – in Transit	F2139	Link to SAP Fiori Apps Reference Library
Overdue Materials – GR Blocked	F2347	Link to SAP Fiori Apps Reference Library
Slow or Non-Moving Materials	F2137	Link to SAP Fiori Apps Reference Library
Dead Stock Analysis	F2899	Link to SAP Fiori Apps Reference Library
Create Outbound Delivery – with Order Reference	VL01N	Link to SAP Fiori Apps Reference Library
Create Outbound Delivery – without Order Reference	VL01NO	Link to SAP Fiori Apps Reference Library
Pick Outbound Delivery	F0868	Link to SAP Fiori Apps Reference Library
Display Outbound Delivery	VL03N	Link to SAP Fiori Apps Reference Library
Outbound Deliveries for Handling Units - Documents due for Delivery	VL10HU	Link to SAP Fiori Apps Reference Library
Schedule Delivery Creation	F2228	Link to SAP Fiori Apps Reference Library
Schedule Goods Issue for Deliveries	F2259	Link to SAP Fiori Apps Reference Library
Analyze Delivery Logs	F0870A	Link to SAP Fiori Apps Reference Library

App	App ID	Additional Information
Create Outbound Deliveries – From Sales Orders	F0869A	Link to SAP Fiori Apps Reference Library
Manage Output Items for Deliveries	F4455	Link to SAP Fiori Apps Reference Library
Schedule Output for Deliveries	F5979	Link to SAP Fiori Apps Reference Library
Reverse Goods Movement – Special Exceptions 3rd Party Warehouse	VL09_EXH	Link to SAP Fiori Apps Reference Library
Change Outbound Delivery – Special Exceptions 3rd Party Warehouse	VL02N_EXH	Link to SAP Fiori Apps Reference Library
Analyze Delivery Performance – Shipped as Planned	F2878A	Link to SAP Fiori Apps Reference Library
Create Delivery Group	VG01	Link to SAP Fiori Apps Reference Library
Display Delivery Group	VG03	Link to SAP Fiori Apps Reference Library
Change Outbound Delivery – Proof of Delivery	VLPOD	Link to SAP Fiori Apps Reference Library
Deliveries Worklist – Proof of Delivery	VLPODL	Link to SAP Fiori Apps Reference Library
Display Outbound Delivery – Proof of Delivery	VLPODA	Link to SAP Fiori Apps Reference Library
Automatic Confirmation – Proof of Delivery	VLPODQ	Link to SAP Fiori Apps Reference Library
General Outbound Delivery List	VL06F	Link to SAP Fiori Apps Reference Library
Outbound Deliveries for Goods Issue	VL06G	Link to SAP Fiori Apps Reference Library
Outbound Deliveries for Loading	VL06L	Link to SAP Fiori Apps Reference Library
Outbound Deliveries for Picking	VL06P	Link to SAP Fiori Apps Reference Library
Change Delivery Group	VG02	Link to SAP Fiori Apps Reference Library
Display Incomplete SD Documents	V_UC	Link to SAP Fiori Apps Reference Library
Change Outbound Delivery	VL02N	Link to SAP Fiori Apps Reference Library
My Sales Order Due for Delivery – Edit User-Specific Criteria	VL10	Link to SAP Fiori Apps Reference Library
My Sales Orders – Due for Delivery	VL10A	Link to SAP Fiori Apps Reference Library
My Purchase Orders – Due for Delivery	VL10B	Link to SAP Fiori Apps Reference Library
My Sales Order Items - Due for Delivery	VL10C	Link to SAP Fiori Apps Reference Library
My Purchase Order Items – Due for Delivery	VL10D	Link to SAP Fiori Apps Reference Library
My Sales Order Schedule Lines – Due for Delivery	VL10E	Link to SAP Fiori Apps Reference Library
My Purchase Order Schedule Lines – Due for Delivery	VL10F	Link to SAP Fiori Apps Reference Library

App	App ID	Additional Information
My Documents – Due for Delivery	VL10G	Link to SAP Fiori Apps Reference Library
My Items - Due for Delivery	VL10H	Link to SAP Fiori Apps Reference Library
My Schedule Lines – Due for Delivery	VL10I	Link to SAP Fiori Apps Reference Library
Split Outbound Delivery	VLSP	Link to SAP Fiori Apps Reference Library
My Collective Run Logs	V_SA	Link to SAP Fiori Apps Reference Library
Outbound Deliveries for Confirmation	VL06C	Link to SAP Fiori Apps Reference Library
Manage Outbound Deliveries	F0867A	Link to SAP Fiori Apps Reference Library
Subsequent Processing – Proof of Delivery	VLPODF	Link to SAP Fiori Apps Reference Library
Manage Outbound Deliveries – VL06O	VL06O	Link to SAP Fiori Apps Reference Library

Feature Comparison for Outbound Deliveries

The following tables compare the features offered by various apps in Delivery Management. The column headings contain links to the app documentation and the app's entry in the SAP Fiori apps reference library.

Single Deliveries

These apps allow you to create single outbound deliveries with or without a preceding document.

Functions	App Name	Create Outbound Delivery – With Order Reference	Create Outbound Delivery – Without Order Reference
	App ID	VL01N	VL01NO
Features			
with sales order reference	Basic ability	Yes	n/a
	w/ serial numbers	Yes	n/a
	w/ batches	Yes	n/a
without reference document	Basic ability	n/a	Yes
	w/ serial numbers	n/a	Yes
	w/ batches	n/a	Yes
Create / change texts		Yes	Yes
Non-Functional			
Custom fields		Yes	Yes

Multiple Deliveries

These apps allow you to create one or more outbound deliveries based on one or more preceding documents. For this, you can select the preceding documents that you want to create deliveries for and Delivery Management creates a outbound delivery based on the data in the document.

Functions	App Name	Create Outbound Deliveries – From Sales Orders	My Sales Orders – Due for Delivery	My Documents – Due for Delivery	My Sales Order Items - Due for Delivery	My Items - Due for Delivery	My Sales Order Schedule Lines – Due for Delivery	My Schedule Lines – Due for Delivery	My Purchase Orders – Due for Delivery	My Purchase Order Items – Due for Delivery	Pu C Sc Li D De
	App ID	F0869A	VL10A	VL10G	VL10C	VL10H	VL10E	VL10I	VL10B	VL10D	V
Features											
Based on sales orders	per sales order	Yes	Yes	Yes	n/a	n/a	n/a	n/a	n/a	n/a	
	per sales order item	No	n/a	n/a	Yes	Yes	n/a	n/a	n/a	n/a	
	per sales order schedule line	No	n/a	n/a	n/a	n/a	Yes	Yes	n/a	n/a	
Based on purchase orders, stock transport orders, or return purchase orders	per purchase order	n/a	n/a	Yes	n/a	n/a	n/a	n/a	Yes	n/a	
	per purchase order item	n/a	n/a	n/a	n/a	Yes	n/a	n/a	n/a	Yes	
	per purchase order schedule line	n/a	n/a	n/a	n/a	n/a	n/a	Yes	n/a	n/a	
Non-Functional											
Key user adaptation supported		Yes	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	
Custom fields		No	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	

Editing

These apps allow you to edit outbound deliveries after creating them. This includes adding and editing items and attachments, confirming picking and packing, and posting and reversing goods issue documents.

Functions	App Name	Change Outbound Delivery	Manage Outbound Deliveries – VL06O	Reverse GI - Outbound Delivery / Reverse GI - Outbound Delivery, Reverse Goods Receipt for Inbound Delivery	Pick Outbound Delivery	Manage Outbound Deliveries	Manage Output Items for Deliveries
	App ID	VL02N	VL06O_CLOUD	VL09	F0868	F0867A	F4455
General							
Perform picking	Basic ability	Yes	n/a	n/a	Yes	n/a	n/a
	w/ serial numbers	Yes	n/a	n/a	Yes	n/a	n/a
	w/ batches	Yes	n/a	n/a	No	n/a	n/a
Make other changes For example: • add / delete items • perform packing • create/change attachments	Basic ability	Yes	n/a	n/a	Yes (limited)	n/a	n/a
	w/ serial numbers	Yes	n/a	n/a	Yes (limited)	n/a	n/a
	w/ batches	Yes	n/a	n/a	No	n/a	n/a
Post Goods Issue	single outbound delivery	Yes	Yes	n/a	Yes	Yes	n/a
	multiple outbound deliveries	n/a	Yes	n/a	n/a	Yes	n/a
Reverse Goods Issue Posting	single outbound delivery	Yes	No	Yes	Yes	Yes	n/a
	multiple outbound deliveries	n/a	No	Yes	n/a	Yes	n/a
Display output items		Yes	n/a	n/a	Yes	No	No
Send or duplicate output items		No	n/a	n/a	No	No	No
Non-functional							
Create / change texts		Yes	n/a	n/a	No	n/a	n/a
Document flow		Yes	n/a	n/a	No	n/a	n/a

Functions	App Name	Change Outbound Delivery	Manage Outbound Deliveries – VL06O	Reverse GI - Outbound Delivery / Reverse GI - Outbound Delivery, Reverse Goods Receipt for Inbound Delivery	Pick Outbound Delivery	Manage Outbound Deliveries	Manage Output Items for Deliveries
	App ID	VL02N	VL06O_CLOUD	VL09	F0868	F0867A	F4455
Key user adaptation supported		No	No	No	No	Yes	No
Custom fields		Yes	No	No	No	No	No

Displaying

These apps allow you to view outbound deliveries after creating them.

Functions	App Name	Display Outbound Delivery	Outbound Delivery	Pick Outbound Delivery	Manage Output Items for Deliveries
	App ID	VL03N	F0233A	F0868	F4455
Features					
Header Level	Full scope of fields	Yes	No	No	n/a
	Selected fields	Yes	Yes	Yes	n/a
Item Level	Full scope of fields	Yes	No	No	n/a
	Selected fields	Yes	Yes	Yes	n/a
Display output items		Yes	Yes	Yes	Yes
Send and duplicate output items		Yes	Yes	Yes	Yes
Non-Functional					
Texts		Yes	No	No	n/a
Attachments		Yes	Yes	Yes	n/a
Document flow		Yes	n/a	n/a	n/a
Business process flow		n/a	Yes	No	n/a
Key user adaptation		No	Yes	No	No
Custom fields		Yes	Yes	No	No

Outbound Delivery

App ID: F0233A (Link to SAP Fiori Apps Reference Library)

You can access this app from the Enterprise Search for outbound deliveries search results. You can search for outbound deliveries based on key attributes, for example, shipping point, material number, or customer. The search results are displayed as a worklist where you can directly access outbound delivery object pages. With the outbound delivery object page, you can display outbound delivery data and navigate to additional relevant information. You can also quickly access related documents and key facts about relevant master data, especially business partners and material characteristics.

Key Features

This app provides the following features:

- Display general attributes of the outbound delivery document (such as shipping point, overall status, and delivery date), as well as contextual information.
- If a product has been marked as relevant for compliance in the product master data, the system performs marketability, dangerous goods, and safety data sheet checks for the corresponding outbound delivery. The results are displayed in the corresponding columns.
- Display the individual items that are to be delivered and their associated quantities, along with the respective plant and storage locations.
- Display the contact details of the business partners involved in the delivery process.
- Navigate to additional information relevant in your business context, such as detailed information about related business partners, material master data, and related sales orders.
- Display the process flow that provides you with a graphical overview of the chain of connected process steps and corresponding business documents (document flow). For example, you can see the sales order that might have been created prior to the outbound delivery, or the material document that recorded the goods issue posting for the outbound delivery.

Supported Device Types

- Desktop
- Tablet
- Smartphone

Feature Comparison Documentation

For a comparison of the features offered by this app and other apps in Delivery Management see: Feature Comparison for Outbound Deliveries

Component for Customer Incidents

LE-SHP-OBP

App Extensibility: Outbound Delivery

You can extend the **Outbound Delivery** app according to your business needs.

Features

Key users can extend the app in the **Custom Fields** app using the business context:

- **Shipping: Delivery**
- **Shipping: Delivery Item**

You can enable usage of your custom fields for the data source **Outbound Delivery** in the **User Interfaces** section of the custom field maintenance and publish your custom fields.

You can add fields to the following UI elements using UI adaption at runtime:

UI Element	Business Context
General Information	Shipping: Delivery
Items	Shipping: Delivery Item

For more information about how to adapt an SAP Fiori UI at runtime, see Making UI Changes (Object Pages).

Related Information

Custom Fields App and Custom Logic App
Key User Extensibility

Searching for an Outbound Delivery

Use

To find a specific outbound delivery, you use the Enterprise Search in the header bar of the SAP Fiori launchpad. Before you enter any attributes in the search input field, you should select the category **Outbound Delivery**.

In the search input field, you can, for example, enter a material number to display all outbound deliveries containing this material number. To narrow down your search results, you can also combine several attributes, such as a material number and a sold-to party. The search takes into account the following fields:

- Delivery document type
- Shipping condition
- Created by
- Shipping point
- Material number
- Ship-to party (ID and name)
- External delivery ID
- Material description
- Customer
- Reference document
- Sold-to party (ID and name)

A list of outbound deliveries is displayed as a result. For the first outbound delivery, the display area includes the header, the summary, and details. For all other outbound deliveries, you only see a header and a summary. To display the details of other outbound deliveries, you can expand the display area.

To further limit your search results to the ones that are relevant to you, you can apply one or more filters (for example, a specific status). You can also sort the search results according to your needs.

More Information

For general information about how to search for an object page, see Search.

Create Outbound Delivery – with Order Reference

App ID: VL01N (Link to SAP Fiori Apps Reference Library)

With this app, if you know the order number, you can manually create a single outbound delivery for one sales order. Only the order items from this order that are due for shipment will be included in the delivery. This app also allows you to change the shipping data if necessary.

> **Note**
> You use this app both to create outbound and customer returns deliveries. The app creates the relevant delivery type based on the sales order type you add in the search field and the underlying configuration settings. For more information on the creation of customer returns deliveries, see Creating Single Customer Returns Deliveries.

Supported Device Types

- Desktop
- Tablet

Component for Customer Incidents

LE-SHP-GF

Related Information

Creating Outbound Deliveries with Order Reference

Create Outbound Delivery – without Order Reference

App ID: VL01NO (Link to SAP Fiori Apps Reference Library)

With this app, you can create outbound deliveries without referring to an order. When you create a delivery without reference to an order, you must enter the ship-to party and the materials manually. Data is copied into the document from the appropriate master records. The system does not check whether the data is complete. Therefore, you can create a delivery without specifying a delivering plant or a storage location, for example.

Supported Device Types

- Desktop

- Tablet

Component for Customer Incidents

LE-SHP-GF

Create Outbound Deliveries – From Sales Orders

App ID: F0869A (Link to SAP Fiori Apps Reference Library)

With this app, you can create outbound delivery runs from a list of sales orders. This app also allows you to display logs with information related to your sales orders or deliveries. For example, if errors occur while you are creating deliveries, the system issues a message to the log. Similarly, the system issues a message to the log if you successfully create a delivery.

This app is available for the role **shipping specialist**.

ℹ Note

You use this app both to create outbound and customer returns deliveries. The app creates the relevant delivery type based on the sales order type you add in the search field and the underlying configuration settings. For more information on the creation of customer returns deliveries, see Creating Single Customer Returns Deliveries.

Key Features

You can use this app to do the following:

- Create outbound deliveries from sales orders with a limited number of items, or navigate to the **Schedule Delivery Creation** app to schedule the mass processing of sales orders.
- Search for sales orders due for delivery using basic filter options. You can also search using additional filter options.
- Sort your table entries by ascending or descending and then select the filter option you want to sort by.
- Personalize your table columns.
- If a product has been marked as relevant for compliance in the product master data, the system performs marketability, dangerous goods, and safety data sheet checks for the corresponding outbound delivery. The results are displayed in the corresponding columns.
- Display a log that provides you with information about the delivery run that you have created.
- Use the **Save as Tile** option to create a personalized tile on the launchpad. This tile will display a fixed snapshot of your current selection criteria. For example, if you filter by the date option **Today**, your personalized tile will only display the number of documents that are relevant for that specific day, as the current date selection is stored as a fixed date.

  ### → Tip

  When saving a tile, include the specific date as the subtitle or description, so that you can easily identify tiles.

  For more information, see SAP Note 2518476.

Supported Device Types

- Desktop

- Tablet
- Smartphone

Related Apps

- Schedule Delivery Creation
- Manage Outbound Deliveries
- Pick Outbound Delivery
- Analyze Delivery Logs

Related Links

For important details related to the **Create Outbound Deliveries – From Sales Orders** app, please refer to the following:

- Selecting Multiple Documents Due for Delivery
- SAP Note 2710216

Component for Customer Incidents

`LE-SHP-FIO`

Selecting Multiple Documents Due for Delivery

The **Create Outbound Deliveries – From Sales Orders** app displays a list of sales orders due for delivery that allows you to select individual or multiple sales documents for delivery creation.

You use the checkboxes in the **Sales Orders Due for Delivery** list to select your sales orders for delivery creation. Please consider the following filter logic that applies for multiple entries:

Multiple Entries for the same Sales Order

If you select an entry for a sales order with multiple entries, the latest delivery creation date is the chose due date. All index entries up to this point for the selected order are considered for delivery creation.

Example

The **Sales Orders Due for Delivery** list shows two entries for **Sales Document 22773** due to the filter criteria you applied. You use the checkbox to select the entry with the latest planned creation date:

Sales Orders Due for Delivery (1)		
Planned Creation Date		**Sales Document**
Sales Document 22773		
☑	03/24/2023	22773
	03/15/2023	22773

Result

The system considers both entries.

Multiple Entries for Multiple Sales Orders

If you select multiple entries for multiple sales orders, the latest delivery creation date of all selected entries is considered as the selected due date. All index entries up to this point for all selected sales orders are considered for delivery creation. This may include entries not explicitly selected with the checkbox in the list if they belong to a sales order for which other entries have been selected.

The following examples show the filter behavior:

Example 1

The Sales Orders Due for Delivery list shows multiple entries for Sales Document 22773 due to the filter criteria you applied. It also shows one entry for Sales Document 22776.

You use the checkbox to select the entry with the latest planned creation date of Sales Document 22773. You leave the entry of Sales Document 22776 unselected.

Sales Orders Due for Delivery (2)		
Planned Creation Date		Sales Document
Sales Document 22773		
☑	03/24/2023	22773
	03/15/2023	22773
Sales Document 22776		
	03/28/2023	22776

Result

- The system considers 03/24/2023 as due date.
- The system considers both entries of Sales Document 22773.
- The system does not consider the entry of Sales Document 22776.

Example 2

The Sales Orders Due for Delivery list shows multiple entries for Sales Document 22773 due to the filter criteria you applied. It also shows one entry for Sales Document 22776.

You use the checkbox to select the entry with the earliest planned creation date of Sales Document 22773 and the entry of Sales Document 22776.

Sales Orders Due for Delivery (2)	
Planned Creation Date	Sales Document
Sales Document 22773	

Sales Orders Due for Delivery (2)		
	03/24/2023	22773
☑	03/15/2023	22773
Sales Document 22776		
☑	03/28/2023	22776

Result

- The system considers 03/28/2023 as due date.
- The system considers both entries of Sales Document 22773.
- The system considers the entry of Sales Document 22776.

ℹ Note

The system logic may consider entries not explicitly selected in the list if they belong to a sales order for which other entries have been selected. It does not consider entries of sales orders without any selected checkbox.

Additionally, you can use the Planned Creation Date filter on the filter bar to specify a single planned creation date or date ranges for which you would like to create deliveries.

If you use the Planned Creation Date filter, the following logic applies:

All entries up to the provided filter date are considered for delivery creation for an order selected via the checkbox.

⚠ Caution

The Planned Creation Date filter value overwrites the due date selected using the checkbox. If you specify a date range filter, the latest date is considered as the due date. All entries of a selected order up to this date are considered.

Example

The Sales Orders Due for Delivery list shows multiple entries for Sales Document 22773 due to the filter criteria you applied. It shows one entry for Sales Document 22776. For this selection, you also set the planned creation date to `March 31, 2023`, using the Planned Creation Date filter.

You use the checkbox to select the entry with the earliest planned creation date of Sales Document 22773. You leave the entry of Sales Document 22776 unselected.

Sales Orders Due for Delivery (2)		
Planned Creation Date		Sales Document
Sales Document 22773		
	03/24/2023	22773
☑	03/15/2023	22773
Sales Document 22776		
	03/28/2023	22776

Result

- The system considers March 31, 2023, as due date.
- The system considers both entries of **Sales Document 22773**.
- The system does not consider the entry of **Sales Document 22776**.

Tip

If you need a list of items or schedule lines to choose from for delivery creation, please use the app **My Sales Order Items – Due for Delivery** (`VL10C`) or **My Sales Order Schedule Lines – Due for Delivery** (`VL10E`).

Change Outbound Delivery

App ID: VL02N (Link to SAP Fiori Apps Reference Library)

With this app, you can make changes to outbound deliveries before any further processing starts. You can access the delivery overview screen directly using the document number.

Using This App for Returns Deliveries

Note

Although the customer returns process is about the physical receipt of goods returned by a customer, it technically uses the apps and processes of the goods issue process. Therefore, you can also make changes to returns deliveries using this app.

Key Features

You can use this app to:

- Change header and item data of an outbound delivery.
- Make structural changes to an outbound delivery. For example, you can delete individual items from the outbound delivery or add new items.

3PL Warehouse Scenarios

For important guidelines for changing deliveries after initial distribution to a 3PL warehouse, please refer to SAP Note 3260042 .

Supported Device Types

- Desktop
- Tablet

Related Information

Changing Outbound Deliveries

Change Outbound Delivery – Proof of Delivery

App ID: VLPOD (Link to SAP Fiori Apps Reference Library)

With this app, you can select individual deliveries and confirm the quantities. After confirming proof of delivery (POD), you can cancel it to discard any entered quantities.

Key Features

You can use this app to:

- Select individual deliveries that you want to update proof of delivery for and confirm the quantities.
- Specify a reason for any overdelivery or underdelivery.
- You can optionally add multiple reasons and quantities for an item, and enter the relevant information.
- After confirming POD, you can cancel it to discard any entered quantities.

> **Note**
>
> You can only cancel the POD confirmation if billing documents have not yet been created and if no subsequent processing steps for value or quantity correction have been taken.
>
> After cancelling, if necessary, you can change the difference quantities and reasons and reconfirm.

Supported Device Types

- Desktop
- Tablet

Component for Customer Incidents

`LE-SHP-DL-POD`

Deliveries Worklist – Proof of Delivery

App ID: VLPODL (Link to Fiori App Library)

With this app, you can create a worklist for proofs of delivery (PODs) that require manual changes after a recipient has confirmed receipt of the delivery. It is important for deliveries in which the delivery quantity varies because of the nature of the goods or for which the exact delivery quantity is unknown from the start. After the delivery has been confirmed, the system creates a worklist for manual subsequent processing for any necessary quantity and value corrections, for example, the cost of sales.

Key Features

This app provides the following key features:

- Create a worklist for PODs that need manual changes based on information such as variance reason, POD date, or POD verification date
- Filter the worklist

- Access the PODs in the work list to make manual corrections
- Set the status of the PODs to complete when you've finished making manual changes

Supported Device Types

- Desktop
- Tablet

Component for Customer Incidents

`LE-SHP-DL-POD`

Automatic Confirmation – Proof of Delivery

App ID: VLPODQ (Link to Fiori Library)

With this app, you can confirm the proof of delivery (POD) information you received from the ship-to party for one or multiple outbound deliveries in a simple way. Once done, the system displays a log with all processed outbound deliveries and success or error messages. In the log, you can navigate to a detailed view for each outbound delivery and verify that the POD status on outbound delivery header level has been changed to **Confirmed** for all successfully processed outbound deliveries. After the proof of delivery has been confirmed, the system creates a worklist for manual subsequent processing for any necessary quantity and value corrections, for example, the cost of sales.

Key Features

This app provides the following key features:

- Define selection criteria for outbound deliveries to be processed
- Perform automatic confirmation of the proof of delivery for all selected outbound deliveries
- Display a log with results for each selected outbound delivery
- Navigate to any outbound delivery in the log to see further details (for example, the resulting POD Status)

Supported Device Types

- Desktop
- Tablet

Component for Customer Incidents

`LE-SHP-DL-POD`

Display Outbound Delivery – Proof of Delivery

App ID: VLPODA (Link to SAP Fiori Apps Reference Library)

With this app, you can display a single outbound delivery and get an overview of the proof of delivery (POD) data.

Key Features

This app provides the following key features:

- Select an outbound delivery by entering the outbound delivery ID.
- View the document flow for the selected delivery.
- View the header data for the selected delivery.
- Pack the selected delivery.
- View the POD data.
- Open the Change Outbound Delivery – Proof of Delivery app to edit the delivery.

Supported Device Types

- Desktop
- Tablet

Component for Customer Incidents

`LE-SHP-DL-POD`

Display Outbound Delivery

App ID: VL03N (Link to SAP Fiori Apps Reference Library)

With this app, you can view a single outbound delivery.

Using This App for Returns Deliveries

ℹ Note

Although the customer returns process is about the physical receipt of goods returned by a customer, it technically uses the apps and processes of the goods issue process. Therefore, you can also view a single returns delivery using this app.

Key Features

This app provides the following key features:

- Select an outbound delivery by entering the outbound delivery ID.
- View the document flow for the selected delivery.
- View the header data for the selected delivery.
- Pack the selected delivery.
- Open the Change Outbound Delivery app to edit the delivery.

Supported Device Types

- Desktop
- Tablet

Component for Customer Incidents

LE-SHP-GF

Schedule Delivery Creation

App ID: F2228 (Link to SAP Fiori Apps Reference Library)

With this app, you can schedule the collective creation of outbound deliveries as a background job for sales and purchase orders which are due for delivery within the next 30 days. The app allows you to select job templates with specific selection criteria for sales and purchase orders to simplify the creation of the application job.

> **Note**
> Although the customer returns process is about the physical receipt of goods returned by a customer, it technically uses the apps and processes of the goods issue process. Therefore, you can also schedule the collective creation of returns deliveries using this app.

Key Features

You can use this app to do the following:

- View the result of the scheduled jobs in the **Jobs** list on the initial screen
- Create a new job for generating outbound deliveries
- Copy an existing job for adapting and reusing
- Navigate to the details of an existing job
- Cancel an existing job that has not yet reached the status **Finished**
- Restart an existing job that has not yet reached the status **Finished**
- Define optional parameters for a job

Supported Device Types

- Desktop
- Tablet

Support

Application Component LE-SHP-GF-JS

Related Information

SAP Note 2710216

Schedule Delivery Creation for Sales Orders

Job template to create outbound deliveries from sales orders.

With the **Schedule Delivery Creation for Sales Orders** template, you can schedule the collective creation of outbound deliveries from sales orders due for delivery as a background job.

> **Note**
> You also use this template to schedule the collective creation of returns deliveries from returns orders.

Schedule Delivery Creation for Purchase Orders

Job template to create outbound deliveries from purchase orders.

With the **Schedule Delivery Creation for Purchase Orders** template, you can schedule the collective creation of outbound deliveries for purchase orders (stock transport orders, return purchase orders) due for delivery as a background job.

Change Outbound Delivery – Special Exceptions 3rd Party Warehouse

App ID: VL02N_EXH (Link to SAP Fiori Apps Reference Library)

With this app, you can change an outbound delivery corresponding to a warehouse number in a connected third-party logistics service provider's system or a connected decentralized warehouse management system. You use this app only in exceptional cases so that you can change an outbound delivery after canceling a goods issue for this delivery with the **Reverse Goods Movement – Special Exceptions 3rd Party Warehouse** app. Please note that changes to delivery data made with this app (quantity, material number, batch, serial number) are not automatically transferred to the connected system and can cause inconsistencies between stock data in SAP S/4HANA Cloud Public Edition and the connected system. These inconsistencies must be manually resolved in the connected system. If you have reversed a goods issue using a method other than the app Reverse Goods Movement – Special Exceptions 3rd Party Warehouse, you cannot use this app to make any changes to the corresponding outbound delivery.

Key Features

You can use this app to do the following:

- Change stock relevant delivery data of an outbound delivery from a connected third-party logistics service provider's system or a connected decentralized warehouse management system.

Reverse Goods Movement – Special Exceptions 3rd Party Warehouse

App ID: VL09_EXH (Link to SAP Fiori Apps Reference Library)

With this app, you can cancel a goods issue for an outbound delivery that was originally processed by a connected third-party logistics service provider's system or a connected decentralized warehouse management system. You use this app only in exceptional cases so that you can use the Change Outbound Delivery – Special Exceptions 3rd Party Warehouse app to edit an outbound delivery and re-post the goods issue with the updated information.

Key Features

You can use this app to do the following:

- Cancel a goods issue for an outbound delivery processed in a connected third-party logistics service provider's system or a connected decentralized warehouse management system.

! Restriction

You cannot use the SOAP service Warehouse Shipping Advice – Receive from Warehouse (B2B) to re-post the goods issue with updated delivery data after using this app to cancel a goods issue. You have to use the Change Outbound Delivery – Special Exceptions 3rd Party Warehouse app to do this.

⚠ Caution

The preferred method for canceling goods issue is with the SOAP service Warehouse Shipping Advice Cancel Request – Receive from Warehouse (B2B).

Use this app only in exceptional cases where you can't cancel goods issue in the third-party warehouse with the corresponding SOAP service. Changes made using this app and the Change Outbound Delivery – Special Exceptions 3rd Party Warehouse app are not updated in the third-party system and can cause inconsistencies between data in SAP S/4HANA Cloud Public Edition and the connected system. Ensure that you synchronize the changes offline with the third-party warehouse.

Pick Outbound Delivery

App ID: F0868 (Link to SAP Fiori Apps Reference Library)

With this app, you pick goods for outbound deliveries. You can also enter the results of the picking process for a particular delivery. The picking process involves taking goods from various storage bins and moving them to a staging area where they can be prepared for loading and shipping. This app shows you how many items are in the delivery, the delivery's overall details, such as the gross weight, picking status, and so on, and the details of individual delivery items. Once you have entered the picking quantity of an item and the system has set the picking status to **Completely Processed**, you can then post a goods issue. With this app, you can also assign serial numbers to delivery items. Assigning serial numbers allows you to track and trace your goods more accurately.

Using This App for Returns Deliveries

i Note

Although the customer returns process is about the physical receipt of goods returned by a customer, it technically uses the apps and processes of the goods issue process. Therefore, you can also put away goods for returns deliveries using this app.

Key Features

You can use this app to the following:

- Enable picking for an outbound delivery item that does not have warehouse management assigned to it and does not contain items with batches
- Assign serial numbers to delivery items in an outbound delivery
- Manage deviations during or after picking

- Delete a delivery
- Post a goods issue
- Reverse a goods issue
- Display output items for deliveries and delivery items

Feature Comparison Documentation

For a comparison of the features offered by this app and other apps in Delivery Management see: Displaying

Manage Outbound Deliveries

App ID: F0867A (Link to SAP Fiori Apps Reference Library)

With this app, you can display a list of all your outbound deliveries and navigate to them for further processing. You can choose each delivery to see its details, such as its weight and volume. Depending on the current goods issue status, you can post a goods issue for a delivery. If you need to make corrections to a delivery, you can easily reverse the goods issue posting and edit the corresponding documents. You can process all selected deliveries at once.

Using This App for Returns Deliveries

ⓘ Note

Although the customer returns process is about the physical receipt of goods returned by a customer, it technically uses the apps and processes of the goods issue process. Therefore, you can also display a list of all your returns deliveries using this app.

Key Features

You can use the app to do the following:

- Search for outbound using basic filter options. You can also search using additional filter options.
- Sort your table entries by ascending or descending and then select the filter option you want to sort by.
- Personalize your table columns.
- If a product has been marked as relevant for compliance in the product master data, the system performs marketability, dangerous goods, and safety data sheet checks for the corresponding outbound delivery. The results are displayed in the corresponding columns.
- Navigate to the **Pick Outbound Delivery** app to perform the picking step.
- Post goods issue for one or multiple outbound deliveries.
- Reverse goods issue postings for one or multiple deliveries.
- Use the **Save as Tile** option to create a personalized tile on the launchpad. This tile will display a fixed snapshot of your current selection criteria. For example, if you filter by the date option **Today**, your personalized tile will only display the number of documents that are relevant for that specific day, as the current date selection is stored as a fixed date.

→ Tip

When saving a tile, include the specific date as the subtitle or description, so that you can easily identify tiles.

For more information, see SAP Note 2518476.

Feature Comparison Documentation

For a comparison of the features offered by this app and other apps in Delivery Management see: Feature Comparison for Outbound Deliveries.

Related Links

Pick Outbound Delivery

Manage Outbound Deliveries – VL06O

App ID: VL06O (Link to SAP Fiori Apps Reference Library)

With this app, you can display and process both completed and open outbound deliveries at different stages of delivery processing. You can create worklists based on your area of responsibility and the organization of your shipping department. This app allows you to perform specific steps as standalone activities, such as picking, verifying and confirming picking data, and posting goods issue. It also allows you to perform various steps in a specific sequence throughout your workday.

Key Features

You can use this app to:

- Display outbound deliveries that are relevant for distribution and confirm that they have been distributed to a warehouse management system.
- Display outbound deliveries that are ready for picking and navigate directly to the **Change Outbound Delivery** app to record the result of the physical picking activities for each document.
- Display outbound deliveries that are ready for confirmation and navigate directly to the **Change Outbound Delivery** app to check and confirm the results of the picking step for each document.
- Display a list of outbound deliveries to be loaded and navigate directly to the **Change Outbound Delivery** app to further process the selected documents.
- Display outbound deliveries that are ready for goods issue and navigate directly to the **Change Outbound Delivery** app to post goods issue for the selected documents.
- Display a list of outbound deliveries based on various criteria and navigate directly to the **Change Outbound Delivery** app to further process the selected document or download the complete list for further analysis.

Feature Comparison Documentation

For a comparison of the features offered by this app and other apps in Delivery Management see: Feature Comparison for Outbound Deliveries.

Display Incomplete SD Documents

App ID: V_UC (Link to SAP Fiori Apps Reference Library)

With this app, you display incomplete sales and distribution documents of open outbound and returns deliveries.

Key Features

This app provides the following key features:

- Display outbound and returns deliveries using basic filter options.
- Sort your table entries by ascending or descending and then select the filter option you want to sort by.
- Personalize your table columns.

In addition, the app supports the following technical features and options:

You use the **Define Status Groups** (ID `102386`) configuration activity to configure the definition of incomplete.

Supported Device Types

- Desktop
- Tablet

Component for Customer Incidents

`LE-SHP-GF`

Schedule Goods Issue for Deliveries

App ID: F2259 (Link to SAP Fiori Apps Reference Library)

With this app you can schedule background jobs for posting goods issue for outbound deliveries and view the log.

Key Features

You can use this app to do the following:

- View the results of the scheduled jobs in the **Jobs** list on the initial screen
- Create a new job or copy an existing job for editing and reusing
- Navigate to the details of an existing job
- Cancel an existing job that has not yet reached the status **Finished**
- Restart an existing job that has not yet reached the status **Finished**
- Define optional parameters for a job

Supported Device Types

- Desktop

- Tablet

Support

Application Component `LE-SHP-GF-JS`

Schedule Goods Issue for Deliveries Using Developer Extensibility

Job template to post goods issues for outbound deliveries with Developer Extensibility.

With the **Schedule Goods Issue for Deliveries** (`SAP_LE_WS_MONITOR_OUTB_DEL_GDS`) template, you can schedule the collective posting of goods issues for outbound deliveries as a background job.

You can also access this template programmatically using the `CL_APJ_RT_API` class of the Job Scheduling framework. This allows you to schedule and manage goods issue posting jobs directly from custom ABAP code via developer extensibility - without using the Schedule Goods Issue for Deliveries app (F2259).

Related Information

Maintaining Application Jobs Using an API

Subsequent Processing – Proof of Delivery

App ID: VLPODF (Link to SAP Fiori Apps Reference Library)

With this app, you can create a worklist for proofs of delivery (PODs) that require manual changes after a recipient has confirmed receipt of the delivery. It is important for deliveries in which the delivery quantity varies because of the nature of the goods or for which the exact delivery quantity is unknown from the start. After the delivery has been confirmed, the system creates a worklist for manual subsequent processing for any necessary quantity and value corrections, for example, the cost of sales.

Key Features

This app provides the following key features:

- Create a worklist for PODs that need manual changes based on information such as variance reason, POD date, or POD verification date
- Filter the worklist
- Access the PODs in the work list to make manual corrections
- Set the status of the PODs to complete when you've finished making manual changes

Supported Device Types

- Desktop
- Tablet

Component for Customer Incidents

Analyze Delivery Logs

App ID: F0870A (Link to SAP Fiori Apps Reference Library)

You can use this app to check system messages that have been logged during the collective creation run of deliveries, regardless of whether you started the creation run online or in the background. If your attempt to create a delivery is unsuccessful, you must correct the issues that are mentioned in the log and then create a new delivery for the respective sales order or purchase order. For each delivery log, you can look up the messages that the system has logged during the creation run. These messages can be related to a sales order or purchase order, an individual delivery item, or to a delivery as a whole. You can also find out the numbers of the deliveries that the system has created.

This app is available for the role **shipping specialist**.

Using This App for Returns Deliveries

ℹ Note

Although the customer returns process is about the physical receipt of goods returned by a customer, it technically uses the apps and processes of the goods issue process. Therefore, you can also view logs for returns deliveries using this app.

Key Features

- Search for your delivery logs by using the basic filter options.
- Sort your table entries by ascending or descending and then select the filter option you want to sort by.
- Personalize your table columns.
- Display a log that provides you with information about the delivery run that you have created.
- Create a new delivery once you have corrected any errors in the log.

  You can select the log and navigate to the **Create Outbound Deliveries – From Sales Orders** app from within the **Analyze Delivery Logs** app.
- Use the **Save as Tile** option to create a personalized tile on the launchpad. This tile will display a fixed snapshot of your current selection criteria. For example, if you filter by the date option **Today**, your personalized tile will only display the number of documents that are relevant for that specific day, as the current date selection is stored as a fixed date.

  ### → Tip

  When saving a tile, include the specific date as the subtitle or description, so that you can easily identify tiles.

  For more information, see SAP Note 2518476.

Navigation Targets

The app enables users to access other apps directly (for example to display detailed information).

ℹ Note

These linked apps have to be already available in your system landscape or you have to implement them along with this app.

- Create Outbound Deliveries – From Sales Orders

Supported Device Types

- Desktop
- Tablet
- Smartphone

Related Apps

- Create Outbound Deliveries – From Sales Orders
- Manage Outbound Deliveries
- Pick Outbound Delivery

Component for Customer Incidents

`LE-SHP-FIO`

Create Delivery Group

App ID: VG01 (Link to SAP Fiori Apps Reference Library)

With this app, you can combine outbound deliveries into groups. An outbound delivery can belong to several groups. All grouped deliveries are identified by a group number. You can use this number to access the group and to find the individual outbound deliveries that belong to that group.

Key Features

You can use this app to do the following:

- Create Groups of outbound deliveries by specifying the delivery document ID
- Define the sorting sequence of the deliveries in the group
- Display a specific sales document in the group

Supported Device Types

- Desktop
- Tablet

Change Delivery Group

App ID: VG02 (Link to SAP Fiori Apps Reference Library)

With this app, you can change a delivery group that has been created with the **Create Delivery Group** app.

Key Features

This app provides the following key features:

- Searching for delivery groups to be changed according to **Group Type**, **Created By**, **Created On**, **Shipping Point**, and **Description**

Supported Device Types

- Desktop
- Tablet

Display Delivery Group

App ID: VG03 (Link to SAP Fiori Apps Reference Library)

With this app, you can display a delivery group that has been created with the **Create Delivery Group** app.

Key Features

This app provides the following key features:

- Searching for delivery groups to be changed according to **Group Type**, **Created By**, **Created On**, **Shipping Point**, and **Description**

Supported Device Types

- Desktop
- Tablet

Outbound Deliveries for Picking

App ID: VL06P (Link to SAP Fiori Apps Reference Library)

With this app, you can select outbound deliveries for which the picking status is A (**Not Yet Picked**) or B (**Partially Picked**).

Using This App for Returns Deliveries

Note

Although the customer returns process is about the physical receipt of goods returned by a customer, it technically uses the apps and processes of the goods issue process. Therefore, you can also use this app to select returns deliveries whose picking status is A (**Not Yet Picked**) or B (**Partially Picked**).

Key Features

You can use this app to do the following:

- Select deliveries that have not yet been picked
- Select deliveries that have been partially picked

Supported Device Types

- Desktop
- Tablet

Outbound Deliveries for Confirmation

App ID: VL06C(Link to SAP Fiori Apps Reference Library)

With this app, you can group manually selected deliveries together which are ready for confirmation.

Key Features

This app provides the following key features:

- Define the criteria for grouping the deliveries that are ready for confirmation.

Supported Device Types

- Desktop
- Tablet

Outbound Deliveries for Goods Issue

App ID: VL06G (Link to SAP Fiori Apps Reference Library)

With this app, you can group manually selected deliveries together which are ready for goods issue.

Key Features

This app provides the following key features:

- Define the criteria for grouping the deliveries that are ready for goods issue.

Supported Device Types

- Desktop
- Tablet

Outbound Deliveries for Loading

App ID: VL06L (Link to SAP Fiori Apps Reference Library)

With this app, you can group manually selected deliveries together which are ready for loading.

Key Features

This app provides the following key features:

- Define the criteria for grouping the deliveries that are ready for loading.

Supported Device Types

- Desktop
- Tablet

General Outbound Delivery List

App ID: VL06F (Link to SAP Fiori Apps Reference Library)

With this app, you can display a list of outbound deliveries which require further processing.

Using This App for Returns Deliveries

> **Note**
> Although the customer returns process is about the physical receipt of goods returned by a customer, it technically uses the apps and processes of the goods issue process. Therefore, you can also display a list of returns deliveries using this app.

Key Features

This app provides the following key features:

- Display a list of outbound deliveries based on various criteria and navigate directly to the **Change Outbound Delivery** app to further process the selected document or download the complete list for further analysis.

Supported Device Types

- Desktop
- Tablet

Split Outbound Delivery

App ID: VLSP (Link to SAP Fiori Apps Reference Library)

With this app, you can adapt outbound deliveries during the shipping process if the planned means of transport cannot accommodate the delivery in full. You can select an outbound delivery and split it into several smaller deliveries. The system assigns new delivery numbers, updates the original documents, and ensures that only the quantities actually loaded are processed. Simulation and highlighting functions support you in previewing and verifying the split before saving.

Key Features

This app provides the following key features:

- Split deliveries into smaller units.
- Apply predefined split profiles.
- Separate items with result numbers.
- Split full or partial item quantities.
- Create new deliveries with unique numbers.
- Highlight items during the split.
- Simulate and preview split results.
- Support for handling-unit-based splits.

Supported Device Types

- Desktop
- Tablet

My Sales Order Due for Delivery – Edit User-Specific Criteria

App ID: VL10 (Link to SAP Fiori Apps Reference Library)

With this app, you can create outbound delivery runs from a list of sales orders.

Key Features

This app provides the following key features:

- Enter search criteria for displaying corresponding sales orders
- Create single or multiple outbound deliveries from the displayed list of sales orders
- Display errors, warnings, and details corresponding to the created outbound deliveries

Creating Deliveries

To create single or multiple outbound deliveries based on the list of sales orders, carry out the following steps:

1. Select one or more sales orders from the list. You have the following options for creating deliveries from these sales orders:
    - **Option 1:** Choose the **Dialog** button to create single or multiple deliveries sequentially in a dialog using the **Create Outbound Delivery** app.
    - **Option 2:** Choose the **Background** button to automatically create single or multiple deliveries in the background.

Supported Device Types

- Desktop
- Tablet

Component for Customer Incidents

LE-SHP-GF

My Sales Orders – Due for Delivery

App ID: VL10A (Link to SAP Fiori Apps Reference Library)

With this app, you can create outbound delivery runs from a list of sales orders.

Key Features

This app provides the following key features:

- Enter search criteria for displaying corresponding sales orders
- Create single or multiple outbound deliveries from the displayed list of sales orders
- Display errors, warnings, and details corresponding to the created outbound deliveries

Creating Deliveries

To create single or multiple outbound deliveries based on the list of sales orders, carry out the following steps:

1. Select one or more sales orders from the list. You have the following options for creating deliveries from these sales orders:
    - **Option 1:** Choose the **Dialog** button to create single or multiple deliveries sequentially in a dialog using the **Create Outbound Delivery** app.
    - **Option 2:** Choose the **Background** button to automatically create single or multiple deliveries in the background.

Supported Device Types

- Desktop
- Tablet

Component for Customer Incidents

LE-SHP-GF

My Purchase Orders – Due for Delivery

App ID: VL10B (Link to SAP Fiori Apps Reference Library)

With this app, you can create outbound delivery runs from a list of purchase orders. (stock transport orders or return purchase orders only)

Key Features

This app provides the following key features:

- Enter search criteria for displaying corresponding purchase orders
- Create single or multiple outbound deliveries from the displayed list of purchase orders

Creating Deliveries

To create single or multiple outbound deliveries, select one or more purchase orders from the list and choose the **Background** button. The system automatically creates deliveries corresponding to the selected purchase orders.

Supported Device Types

- Desktop
- Tablet

Component for Customer Incidents

LE-SHP-GF

My Sales Order Items - Due for Delivery

App ID: VL10C (Link to SAP Fiori Apps Reference Library)

With this app, you can create outbound delivery runs from a list of sales order items.

Key Features

This app provides the following key features:

- Enter search criteria for displaying corresponding sales order items
- Create single or multiple outbound deliveries from the displayed list of sales order items
- Display errors, warnings, and details corresponding to the created outbound deliveries

Creating Deliveries

To create single or multiple outbound deliveries from the list of sales order items, carry out the following steps:

1. Select one or more sales order items from the list. You have the following options for creating deliveries from these sales order items:
    - **Option 1:** Choose the **Dialog** button to create single or multiple deliveries sequentially in a dialog using the **Create Outbound Delivery** app.
    - **Option 2:** Choose the **Background** button to automatically create single or multiple deliveries in the background.

Supported Device Types

- Desktop
- Tablet

Component for Customer Incidents

LE-SHP-GF

My Purchase Order Items – Due for Delivery

App ID: VL10D (Link to SAP Fiori Apps Reference Library)

With this app, you can create outbound delivery runs from a list of purchase order items. (items from stock transport orders or return purchase orders only)

Key Features

This app provides the following key features:

- Enter search criteria for displaying corresponding purchase order items
- Create single or multiple outbound deliveries from the displayed list of purchase order items

Creating Deliveries

To create single or multiple outbound deliveries, select one or more purchase order items from the list and choose the **Background** button. The system automatically creates deliveries corresponding to the selected purchase order items.

Supported Device Types

- Desktop
- Tablet

Component for Customer Incidents

LE-SHP-GF

My Sales Order Schedule Lines – Due for Delivery

App ID: VL10E (Link to SAP Fiori Apps Reference Library)

With this app, you can create outbound delivery runs from a list of sales order schedule lines.

Key Features

This app provides the following key features:

- Enter search criteria for displaying corresponding sales order schedule lines
- Create single or multiple outbound deliveries from the displayed list of sales order schedule lines
- Display errors, warnings, and details corresponding to the created outbound deliveries

Creating Deliveries

To create single or multiple outbound deliveries from the list of sales order schedule lines, carry out the following steps:

1. Select one or more sales order schedule lines from the list. You have the following options for creating deliveries:
    - **Option 1:** Choose the **Dialog** button to create single or multiple deliveries sequentially in a dialog using the **Create Outbound Delivery** app.
    - **Option 2:** Choose the **Background** button to automatically create single or multiple deliveries in the background.

Supported Device Types

- Desktop
- Tablet

Component for Customer Incidents

LE-SHP-GF

My Purchase Order Schedule Lines – Due for Delivery

App ID: VL10F (Link to SAP Fiori Apps Reference Library)

With this app, you can create outbound delivery runs from a list of purchase order schedule lines.

Key Features

This app provides the following key features:

- Enter search criteria for displaying corresponding purchase order schedule lines
- Create single or multiple outbound deliveries from the displayed list of purchase order schedule lines
- Display errors, warnings, and details corresponding to the created outbound deliveries

Creating Deliveries

To create single or multiple outbound deliveries, select one or more purchase order schedule lines from the list and choose the **Background** button. The system automatically creates deliveries from the selected purchase order schedule lines.

Supported Device Types

- Desktop
- Tablet

Component for Customer Incidents

LE-SHP-GF

My Documents – Due for Delivery

App ID: VL10G (Link to SAP Fiori Apps Reference Library)

With this app, you can create outbound delivery runs from a list of sales and purchase orders.

Key Features

This app provides the following key features:

- Enter search criteria for displaying corresponding sales and purchase orders
- Create single or multiple outbound deliveries from the displayed list of sales and purchase orders
- Display errors, warnings, and details corresponding to the created outbound deliveries

Creating Deliveries

To create single or multiple outbound deliveries from the list of sales and purchase orders, carry out the following steps:

1. Select one or more sales or purchase orders from the list. You have the following options for creating deliveries:
    - **Option 1:** Choose the **Dialog** button to create single or multiple deliveries sequentially in a dialog using the **Create Outbound Delivery** app.
    - **Option 2:** Choose the **Background** button to automatically create single or multiple deliveries in the background.

Supported Device Types

- Desktop
- Tablet

Component for Customer Incidents

LE-SHP-GF

My Items - Due for Delivery

App ID: VL10H (Link to SAP Fiori Apps Reference Library)

With this app, you can create outbound delivery runs from a list of sales order and purchase order items.

Key Features

This app provides the following key features:

- Enter search criteria for displaying corresponding sales order and purchase order items
- Create single or multiple outbound deliveries from the displayed list of sales order and purchase order items
- Display errors, warnings, and details corresponding to the created outbound deliveries

Creating Deliveries

To create single or multiple outbound deliveries from the list of sales order and purchase order items, carry out the following steps:

1. Select one or more sales order or purchase order items from the list. You have the following options for creating deliveries:
    - **Option 1:** Choose the **Dialog** button to create single or multiple deliveries sequentially in a dialog using the **Create Outbound Delivery** app.
    - **Option 2:** Choose the **Background** button to automatically create single or multiple deliveries in the background.

Supported Device Types

- Desktop
- Tablet

Component for Customer Incidents

LE-SHP-GF

My Schedule Lines – Due for Delivery

App ID: VL10I (Link to SAP Fiori Apps Reference Library)

With this app, you can create outbound delivery runs from a list of sales and purchase order schedule lines.

Key Features

This app provides the following key features:

- Enter search criteria for displaying corresponding sales and purchase order schedule lines
- Create single or multiple outbound deliveries from the displayed list of sales and purchase order schedule lines
- Display errors, warnings, and details corresponding to the created outbound deliveries

Creating Deliveries

To create single or multiple outbound deliveries from the list of sales and purchase order schedule lines, carry out the following steps:

1. Select one or more sales or purchase order schedule lines from the list. You have the following options for creating deliveries:
    - **Option 1:** Choose the **Dialog** button to create single or multiple deliveries sequentially in a dialog using the **Create Outbound Delivery** app.
    - **Option 2:** Choose the **Background** button to automatically create single or multiple deliveries in the background.

Supported Device Types

- Desktop

- Tablet

Component for Customer Incidents

LE-SHP-GF

My Collective Run Logs

App ID: V_SA (Link to SAP Fiori Apps Reference Library)

With this app, you can display lists of collective run logs related to sales, billing, or delivery documents.

Key Features

This app provides the following key features:

- Enter search criteria for the corresponding collective runs to be displayed
- Navigate to other apps for displaying and editing documents
- Display document flows
- Display details and status overviews for documents in the document flow
- Display error logs for the collective runs that contain errors

Navigation Targets

The app enables users to access other apps directly (for example to display and edit documents).

> **Note**
> To edit specific SD documents that you can display with this app, the corresponding business roles are required.

Displaying the Error Log

To display the error log, select the corresponding collective run in the list that contains errors and choose the **Notes** button.

Displaying Documents

To display SD documents, carry out the following steps:

1. Select the corresponding collective run in the list and choose the **Documents** button. The system displays a list of **Group** numbers and **SD Documents**.
2. Select the corresponding document ID in the list and choose the **Disp. doc.** button. The system navigates to the corresponding app and displays the document. If no document is available for the collective run due to an error, no document is displayed. In this case, choose the **Notes** button to display the error log.

Displaying Document Flows

To display document flows, carry out the following steps:

1. Select the corresponding collective run in the list and choose the **Documents** button. The system displays a list of **Group** numbers and **SD Documents**.
2. Select the corresponding document ID in the list and choose the **Document flow** button. The system displays the document flow for the collective run.

The following table describes the functions of the buttons available in the **Document Flow**:

Button	Function
Details	Displays details about a selected document.
Status Overview	Displays relevant statuses of a selected document.
Display Document	Navigates to the corresponding app and displays the selected document.
Service Documents	Navigates to the corresponding app and displays the selected order or service document.
Additional Links	Displays additional links related to the selected document.

Supported Device Types

- Desktop
- Tablet

Manage Output Items for Deliveries

App ID: F4455 (Link to SAP Fiori Apps Reference Library)

With this app, you can display a list of output items for delivery headers and items. According to your filter criteria, it lets you view and process one or more outputs. You can also display output logs and navigate to document previews for selected output items.

Key Features

You can use this app to do the following:

- Display a list of output items for outbound delivery headers and items
- Check the status of output items
- Mass process output items
- Send output items
- Duplicate completed output items
- Retry to send output items
- Set output items to completed
- Display output logs and document previews

Supported Device Types

- Desktop
- Tablet
- Smartphone

Feature Comparison Documentation

For a comparison of the features offered by this app and other apps in Delivery Management see: Feature Comparison for Outbound Deliveries

Schedule Output for Deliveries

App ID: F5979 (Link to SAP Fiori Apps Reference Library)

With this app, you can schedule the creation of output for outbound deliveries. Creating output describes the production of any format of a delivery document that is visible outside of SAP S/4HANA Cloud Public Edition, for example, PDF's for printing or emails for sending.

Key Features

This app provides the following key features:

- Schedule when output for deliveries is created
- Quickly set up single and recurring jobs
- Copy and modify existing jobs
- Select the delivery document output to be processed by specifying selection criteria, such as output channel and organizational data
- Use the job log to check whether the sending of output was successful or to diagnose issues

Supported Device Types

- Desktop
- Tablet

Inbound Delivery

App ID: F0232A (Link to SAP Fiori Apps Reference Library)

You can access this app from the Enterprise Search for inbound deliveries search results. You can search for inbound deliveries based on key attributes, for example, receiving point, material number, or supplier. The search results are displayed as a worklist where you can directly access inbound delivery object pages. With the inbound delivery object page, you can display inbound delivery data and navigate to additional relevant information. You can also quickly access related documents and key facts about relevant master data, especially business partners and material characteristics.

Key Features

- Display general attributes of the inbound delivery document (such as receiving point, overall status, and delivery date), as well as contextual information.
- Display the individual items that are to be received and their associated quantities, along with the respective plant and storage locations.
- Display the contact details of the business partners involved in the delivery process.
- Navigate to additional information relevant in your business context, such as detailed information about related business partners, material master data, and related purchase orders.
- Display the process flow that provides you with a graphical overview of the chain of connected process steps and corresponding business documents (document flow). For example, you can see the purchase order that might have been created prior to the inbound delivery, or the material document that recorded the goods receipt posting for the inbound delivery.

Supported Device Types

- Desktop
- Tablet
- Smartphone

App Extensibility: Inbound Delivery

Definition

You can extend the **Inbound Delivery** app according to your business needs.

Features

Key users can extend the app in the **Custom Fields** app using the following Business Contexts:

- **Delivery**
- **Delivery Item**

You can add fields to the following UI elements using UI adaptation at runtime:

UI Element	Business Context
General Information	Delivery
Items	Delivery Item

For more information about how to adapt an SAP Fiori UI at runtime, see http://help.sap.com/nw75 **Application Help** > **UI Technologies in SAP NetWeaver with SAP_UI 750** > **SAP Fiori Launchpad** > **Using the Launchpad** > **Running an SAP Fiori App** > **Adapting the UI of an SAP Fiori App at Runtime** > **Making UI Adaptations** >.

Related Information

Custom Fields App and Custom Logic App

Manage Inbound Deliveries – VL06I

App ID: VL06I (Link to SAP Fiori Apps Reference Library)

With this app, you can display and process both completed and open inbound deliveries at different stages of delivery processing. You can create worklists based on your area of responsibility and the organization of your receiving department. This app allows you to perform specific process steps as stand-alone activities, such as putaway, verifying and confirming putaway data, and posting goods receipt. It also allows you to perform various steps in a specific sequence throughout your workday.

Key Features

You can use this app to:

- Display inbound deliveries that are relevant for distribution and confirm that they have been distributed to a warehouse management system.
- Display inbound deliveries that are ready for putaway and navigate directly to the **Change Inbound Delivery** app to record the result of the physical putaway activities for each document.
- Display inbound deliveries that are ready for goods receipt and post goods receipt for the selected documents.
- Display a list of inbound deliveries based on various criteria and navigate directly to the **Change Inbound Delivery** app to process the selected document or download the complete list for further analysis.

Supported Device Types

- Desktop
- Tablet

Schedule Inbound Delivery Creation

App ID: F2798 (Link to SAP Fiori Apps Reference Library)

With this app you can schedule the collective creation of inbound deliveries as a background job for purchase orders which are due for delivery.

Key Features

You can use this app to do the following:

- View the results of the scheduled jobs in the **Jobs** list on the initial screen
- Create a new job or copy an existing job for editing and reusing
- Navigate to the details of an existing job
- Cancel an existing job that has not yet reached the status **Finished**
- Restart an existing job that has not yet reached the status **Finished**
- Define optional parameters for a job

Supported Device Types

- Desktop
- Tablet

Support

Application Component LE-SHP-GF-JS

Related Information

Application Jobs

Schedule Goods Receipt for Inbound Deliveries

App ID: F2776 (Link to SAP Fiori Apps Reference Library)

With this app you can schedule background jobs for posting goods receipts for inbound deliveries and view the log.

Key Features

You can use this app to do the following:

- View the results of the scheduled jobs in the **Jobs** list on the initial screen
- Create a new job or copy an existing job for editing and reusing
- Navigate to the details of an existing job
- Cancel an existing job that has not yet reached the status **Finished**
- Restart an existing job that has not yet reached the status **Finished**
- Define optional parameters for a job

Supported Device Types

- Desktop
- Tablet

Support

Application Component LE-SHP-GF-JS

Related Information

Application Jobs

Expected Goods Receipt

App ID: F3298 (Link to SAP Fiori Apps Reference Library)

With this app, you can schedule background jobs for data transfer of expected goods receipts (EGRs) for purchase orders and production orders to SAP Extended Warehouse Management (SAP EWM) and view the related log.

Key Features

You can use this app to do the following:

- View the results of the scheduled jobs in the **Jobs** list on the initial screen
- Create a new job or copy an existing job for editing and reusing
- Navigate to the details of an existing job
- Cancel an existing job that has not yet reached the status **Finished**
- Restart an existing job that has not yet reached the status **Finished**
- Define optional parameters for a job

Supported Device Types

- Desktop
- Tablet

Support

Application Component `LE-SHP-GF-JS`

Related Information

Application Jobs

General Inbound Delivery List

App ID: VL06IF (Link to SAP Fiori Apps Reference Library)

With this app, you can display a list of inbound deliveries that meet the selection criteria that you enter and process them.

Key Features

This app provides the following key features:

- Select inbound deliveries based on the following criteria:
    - Organizational data
    - Purchase order data
    - Document editing data
    - Delivery date
    - Putaway data
    - Partner data

    - Document data
    - Returns data
    - Material data
    - Shipment data
    - Status
    - Display options
    - Output proposals
- Post goods receipts for the selected inbound deliveries.
- Confirm putaway for the selected deliveries.
- Display error logs for the selected inbound deliveries.
- Display selected inbound deliveries.
- Change selected inbound deliveries.
- Output the selected inbound deliveries.

Supported Device Types

- Desktop
- Tablet

Create Inbound Delivery

App ID: VL31N (Link to SAP Fiori Apps Reference Library)

With this app, you create single inbound deliveries based on a purchase order.

Key Features

You can use this app to do the following:

- Select a purchase order and supplier as the preceding document for the inbound delivery.
- Create an inbound delivery.

Supported Device Types

- Desktop
- Tablet

Related Information

Creating Inbound Deliveries

Change Inbound Delivery

Using this app, you change inbound deliveries.

App ID: VL32N (Link to SAP Fiori Apps Reference Library)

With this app, you can change a single inbound delivery. You can also navigate to other apps and process the delivery.

Key Features

You can use this app to do the following:

- Select a single inbound delivery by entering the inbound delivery ID
- Change data for the selected delivery
- View the document flow for the selected delivery
- View the header data for the selected delivery
- Post a goods receipt for the selected delivery
- Pack the selected delivery

3PL Warehouse Scenarios

For important guidelines for changing deliveries after initial distribution to a 3PL warehouse, please refer to SAP Note 3260042 .

Supported Device Types

- Desktop
- Tablet

Related Information

Changing Inbound Deliveries
Deleting Inbound Deliveries

Display Inbound Delivery

Using this app, you view inbound deliveries.

App ID: VL33N (Link to SAP Fiori Apps Reference Library)

With this app, you can view a single inbound delivery.

Key Features

This app provides the following key features:

- Select an inbound delivery by entering the inbound delivery ID.

- View the document flow for the selected delivery.
- View the header data for the selected delivery.
- Pack the selected delivery.
- Open the Change Inbound Delivery app to edit the delivery.

Supported Device Types

- Desktop
- Tablet

Related Information

Displaying Inbound Deliveries

Post Goods Receipt for Inbound Delivery

App ID: F2502 (Link to SAP Fiori Apps Reference Library)

With this app, you can post a receipt of goods with reference to an inbound delivery.

The app proposes a selection of inbound deliveries from the supplier that can be posted. This app supports you, for example, as a receiving specialist in your daily work.

Key Features

Central Use

→ Tip

To handle more user-specific information on the main page, you can use the view settings dialog. Individual adjustments can be carried out quickly by flagging the checkboxes under **Show in Table**. With these setting options, you can display all important fields together on the main page without having to navigate to the detail page.

- Post the receipt of goods with reference to an inbound delivery, which contains order items for stock material (procurement for stock).
- Post the receipt of goods with reference to an inbound delivery, which contains order items with single or multiple account assignments (procurement for direct consumption). With multiple account assignments, you can distribute the purchasing cost among different cost centers or projects, for example.

  ### ℹ Note

  You defined the accounting assignment distribution (with accounting information and ordered quantity) for the purchase order items – for the valuated goods receipt as well as for the non-valuated goods receipt.

  Note that the app only displays the account assignments as a further information. In the case of partial goods receipt the app do not re-calculate the distributed quantity.

- Post the receipt of goods with reference to vendor consignment stock.

- Post the receipt of goods with reference to a delivery, which contains order items with shelf life expiration date and production date. Before you post the goods receipt, the app supports a check of the shelf life expiration date.

  > **Note**
  >
  > If the shelf life expiration date check is active and at least one of the fields **Minimum Remaining Shelf Life** or **Total Shelf Life** are maintained in the material master, you have to enter the shelf life expiration date or the production date of the material at the time of the goods receipt. The app then checks whether the remaining shelf life is sufficient. Note that the minimum remaining shelf life is a minimum amount of time up to which the material must be in good condition so that the system will accept the goods receipt. Depending on your settings, the fields are displayed and/or are ready for input.
  >
  > Note that you can only enter the shelf life expiration date of a material if the shelf life expiration date check is active for the plant, active for the used movement type and the minimum remaining shelf life is maintained in the material master record or in the purchase order item.

- Post the receipt of goods with reference to a reason code. You can store a text that gives a reason for the movement (for example, reason for return delivery).
- Create and edit batches out of the post goods receipt processing.

  > **Note**
  >
  > Depending on your role, you need to assign the dependent business catalog **Batch Management - Batches** (`SAP_SCM_BC_BATCH_MGMT_MC`) to the app-specific business catalogs **Materials Management - Warehouse Processing** (`SAP_MM_BC_IM_PROCESS`) and **Materials Management - Goods Receipt Processing** (`SAP_MM_BC_IM_GR_PROCESS_PC`). Based on this assignment, the app shows a further button for creating and editing batches.

- Post the receipt of goods using a standard movement type or a custom movement type that you have created.

  > **Note**
  >
  > If you have created custom movement types based on the default ones, these custom movement types are only supported in the Goods Receipt apps. After creating your custom movement type, implement the Business Add-In (BAdI) MMIM_GR4XY_PROVIDE_DATA. This step is required to use your new movement type in the goods receipt apps. For more details on this BAdI, see App Extensibility: Post Goods Receipt for Inbound Delivery.

Split Valuation

- The app supports materials with split valuation. It is possible to either select a valuation type or show the assigned valuation type for the split valuated material.

  > **Note**
  >
  > If you do not specify a valuation type in the purchasing document, the valuation type remains initial in the inbound delivery document, and the system defines the valuation type at goods receipt. Afterwards, the system updates the valuation type in the purchasing document but not in the inbound delivery document.
  >
  > In order to keep the valuation type of the item between purchasing document and inbound delivery document consistent, you have to provide the valuation type in the purchasing document or inbound delivery document before you execute the goods receipt.

> **Note**
>
> In your configuration environment, use the search function to open the following activities: **Activate Split Valuation for Global Data** and **Activate Split Valuation for Valuation Area**. For more information, see Configuration Environment of SAP S/4HANA

Cloud Public Edition.

Subcontracting

- For **subcontracting**, the app is enhanced with a component information area.
    - The app has a component information area for the subcontracting process for inbound deliveries.
    - The app displays for each material whether components exist.
    - By default, all components are considered for posting.
    - Serial numbers are supported at both header and component level.
    - A quantity change on material header level on the main screen leads to a proportional adaption of the quantity of the component materials (item level).
    - You can decide if you want to go further with posting of the goods receipt for the complete material (header level) or to navigate to the detailed page and edit each component individually (item level).
    - You can make changes for all components of a material. You can change the quantity or the unit of measure for each component separately or deselect components for posting.

      Note that a quantity change on material header level on the detail screen does not trigger proportional adaption for component materials.
    - Support different process variants that can occur at the component level such as shelf life expiration date and reason code.
    - For additional information on how to handle subcontracting, see Subcontracting Cockpit.

> **Note**
>
> Posting of subcontracting with special stock E (Sales Order Stock) and Q (Project Stock) are not supported in the cloud environment.

Serial Numbers

- Assign, change, or remove serial numbers for materials and components serialized at document level. For additional information on serial numbers, see Serial Number Management.

> **Note**
>
> If you work with serial numbers, you must always post the entire quantity of a line item.

Integration of Mobile Scanner and Barcode Scanning

- Support different options for barcode processing to scan inbound delivery document numbers with internal and external scanner devices. Ensure fast and reliable data collection and mapping to the appropriate inbound delivery documents.
- Depending on your chosen scanning options for barcodes, you can scan the following barcodes:
    - *Simple barcode*: You can scan an inbound delivery document number.
    - *Composite barcode*: You can scan a GS1-standard-conform barcode with an inbound delivery document number.

  For more information, see Barcode Scanning in Inventory Management.
- Support manual barcode entry if necessary.

Others

- Print slips and labels during the entry of a goods receipt manually or automatically. For more details, see Printing Slips and Labels.
- Extend the app according to your business needs (app extensibility).
- Display detailed information about the level on which a material is stored: plant, storage location, or storage bin. Note that the availability of these fields depends on your configuration settings.

> **Note**
>
> The app calculates the stock values based on the stock quantity of the selected reporting date multiplied by the current material price. That means that the app calculates always with the current material price independent of the reporting date you have selected. This calculation method for the stock value is common to all Fiori apps in Inventory Management.

In addition, the app supports the following technical features and options:

- This app supports the output management and attachment service.
- To start a conversation and to share your comments on the selected material, use chat in **Microsoft Teams** or **Send Email**. For more information, see Integrating Microsoft Teams.

Supported Device Types

- Desktop
- Tablet
- Smartphone

See also

The following apps are related to the **Post Goods Receipt for Inbound Delivery** app:

- Post Goods Receipt for Purchasing Document
- Post Goods Receipt for Production Order
- Post Goods Receipt without Reference

Information for Key Users

The SAP Fiori apps reference library has details about the content necessary for giving users access to an app on the **SAP Fiori launchpad**. To see this app's Fiori content, go to the SAP Fiori apps reference library and search for the app. Then select the product. On the **Implementation Information** tab, select the correct release. The details are in the **Configuration** section.

App entry in the Fiori apps reference library

Feature Comparison Documentation

For a comparison of the features offered by this app and other apps in the area of inventory management, see:

Feature Comparison for Goods Movement

Related Information

App Extensibility: Post Goods Receipt for Inbound Delivery
Barcode Scanning in Inventory Management
How to Personalize the View Settings in Post Goods Receipt Apps

Inbound Deliveries for Goods Receipt

Using this app, you view a worklist of inbound deliveries ready for goods receipt posting.

App ID: VL06IG (Link to SAP Fiori Apps Reference Library)

With this app, you can display a list of inbound deliveries that meet the selection criteria that you enter and process them.

Key Features

This app provides the following key features:

- Select inbound deliveries based on the following criteria:
  - Purchase order data
  - Delivery date
  - Putaway data
  - Partner data
  - Document data
  - Returns data
  - Material data
  - Shipment data
  - Display options
- Post goods receipts for the selected inbound deliveries.
- Confirm putaway for the selected deliveries.
- Display error logs for the selected inbound deliveries.
- Display selected inbound deliveries.
- Change selected inbound deliveries.
- Output the selected inbound deliveries.

Supported Device Types

- Desktop
- Tablet
- Smartphone

Inbound Deliveries for Purchase Orders

Using this app, you create inbound deliveries from purchase orders.

App ID: VL34 (Link to SAP Fiori Apps Reference Library)

With this app, you can select purchase orders that meet the selection criteria that you enter and create inbound deliveries form them. You can also do a test run to check how many inbound deliveries the app creates.

Key Features

This app provides the following key features:

- Select purchase orders with a delivery date up to a given date
- Create inbound deliveries from these purchase orders
- Simulate the delivery creation

Supported Device Types

- Desktop
- Tablet

Inbound Deliveries for Putaway

Using this app, you view inbound deliveries that are relevant for putaway.

App ID: VL06IP (Link to SAP Fiori Apps Reference Library)

With this app, you can display a list of inbound deliveries that meet the selection criteria that you enter and process them.

Key Features

This app provides the following key features:

- Select inbound deliveries based on the following criteria:
    - Purchase order data
    - Delivery date
    - Putaway data
    - Partner data
    - Document data
    - Returns data
    - Shipment data
    - Status
    - Display options

- Post goods receipts for the selected inbound deliveries.
- Confirm putaway for the selected deliveries.
- Display error logs for the selected inbound deliveries.
- Display selected inbound deliveries.
- Change selected inbound deliveries.
- Output the selected inbound deliveries.

Supported Device Types

- Desktop
- Tablet

Reverse GI - Outbound Delivery, Reverse Goods Receipt for Inbound Delivery

App ID: VL09 (Link to SAP Fiori Apps Reference Library)

With this app, you can reverse goods movements for inbound, outbound, or returns deliveries.

Key Features

This app provides the following key features:

- Filter for inbound, outbound, or returns deliveries.
- Cancel goods movements for the selected deliveries.

Supported Device Types

- Desktop
- Tablet

Monitor Handling Units

App ID: F4557 (Link to Fiori App Library)

With this app, you can get an overview of handling units (HUs) created in your plant, warehouse, and storage location.

The app supports you, for example, as a shipping specialist, to monitor the path of an HU through your business transactions. The app displays all handling unit information such as status, referenced documents, handling unit hierarchy ancestors, and other properties.

The app provides a list view as a starting point with different filter criteria so that you can fine-tune your search result step by step. From this result list, you can navigate to further details of the selected HU, grouped in sections like **Packaged Products**, **Measurements** or **General Information**. If JIT calls are referring to the handling unit, this information is displayed in the additional **JIT Calls** section.

For example, the status includes whether an HU is shipped or planned, whether it is in the warehouse or whether it was posted as a goods issue. Or you can check which HUs are part of a delivery.

Additionally, you can navigate to referenced documents of the selected handling unit.

Key Features

This app provides the following key features:

- Multiple filter criteria
    - You can use multiple filter criteria to better search for the HUs you are looking for.
    - Filter criteria examples: Packaging material, product, storage location, plant
- Step-by-step navigation
    - You can navigate through all available handling units, from multiple nested HUs to the original product.
    - You can navigate from every step to the HUs or products involved.
- Detailed display of all HU relevant information
    - The app displays the search result in a list view. This list view is the starting point from which you can navigate and filter down handling units.
    - The app structures important information divided into sections with the same content. The app displays the information at a glance, like weight and value data for an HU. The **JIT Calls** section is only visible if JIT calls actually refer to a handling unit.
- Support for follow-on activities
    - You can navigate to referenced documents of the selected handling unit to carry out further investigations or to initiate direct actions, provided you have the appropriate authorization.

Situation Handling

This app uses Situation Handling, which automatically informs users responsible about matters that require their attention. For more information, see Situation Handling.

Supported Device Types

- Desktop
- Tablet

Related Information

App Extensibility: Monitor Handling Units
Handling Unit Situation

Outbound Deliveries for Handling Units - Documents due for Delivery

App ID: VL10HU (Link to Fiori App Library)

Use

You can create outbound deliveries for handling units that have already been created, for example, in production and also assign existing handling units to open deliveries.

Example Scenario:

The shipping specialist monitors the documents that are due for delivery and creates outbound deliveries at the appropriate time. Handling units have already been created in production and assigned to the customer requirements. The shipping specialist calls up a list containing both the delivery-due documents and the handling units already created and assigned to them, and creates the deliveries accordingly.

Key Features

With this app, you can do the following:

- Call up a list of documents that are due for delivery, handling units that have already been created for these documents (with status **GR-posted**), and open deliveries.
- Select handling units and a delivery-due document and have the system create a delivery for them.

  The system creates a delivery containing the cumulated material items, and assigns the handling units to the delivery. For more information, see How to Create an Outbound Delivery for Handling Units.
- Assign handling units to open deliveries.

  You select handling units for orders and assign them directly to open deliveries. The system updates the delivery with the cumulated material items and assigns the handling units to the delivery. For more information, see How to Assign Handling Units to Existing Deliveries.
- Change the assignment of a handling unit, that is, assign handling units to a different order or delivery.

  You can also delete all existing references. For more information, see How to Change References of Handling Units.
- Assign non-assigned handling units to orders or deliveries.

  You can call up a list of appropriate, non-assigned handling units for any order or delivery, and assign them to your order or delivery from this list. For more information, see How to Assign Non-Assigned Handling Units to Deliveries or Orders.
- Change handling units.
- Delete handling units as long as goods receipt is not posted.

Supported Device Types

- Desktop
- Tablet

Receive Replacement Materials – From Vendor

App ID: MSR_VRM_GR (Link to SAP Fiori Apps Reference Library)

Use this app to receive materials from vendors as part of the supplier returns process. This process replaces returned materials. Enter the replacement materials you receive for a returned item into the system. When you save the data, the system automatically creates a purchase order for the replacement materials and a corresponding inbound delivery.

The system generates these and all subsequent documents with reference to the supplier returns process. If the supplier returns process follows a customer return, the documents are follow-up documents linked to the customer returns process.

When you post the goods receipt for the inbound delivery in the next step, the system creates a document in invoice verification. This document serves as both a credit memo with a value of zero for returned materials and an invoice with a value of zero for replacement materials.

In the My Follow up Activities – For Replacement Materials app, you can specify logistical follow-up activities for the replacement materials.

Features

You can use this app to do the following:

- Enter replacement materials received from a vendor for returns purchase order items (returned materials)
- Monitor the status of supplier returns and all involved documents
- Post goods receipt for the inbound delivery

Further Details

Entering Replacement Materials

- The replacement material you receive may differ from the returned material. You can relate the quantity of replacement material to the quantity of returned material. For example, you might receive 7 pieces of material B to replace 5 pieces of material A. This allows you to enter replacements for subitem quantities of the returned material.
- You can specify that the vendor replaced only part of the returned material and will refund the remaining quantity with a credit memo.
- The plant that receives the replacement can be different from the plant from which you returned the material.
- You can only enter replacement material deliveries for returns purchase orders if a goods issue has not been posted or has been partially posted.
- If a supplier return results from a follow-up activity of a customer return, and you have posted the goods receipt to non-valuated returns stock without providing customer compensation, the system displays the parameter **E** (**Order Stock**) in the **Special Stock** field for the corresponding item. This indicates that the goods receipt is posted to non-valuated returns order stock.
- The **Complete Logistical Process with Goods Receipt** checkbox indicates that you do not need to define follow-up activities after posting the goods receipt for replacement material. The goods receipt completes the logistical process. You cannot edit this checkbox if the goods receipt is posted to non-valuated returns stock.
- You can enter different replacement materials for a single returns purchase order item by copying the item multiple times and assigning a replacement material to each line for the subitem quantities.

Posting Goods Receipt

- When you post a goods receipt, the system creates a document in invoice verification. This document serves as both a credit memo with a value of zero for the returned material and an invoice with a value of zero for the received material. If the replacement material differs from the returned material, or if the value of the received material is less than the returned material, the document includes additional posting items. These posting items correct the stock value lines and post the value difference to a loss account. The document is not posted in invoice verification until the goods issue for the returns purchase order item and the goods receipt for the replacement material are completed.

My Follow up Activities – For Replacement Materials

App ID: MSR_VRM_LFU (Link to SAP Fiori Apps Reference Library)

Use

If necessary, you can specify logistical follow-up activities for replacement materials. You specify these activities by referring to the previously created inbound deliveries. For more information, see Receive Replacement Materials – From Vendor.

Features

You can use this app to do the following:

- Enter logistical follow-up activities for subitem quantities of the delivery quantity by splitting the delivery item
- Enter the quantity and unit of measure of the replacement material for which you want to define a logistical follow-up activity
- Postpone the creation of follow-up documents to a later time by setting a follow-up block
- View the processing status of all returns steps, relevant documents, and follow-up activities by choosing the **Returns Overview** button
- Save and confirm all specified logistical follow up activities

Note

If the supplier return is a logistical follow-up activity of a customer return, the system displays the returns order and the relevant item. If you have already given the customer a replacement material or a credit memo, the system automatically sets the indicators **Replacement Exists Already** or **Refund Exists Already**.

Enter Inspection Results - From Warehouse

App ID: MSR_INSPWH

Use

With this app, you (for example, as a receiving specialist) can record the results of material inspections carried out on returned goods received into your warehouse. Based on these results, you can specify logistical follow-up activities and trigger refunding. The system then creates the necessary follow-up documents.

Key Features

- Select a worklist of returns deliveries that require inspection
- Enter inspection results for the returns delivery items
- Create inspection splits and enter inspection results for the item subquantities if the entire quantity of the returns delivery item does not have the same quality status
- Enter inspection data for bills of material at either the main item level or subitem level
- Specify logistical follow-up activities for the inspected returns delivery items or inspection split items

- Confirm the specified logistical follow-up activities for the system to create logistical follow-up documents
- Release the customer compensation by triggering the creation of a credit memo request or a subsequent delivery free of charge

Prerequisites

- You have created a returns order and posted the goods receipt for the corresponding returns delivery
- You have physically inspected the returned goods

Functional Details

Product Inspection

You record the results of an inspection that you carried out for returned materials that were received in your warehouse. You enter the inspection results with reference to the returns delivery that was created for the returns order. You can select more than one returns delivery that requires inspection and put them in a worklist. Inspection data includes the inspection code, an inspection comment, the inspector, and the inspection date. You can enter inspection data at header or item level, or you can split delivery items if different results are to be recorded for subquantities.

Inspection results can be used as the basis for determining the appropriate logistical follow-up activity and triggering the customer refund.

Defining Logistical Follow-Up Activities

The logistical follow-up activity code allows you to specify the next step in the logistical processing of the return (for example, `0011` **Transfer to Free Available Stock** or `0005` **Ship to Supplier**). Depending on the logistical follow-up activity that you select, you may have to specify additional information such as the next plant or a supplier.

You can save your data with or without confirming the logistical follow-up activities. If you just save the data, the system creates the inspection documents but does not create any logistical follow-up documents. You can still change the data at this point. If you confirm a logistical follow-up activity, the system creates the relevant logistical follow-up documents in the background.

For the detailed explanation on each logistical follow-up activity that is available in this app, see Logistical Follow-Up Activities

Compensation

In this app, you have the option of releasing a credit memo request or a subsequent delivery free of charge (SDF) for the delivery item or inspection split item of a returned material:

- Credit memo request

  Specify the credit memo request quantity and, if necessary, a refund code. If you do not want the customer to receive any money, enter a rejection reason.

- Replacement material

  For the replacement material delivery, enter the replacement material, the replacement quantity, the supplying plant, and the replaced quantity. You can specify a different replacement material than the returned material. The supplying plant can be different than the plant that received the returned material.

  The replacement material function for customer returns is linked to the function for non-valuated returns stock. When you release the replacement material delivery in the material inspection, the material valuation is triggered if the returned material was originally non-valuated.

In this app, if you specify a logistical follow-up activity for non-valuated returns materials that signals a change in ownership, or if you trigger compensation for the customer by releasing a credit memo request or an SDF, the system posts a goods movement for valuating the material first. The valuated stock then already serves as the basis for subsequently posting the logistical follow-up activity.

The logistical follow-up activities with a change in ownership are as follows:

- `0011` **Transfer to Free Available Stock**
- `0012` **Transfer to Scrap**
- `0015` **Transfer to Specified Stock**
- `0031` **No Further Activities**

Related Information

Product Inspection
Enter Inspection Results - From Warehouse
Creating Inspection Splits
Configuration Environment of SAP S/4HANA Cloud Public Edition

Analyze Delivery Performance – Shipped as Planned

App ID: F2878A (Link to SAP Fiori Apps Reference Library)

This app allows you to monitor the current delivery performance within your organization, so you can keep track of customer satisfaction and retention. If delivery performance is below what you expect it to be (meaning if planned dates for issuing goods to internal and external customers are not observed properly), you can drill down to individual outbound deliveries that were processed with delay and navigate to the related sales orders or stock transport orders for more detailed information. You can then use this information to help to resolve the issues that caused these delays in the first place.

Key Features

You can use this app to:

- Display, in percent, a graphical overview of current and past delivery performance.
- Display visual filters or compact filters to help you analyze your delivery performance. You can switch between both types of filters depending on your preference.
- View your delivery performance results in various chart forms, for example, column chart, bar chart, and so on, and drill down on further dimensions, such as customer, product group, or plant, using the chart filters.
- Analyze the delivery performance from an organizational view, a material view, or a customer view.
- Display underlying business documents in a table. You can also download this table to a spreadsheet, which allows you to easily share information with others.
- Sort and personalize the table to suit your business needs.
- Navigate to relevant outbound deliveries and master data object pages. Here, you can focus on specific information, for example, outbound deliveries with the longest delay.

This app uses the following CDS views:

- `I_DelivItmPerfAnlytsCube`
- `C_DelivItmPerfAnlytsQry`

Supported Device Types

- Desktop
- Tablet

Stock - Single Material

App ID: F1076

Use

With this app, you can get an overview of your material stock and review it by plant and storage location for which you are responsible. The app can display the stock information as a table or a diagram. This app supports you, for example, as an inventory manager in your daily work.

Key Features

Control and React

- Display a stock overview of a single material.
- Personalize your data view
    - You can configure table settings to control which columns and rows appear. This allows you to display all necessary information in material documents and stock analytics. For example, you can display reservation-related fields (`Open Receipt Quantity` or `Open Issue Quantity`) directly in the stock table and view reservation quantities at the plant and storage location level.
- Compare the stock information with the safety stock and minimum safety stock values defined for the material.
- Detailed *stock history* for each plant or storage location for the last 11 periods (period-end closing) and the stock balance at the reporting date.

    > **Note**
    > Before you can use the stock history, you have to activate **Show in Stock History** in the view settings for one or multiple entries.

    The stock history is structured as an interactive chart that allows you to analyze the stock history with the following main features:

    - Navigate directly from the micro chart (*thumbnail*) to a *full-scale* chart.
    - The line chart displays 12 months on the x-axis and the stock quantity on the y-axis. The chart contains a line for each stock type with data points for each month. You can check the quantities here, for example.
    - Select a data point and, if desired, drill down for more detailed stock level information on a monthly or even a daily level. You can use the **Zoom In** button to display the data points in greater detail.
    - A waterfall chart shows all goods movements on a daily basis. For each selected day, the chart displays all material quantity changes (y-axis) from the start of the day to the end of the day (x-axis), including all goods receipts and

goods issues that have taken place during this day. This means that the chart displays for each material document all daily changes of the goods receipts and goods issues so that you can see the progress between the day's opening stock and the day's closing stock.

> **Note**
>
> You might have to drill-down several times until you reach the waterfall chart.
>
> If no material documents are displayed in the chart, check if you selected a day, on which the quantity changed.

- Presentation of a first-level drill down to stock history (day-end closing) as a time series.
- Presentation of a second-level drill down to material document item postings for each day showing the key figures of each posting.
- Personalization of stock types and display of the safety stock.

> **Restriction**
>
> The following restrictions apply:
>
> - If you change the reporting date, the **Open Receipt Quantity** and **Open Issue Quantity** remain unchanged because reservation quantities are not tracked historically.
> - **The system cannot calculate stock in the past for the Non-Valuated GR Blocked Stock.**

Split Valuation

- The app supports materials with split valuation. It is possible to either select a valuation type or show the assigned valuation type for the split valuated material.

> **Note**
>
> In your configuration environment, use the search function to open the following activities: **Activate Split Valuation for Global Data** and **Activate Split Valuation for Valuation Area**. For more information, see Configuration Environment of SAP S/4HANA Cloud.

Integration of Mobile Scanner and Barcode Scanning

- Support barcode scanning of material numbers with internal and external scanner devices. Ensure fast and reliable data collection for stock information of your material.
- Depending on your chosen scanning options for barcodes, you can scan the following barcodes:
  - *Simple barcode*: You can scan a material number or GTIN code.
  - *Composite barcode*: You can scan a GS1-standard-conform barcode with a material number or GTIN code.

  For more information, see Barcode Scanning in Inventory Management.
- Support manual barcode entry if required.

> **Note**
>
> Barcode scanning in Inventory Management apps currently only supports GTINs of the base unit of measure.

In addition, the app supports the following technical features and options:

- You can send a link to the stock overview of the selected material via e-mail.
- To start a conversation and to share your comments on the selected material, use chat in **Microsoft Teams** or **Send Email**. For more information, see Integrating Microsoft Teams.

How to Review Stock

1. Select a material.

   The app displays your stock of the material by plant and by storage location as a table. It displays the stock history of the last 11 calendar months and the current stock in a curve diagram.

   The app uses today's date as the reporting date. You can change the reporting date. If you leave the **Reporting Date** field empty, the system displays the stock of material for the current period (end of period).

   ### ℹ Note

   The **Stock History** column or some stock types might be hidden. In the personalization area, you can specify the columns that you want the system to display in the results list. You access the stock history of all visible stock in the table by clicking on the micro chart.

   ### ℹ Note

   The value help for the **Unit of Measure** supports you in selecting all available units of measure of a material that you want to display more easily. The system groups the available units of measure according to the standard units and derived units, in other words, units of the same dimension as the standard units.

2. To display your stock by plant and by storage location in a bar diagram, choose the graphical view.

   Note that you only can get access to the safety stock in the stock history. The app displays the corresponding safety stock and minimum safety stock values. The safety stock is the quantity of stock whose purpose it is to satisfy unexpectedly high demand in the coverage period. The risk of shortfalls is reduced by having a safety stock. The minimum safety stock defines the lower limit for the safety stock. The safety stock cannot fall below the minimum safety stock.

3. To perform further tasks, choose a plant, and then choose one of the following apps:
   - Manage Stock
   - Transfer Stock - In-Plant
   - Transfer Stock - Cross-Plant
   - Material Documents Overview
   - Stock - Multiple Materials
   - Display Serial Numbers

Supported Device Types

- Desktop
- Tablet
- Smartphone

Information for Key Users

The SAP Fiori apps reference library has details about the content necessary for giving users access to an app on the **SAP Fiori launchpad**. To see this app's Fiori content, go to the SAP Fiori apps reference library and search for the app. Then select the product. On the **Implementation Information** tab, select the correct release. The details are in the **Configuration** section.

App entry in the Fiori apps reference library

Related Information

Barcode Scanning in Inventory Management

Stock - Multiple Materials

App ID: F1595

Use

With this app, you can get an overview of your material stocks. You can review your stock by the plants and storage locations for which you are responsible. This app supports you, for example, as an inventory manager in your daily work. The app can display the stock information as a table.

Key Features

Search and Control

- Advanced Search Capabilities

  Various filter options for searching large amounts of data specifically for the required materials. With these filter options you can better investigate your *active* materials (that is, for which goods movements have taken place) and materials without a deletion indicator in more detail.

- Display a stock overview of one or multiple materials.
- Display the value of your stock.
- Display additional storage location information, for example, storage bin.
- Display the carbon footprint of the materials in your plant.

  ℹ Note

  A prerequisite for using this feature is activation of the following business feature: *SAP Sustainability Footprint Management Integration into Core Processes* `PFM_PROCESSINTEGRATION` For more information, see Product Footprint Management (SUS-PFM-INT).

Schedule Export Options

- You can define and schedule jobs for the creation of background jobs on inventory data.
- Cancel the job.
- Manage your own job templates.

  You can define your own job templates or use the predefined ones.

  ℹ Note

The app is delivered with a predefined job template: **Export of Inventory Analytics: Stock - Multiple Materials**

Note that for this **Export of Inventory Analytics: Stock - Multiple Materials** template, stock identifying fields are automatically selected as part of the Excel output.

Note that you can delete your own templates, but you cannot delete the templates provided by SAP.

- Check your entries, before you finally schedule your jobs. The system issues a message to inform you about the accuracy and completeness of your entries.

ⓘ Note

The app calculates the stock values based on the stock quantity of the selected reporting date multiplied by the current material price. This means that the app always calculates using the current material price, irrespective of the reporting date you have selected. This calculation method for the stock value is common to all Fiori apps in Inventory Management.

In addition, in the **Stock - Multiple Materials** app, you can display the result of a calculation method based on the *historical prices* at the time of the related material postings.

Note that these two calculation methods lead to different results. SAP recommends that you use the *main* calculation method based on the current material price, because this method leads to faster results and is optimized for stock analysis.

In the view settings you can specify the desired field selection for each calculation method: The associated fields begin with **Current value of ...** and **Value of ...**.

ⓘ Note

Depending on your authorization settings, the app might only display zero in all the stock value columns in the results table. This type of financial information is often only available to a specific group of users. If only zero is displayed, you can hide these irrelevant value columns in the table display or get in contact with your administrator to clarify what you are authorized to display.

In addition, the app supports the following technical features and options:

- You can export the result list to a spreadsheet.
- You can send a link to the material document details by e-mail.
- To start a conversation and to share your comments on the selected search result, use chat in **Microsoft Teams** or **Send Email**. For more information, see Integrating Microsoft Teams.

How to Review Stock

1. Select one or multiple materials using the filter options.
2. Select the **Reporting Date**. If you leave the **Reporting Date** field empty, the system displays the stock of material for the actual period (end of period).

   The app displays your stocks by plant and by storage location as a table.

   Note that maybe one or more columns are hidden. In the personalization area, you can specify the columns that the system displays in the results list.

Supported Device Types

- Desktop

- Tablet
- Smartphone

Information for Key Users

The SAP Fiori apps reference library has details about the content necessary for giving users access to an app on the **SAP Fiori launchpad**. To see this app's Fiori content, go to the SAP Fiori apps reference library and search for the app. Then select the product. On the **Implementation Information** tab, select the correct release. The details are in the **Configuration** section.

App entry in the Fiori apps reference library

Related Information

App Extensibility: Stock - Multiple Materials

Overdue Materials - Stock in Transit

App ID: F2139

With this app, you can get an overview of the overdue materials for which a stock transport order has been created. You can identify these materials to complete the stock transfer process or investigate potential problems.

This app supports you, for example, as a warehouse clerk in your daily work.

Key Features

Display Options

- Use predefined movement types as a display option for goods receipt and goods movement delivered by SAP.

  > **Note**
  > You can use user-specific movement types as a copy of the predefined movement types.

- Use the **GI/GR Document Selection** field to specify whether all available goods receipts and goods issues for a purchase order item are to be displayed or only the latest posting. Your selection for this display option can reduce the number of documents to be displayed and therefore simplify your analysis.

Transfer Stock

- Check all stock transport orders where materials are scheduled to arrive in a plant by, for example, filtering according to the goods issue date.
- Select the **Stock in Transit Process Type** in a selection field.
- Display a list of stock transport orders where the stock transfer process has not yet been completed by, for example, filtering according to days since posting date.
- Check items where the goods receipt process is still open, or has already been completed, using the three predefined tabs with selection variants for stock transport order (STO) items. The following information is displayed on the tabs:
  - The incomplete or open STO items
  - The complete STO items

- The complete and incomplete STO items together

ℹ Note

The system displays all processes for stock transfer orders, including process types for the advanced intercompany stock transfer.

ℹ Note

Depending on your authorization settings, the app might only display zero in all the stock value columns in the results table. This type of financial information is often only available to a specific group of users. If only zero is displayed, you can hide these irrelevant value columns in the table display or get in contact with your administrator to clarify what you are authorized to display.

In addition, the app supports the following technical features and options:

- Send a link of selected search results by e-mail.
- To start a conversation and to share your comments on the selected search result, use chat in **Microsoft Teams** or **Send Email**. For more information, see Integrating Microsoft Teams.

How to Identify Stock Transfers with Overdue Materials with the predictive analytics model (Deprecated)

1. To search for overdue materials, enter your filter criteria, such as **Delivery completed** (selection `No`) and **Forecast Delivery Date** (selection before actual date) and choose `Go`.

   Depending on your filter criteria, the app displays a list of purchase orders for which the stock transfer process is not yet finished and that have already passed the expected forecast delivery date.

   The corresponding `Shipping Duration` micro chart displays the delivery situation:

   Depending on the current date, you can see the posting date of the purchase order (zero point) in relation to the forecast delivery date.

2. To perform further tasks, choose a purchase order or a material from the results list, and then choose one of the following apps, for example:

   - **Post Goods Receipt for Purchase Order**
   - **Stock - Multiple Materials**
   - **Stock - Single Materials**

Supported Device Types

- Desktop
- Tablet
- Smartphone

Information for Key Users

The SAP Fiori apps reference library has details about the content necessary for giving users access to an app on the **SAP Fiori launchpad**. To see this app's Fiori content, go to the SAP Fiori apps reference library and search for the app. Then select the

product. On the **Implementation Information** tab, select the correct release. The details are in the **Configuration** section.

App entry in the Fiori apps reference library

Situation Handling

This app uses Situation Handling, which automatically informs users responsible about matters that require their attention. For more information, see Situation Handling.

Related Information

Optimized Predictive Analytics Settings (Deprecated)
App Extensibility: Overdue Materials - Stock in Transit
Stock Transport Order Overdue

Overdue Materials - GR Blocked Stock

App ID: F2347

Use

With this app you can get an overview of the overdue materials for which a goods receipt (GR) was posted into the non-valuated GR blocked stock. You, as a warehouse clerk, can identify these overdue materials to complete the goods receipt process or investigate potential problems.

You can search for existing overdue materials, and sort and filter them according to their **Days since Posting Date (GR)** status, for example. Depending on your filter criteria, the app displays a list of purchase orders for which stock is available in the non-valuated GR blocked stock. Some columns in the results list might be hidden. In the personalization area, you can specify which columns you want to see in the results list. You can also organize your search results by choosing further view settings, such as grouping criteria.

For each purchase order or material from the results list you can immediately perform further tasks with linked apps, for example:

- Post Goods Receipt for Purchasing Document
- Stock - Multiple Materials
- Stock - Single Material

Key Features

- Check all purchase orders where a goods receipt was already posted into non-valuated GR blocked stock by filtering according to the days since first goods receipt, for example
- Display a list of purchase order items where stock is present in the non-valuated GR blocked stock regardless of whether the delivery completed indicator of the purchase order item was set or not.

ℹ Note

Depending on your authorization settings, the app might only display zero in all the stock value columns in the results table. This type of financial information is often only available to a specific group of users. If only zero is displayed, you can hide these

irrelevant value columns in the table display or get in contact with your administrator to clarify what you are authorized to display.

In addition, the app supports the following technical features and options:

- Send a link of selected search results by e-mail.
- To start a conversation and to share your comments on the selected search result, use chat in **Microsoft Teams** or **Send Email**. For more information, see Integrating Microsoft Teams.

Supported Device Types

- Desktop
- Tablet
- Smartphone

Information for Key Users

The SAP Fiori apps reference library has details about the content necessary for giving users access to an app on the **SAP Fiori launchpad**. To see this app's Fiori content, go to the SAP Fiori apps reference library and search for the app. Then select the product. On the **Implementation Information** tab, select the correct release. The details are in the **Configuration** section.

App entry in the Fiori apps reference library

Related Information

App Extensibility: Overdue Materials - GR Blocked Stock

Dead Stock Analysis

App ID: F2899

With this app, you can monitor and make time-dependent investigations of your dead stock.

Dead stock represents a combination of locations and materials where the stock value has increased over a defined analysis period despite existing consumption postings within that period.

The analysis of the dead stock identifies materials that could potentially have a negative impact on the company's cash flow and profitability. Based on these results, you can react immediately with follow-on activities and decide whether a material should, for example, be discontinued, discounted, or transferred to a different location.

The **Dead Stock Analysis** app supports you in your daily work, for example, as an inventory manager, to achieve maximum inventory accuracy and subsequently to optimize the inventory situation in your stock.

Key Features

- Monitor the dead stock with a focus on, for example, plants for which you are responsible.
- Select a display currency and track dead stock for several plants in different countries.
- Monitor how much capital is tied up by dead stock.

- Monitor important key performance indicators (KPIs) and key figures, such as the number of occurrences of dead stock (displayed as a value).
- Define follow-up actions such as scrapping or stock transfers, or inform colleagues about the stock situation.
- Save variants of the app with regard to the dead stock for which you are responsible.

  Note

  The app is predefined as follows:

  - The app considers materials that have a higher inventory values at the end of the analysis period than at the beginning.
  - The app only considers materials that have at least one consumption posting in the selected analysis period.
  - The app does not consider negative stock.

Note

The app calculates the stock values based on the stock quantity of the selected reporting date multiplied by the current material price. This means that the app always calculates using the current material price, irrespective of the reporting date you have selected. This calculation method for the stock value is common to all Fiori apps in Inventory Management.

Note

Depending on your authorization settings, the app might only display zero in all the stock value columns in the results table. This type of financial information is often only available to a specific group of users. If only zero is displayed, you can hide these irrelevant value columns in the table display or get in contact with your administrator to clarify what you are authorized to display.

Note

The value helps of business attributes, such as plant or storage location, only display values that have been previously posted in a material document.

This means that if a specific value does not appear in the value help, it indicates that it has never been used in any material document posting.

Supported Device Types

- Desktop
- Tablet

Information for Key Users

The SAP Fiori apps reference library has details about the content necessary for giving users access to an app on the **SAP Fiori launchpad**. To see this app's Fiori content, go to the SAP Fiori apps reference library and search for the app. Then select the product. On the **Implementation Information** tab, select the correct release. The details are in the **Configuration** section.

App entry in the Fiori apps reference library

Related Information

Slow or Non-Moving Materials

App ID: F2137

Use

With this app, you can monitor and make time-dependent investigations of the slow or non-moving materials in your stock. Based on these results you can react immediately with follow-on activities such as scrapping or stock transfers.

Non-moving materials represent a combination of locations and materials where the stock level is not zero and no consumption postings were posted for a defined period of time. It is also possible to track slow-moving materials.

This app supports you in your daily work, for example, as an inventory manager, to achieve maximum inventory accuracy and subsequently to optimize the inventory situation in your stock.

Key Features

Monitor Slow or Non-Moving Materials

- Monitor slow-moving materials for which you are responsible.
- Monitor how much capital is tied up by the slow-moving items, for example, by checking the total value in your chosen display currency.
- Monitor materials with *low consumption* based on the consumption to stock ratio with the **Slow-Moving Indicator**. The **Slow-Moving Indicator** makes it possible to compare different materials with regards to (slow-moving) consumption. Therefore, you can identify and compare critical materials based on your individual criteria.
- Monitor materials *without consumption* (**Slow-Moving Indicator** = 0) based on an analysis for an individual time period. As a result, the app displays a list of materials without consumption that fit your filter criteria.
- Consider only your specific consumption-relevant goods movements using the **Inventory Consumption Group** filter.

  The **Inventory Consumption Group** filter only considers those material documents as a consumption-relevant goods movement where the related movement types are part of your specific inventory consumption group. You can make the necessary settings using the corresponding SSC UI **Define Inventory Consumption Groups**. Here you can create groups with movement types that you consider to be consumption. All material documents connected to the movement types that are part of your specific inventory consumption group are then defined as consumption.
- Define follow-up actions such as scrapping or stock transfers, or inform colleagues about the stock situation.
- Save variants of the tile regarding materials for which you are responsible. For example, a tile for materials in a specific plant where inventory value is close to zero for 90 days or longer.

ℹ Note

The app calculates the stock values based on the stock quantity of the selected reporting date multiplied by the current material price. This means that the app always calculates using the current material price, irrespective of the reporting date you have selected. This calculation method for the stock value is common to all Fiori apps in Inventory Management.

ℹ Note

Depending on your authorization settings, the app might only display zero in all the stock value columns in the results table. This type of financial information is often only available to a specific group of users. If only zero is displayed, you can hide these irrelevant value columns in the table display or get in contact with your administrator to clarify what you are authorized to display.

- Analysis support with charts on detail page

  With the **Slow or Non-Moving Materials** detail page you can obtain more information on material / plant level for the selected material. The **Slow-Moving Indicator over Time** graphic visualizes the evolution of the monthly slow-moving indicator for the past 12 months.

  For more information about analysis support for slow or non-moving materials, see Analysis Options on Detail Level.

Slow-Moving Indicator

$$Slow - Moving\ Indicator = \frac{QCons * 100 * 365}{QStock * NODays}$$

Slow-Moving Indicator Calculation in the Selected Period

This indicator is defined as the total consumption quantity (**QCons**) for the analysis period divided by the product of stock quantity (**QStock**) on the reference date and the period length in number of days (**NODays**). In addition, the indicator is normalized for the year by multiplying by 365 days and 100 percent, because an analysis of slow-moving materials typically considers periods of a few months. By taking time and percentage into account, the normalized result makes different analysis periods comparable.

ℹ Note

The **Slow-Moving Indicator** takes only those stocks into account that already existed at the beginning of the selected reference period.

ℹ Note

A slow-moving indicator = 100 means that 100 % of the current stock will be consumed within one year.

A slow-moving indicator = 1 means that 1 % of the current stock will be consumed within one year.

ℹ Note

NODays is the number of days in the analysis period and determined from the **Days with Low Consumption** parameter + *1. 1* stands for the reference date, which is included in the period.

Predict Slow-Moving Indicator (Deprecated)

- Use predictive analytics models (Deprecated) to monitor the development of the **Slow-Moving Indicator** over the selected time period. If the predictive model (Deprecated) is active, the system calculates a predicted future trend, based on your company's historical data, on how the slow-moving indicator will develop in the course of the next 3 month.

ℹ Note

The value helps of business attributes, such as plant or storage location, only display values that have been previously posted in a material document.

This means that if a specific value does not appear in the value help, it indicates that it has never been used in any material document posting.

Supported Device Types

- Desktop
- Tablet

Information for Key Users

The SAP Fiori apps reference library has details about the content necessary for giving users access to an app on the **SAP Fiori launchpad**. To see this app's Fiori content, go to the SAP Fiori apps reference library and search for the app. Then select the product. On the **Implementation Information** tab, select the correct release. The details are in the **Configuration** section.

App entry in the Fiori apps reference library

Related Information

How to Monitor Slow or Non-Moving Materials
How to Monitor Slow-Moving Materials with the Predictive Analytics Model (Deprecated)
Analysis Options on Detail Level
App Extensibility: Slow or Non-Moving Materials

Returns Delivery

App ID:F0234A (Link to SAP Fiori Apps Reference Library)

You can access this app from the Enterprise Search for returns deliveries search results. You can search for returns deliveries based on key attributes, for example, receiving point, material number, or customer. The search results are displayed as a worklist where you can directly access returns delivery object pages. With the returns delivery object page, you can display returns delivery data and navigate to additional relevant information. You can also quickly access related documents and key facts about relevant master data, especially business partners and material characteristics.

Note

You can only use this app to display customer returns deliveries. To process customer returns deliveries, use the apps described in Creating and Processing Customer Returns Deliveries.

Key Features

You can use this app to:

- Display general attributes of the returns delivery document (such as receiving point, overall status, and delivery date), as well as contextual information.
- Display the individual items that are to be sent back and their associated quantities along with the respective plant and storage locations.
- Display the contact details of the business partners involved in the returns delivery process.
- Navigate to additional information relevant in your business context, such as detailed information about related business partners, material master data and related returns orders.
- Display the process flow that provides you with a graphical overview of the chain of connected process steps and corresponding business documents (document flow). For example, you can see the customer returns document that might

have been created prior to the returns delivery, or the material document that recorded the goods receipt posting for the returns delivery.

Supported Device Types

- Desktop
- Tablet
- Smartphone

Related Information

Creating and Processing Customer Returns Deliveries

App Extensibility: Returns Delivery

You can extend the **Returns Delivery** app according to your business needs.

Features

Key users can extend the app in the **Custom Fields** app using the following business contexts:

- **Shipping: Delivery**
- **Shipping: Delivery Item**

You can enable usage of your custom fields for the data source **Returns Delivery** in the **UIs and Reports** section of the custom field maintenance and publish your custom fields.

You can add fields to the following UI elements using UI adaption at runtime:

UI Element	Business Context
General Information	Shipping: Delivery
Items	Shipping: Delivery Item

For more information about how to adapt an SAP Fiori UI at runtime, see Making UI Changes (Object Pages).

Related Information

Custom Fields App and Custom Logic App
Key User Extensibility

Extensibility

You can extend Delivery Management apps, code, and processes according to your business needs.

ℹ Note

For information about extensibility for a specific function, see the documentation for that function. For example, the output extensibility documentation is found under Generating Output for Outbound Deliveries.

See the related links for a list of available extensibility documentation in Delivery Management.

Related Information

Services (APIs)
App Extensibility for Delivery Documents
Process Extensibility for Delivery Documents
Output Extensibility
Business Add-Ins (BAdIs) for Delivery Management
Developer Extensibility

Services (APIs)

Documentation for Delivery Management APIs is located under Supply Chain.

To view the documentation for Delivery Management APIs, see APIs for Delivery and Transportation.

Business Events

SAP provides business event handling for deliveries

You can handle business events related to the delivery SAP object types **Outbound Delivery** (`OutboundDelivery`), **Inbound Delivery** (`InboundDelivery`), and **Customer Returns Delivery** (`CustomerReturnsDelivery`).

You can consume these events in apps built on SAP Cloud Platform Foundry using SAP Enterprise Messaging.

Related Information

Business Event Handling
Integrating Enterprise Event Enablement
Outbound Delivery Events
Inbound Delivery Events
Customer Returns Delivery Events

CDS Views

Documentation for Delivery Management CDS views is located under in the SAP S/4HANA Cloud Public Edition extensibility documentation.

To view the documentation for Delivery Management CDS Views, see CDS Views for Delivery Management.

Developer Extensibility

Delivery Management provides business objects (BO) that you can extend as needed. These objects are part of the ABAP RESTful Application Programming (RAP) model.

The following RAP BOs are available in Delivery Management. For more information about each business object, click one to go to the SAP Business Accelerator Hub.

Business Object	Description
I_OUTBOUNDDELIVERYTP	Outbound Delivery
I_INBOUNDDELIVERYTP	Inbound Delivery
I_CUSTOMERRETURNSDELIVERYTP	Customer Returns Delivery

App Extensibility for Delivery Documents

In Delivery Management it is possible to extend apps by adding custom fields to the delivery documents you manage in them.

You can extend the following apps according to your business needs:

App	App ID
Create Outbound Delivery – with Order Reference	VL01N
Create Outbound Delivery – without Order Reference	VL01NO
Change Outbound Delivery	VL02N
Display Outbound Delivery	VL03N
Create Inbound Delivery	VL31N
Change Inbound Delivery	VL32N
Display Inbound Delivery	VL33N
Outbound Delivery	F0233A
Inbound Delivery	F0232A
Returns Delivery	F0234A

Features

Key users can extend the apps in the **Custom Fields** app using the following business contexts:

- **Shipping: Delivery**
- **Shipping: Delivery Item**
- **Handling Unit**

After enabling the extension fields for the **Delivery GUI Application** or the **Delivery Item GUI Application** in the **UIs and Reports** section of the custom field maintenance, the custom fields are shown in the **Custom Fields** section in the header or item details of the delivery document.

Extension fields relating to handling units are shown in the **Custom Fields** section of the **Detailed Data for Handling Unit** screen. For more information, see App Extensibility for Handling Units (Key User).

The custom fields will be open for input when creating or changing delivery documents and will be in read-only mode when displaying delivery documents.

Restrictions

The number of custom fields to be displayed in the **Custom Fields** section on either header or item level is limited to 170.

Process Extensibility for Delivery Documents

In Delivery Management it is possible to extend apps by adding custom fields for business scenarios.

You can extend the following apps according to your business needs:

- **Create Outbound Delivery With Order Reference**
- **Change Outbound Delivery**
- **Display Outbound Delivery**
- **Create Inbound Delivery**
- **Change Inbound Delivery**
- **Display Inbound Delivery**

Features

You can make custom fields available in several apps including handing over data by using business scenarios. After enabling the custom field for a business scenario, the value of this field will be copied between the corresponding documents.

For the business context **Shipping: Delivery**, the following business scenarios are available:

- **Sales Document to Delivery on Header Level**
- **Delivery to Billing Document on Header Level**
- **Purchase Order to Delivery on Header Level**
- **Sales Document Item to Delivery on Header Level**
- **Purchase Order Item to Delivery on Header Level**

ℹ Note

After enabling the custom field for one of the following business scenarios, a delivery split is enforced based on the predecessor's custom field value:

- **Sales Document Item to Delivery on Header Level**
- **Purchase Order Item to Delivery on Header Level**

Please also note that custom fields without values do not trigger a delivery split. If a custom field is filled in one item of a sales document or purchase order but left empty in another item, both items can still be part of the same new outbound delivery.

For the business context **Shipping: Delivery Item**, the following business scenarios are available:

- **Sales Document to Delivery on Item Level**
- **Delivery to Billing Document on Item Level**
- **Purchase Order to Delivery on Item Level**

Restrictions

The number of extension fields to be displayed in the **Custom Fields** section for the **Sales Document GUI Application** on either header or item level is limited to 170.

Related Information

Custom Fields App and Custom Logic App
Key User Extensibility

Business Add-Ins (BAdIs) for Delivery Management

With the **Custom Logic** app, you can create and maintain implementations of custom logic for the following Business Add-Ins (BAdIs):

BAdI Description	Business Context	Business Use
Message Validation from 3rd Party Logistics Integration (3PL) (LE_SHP_3PL_VETO_CHECK)	Shipping: Delivery Shipping: Delivery Item	You can use this BAdI to implement customer-specific validation checks when a delivery document is sent by a third-party logistics provider (3PL) warehouse system. These implemented customer-specific validation checks are executed before the delivery documents are saved to the system. The following interface is used in this BAdI: IF_LE_SHP_3PL_VETO_CHECK
Control Distribution of Delivery Documents (LE_SHP_CHECK_DISTRIBUTION)	Shipping: Delivery Shipping: Delivery Item	You can use this BAdI to control the distribution of delivery documents immediately before saving the delivery. You can explicitly prevent the distribution of the delivery document by setting the changing parameter DISTRIBUTION_NOT_ALLOWED_OUT. The following interface is used in this BAdI: IF_LE_SHP_CHECK_DISTRIBUTION
Last Checks Before Saving the Delivery (LE_SHP_DELIVERY_FINAL_CHECK)	Shipping: Delivery Shipping: Delivery Item	You can use this BAdI to perform final checks of the delivery data immediately before saving the delivery. You can prevent a delivery from being created or saved by returning an error message. The following interface is used in this BAdI: IF_LE_SHP_DELIVERY_FIN_CHECK

BAdI Description	Business Context	Business Use
Determine EWM-Relevant Custom Fields for Delivery (LE_SHP_EWM_RELEVANT_DETERMINE)	Shipping: Delivery Shipping: Delivery Item	You can use this BAdI to determine which custom fields on delivery header and delivery item level you want to distribute to an on-premise EWM system. The system calls the BAdI shortly before a delivery is distributed to EWM. The following interface is used in this BAdI: IF_LE_SHP_EWM_REL_DETERMINE
Set Custom Field Properties for Delivery Header (LE_SHP_FIELDPROP_HEAD)	Shipping: Delivery	You can use this BAdI to set properties for custom fields on delivery document header level. For example, you can make a field mandatory, so that a user must populate it. The following interface is used in this BAdI: IF_LE_SHP_FIELDPROP_HEAD
Set Custom Field Properties for Delivery Item (LE_SHP_FIELDPROP_ITEM)	Shipping: Delivery Item	You can use this BAdI to set properties for custom fields on delivery document item level. For example, you can make a field mandatory, so that a user must populate it. The following interface is used in this BAdI: IF_LE_SHP_FIELDPROP_ITEM
BAdI to Modify the Advance Shipping Notification (LE_SHP_MODIFY_ASN_FIELDS)	Shipping: Delivery	You can use this BAdI to pass the values for the fields BuyerSystemID and SupplierSystemID while sending an ASN to a Buyer/Supplier System. The following interface is used in this BAdI: IF_LE_SHP_MODIFY_ASN_FIELDS
Modify Delivery Header (LE_SHP_MODIFY_HEAD)	Shipping: Delivery	You can use this BAdI to modify standard and custom header fields. This BAdI is called after header and/or items have been changed, or items have been deleted by an end user. The following interface is used in this BAdI: IF_LE_SHP_MODIFY_HEAD
Modify Delivery Item (LE_SHP_MODIFY_ITEM)	Shipping: Delivery Item	You can use this BAdI to modify a changed or newly created item. You can also change delivery document header fields. This BAdI is called after one or more items have been changed or newly created. The following interface is used in this BAdI: IF_LE_SHP_MODIFY_ITEM

BAdI Description	Business Context	Business Use
Modify Delivery Item Weight and Volume (LE_SHP_MODIFY_ITEM_WEIGHT_VOL)	Shipping: Delivery Item	You can use this BAdI to modify the weight and volume of a delivery item. The new weight and volume are then considered in the delivery header. This BAdI is called for each delivery item. The following interface is used in this BAdI: IF_LE_SHP_MODIFY_ITEM_WEIGHT
Modification of Material Document for Posting Goods Movement (LE_SHP_MODIFY_POST_GM)	Shipping: Delivery Shipping: Delivery Item	You can use this BAdI to influence data at the interface between delivery processing and goods movement (goods issue/goods receipt). These changes have no effect on the fields in the delivery document but affect the processing of the material document. The BAdI is called when posting goods issue or goods receipt of a delivery document. The following interface is used in this BAdI: IF_LE_SHP_MODIFY_POST_GM
Modify Items for Picklist (LE_SHP_PICKLIST_MODIFY_ITEMS)	Shipping: Delivery Item	You can use this BAdI to modify standard and custom fields on a delivery picklist. The following interface is used in this BAdI: IF_LE_SHP_PICKLST_MODIFY_ITEMS
Modify Standard and Custom Fields for Delivery Header and Items (LE_SHP_SAVE_DOCUMENT_PREPARE)	Shipping: Delivery Shipping: Delivery Item	You can use this BAdI to modify standard and custom fields on header and item level. This BAdI is called shortly before saving the delivery document to the database. The following interface is used in this BAdI: IF_LE_SHP_SAVE_DOC_PREPARE
Storage Location Determination (LE_SHP_STORAGE_LOC_DETERMINE)	Shipping: Delivery Item	You can use this BAdI to find a new storage location on delivery item level. This BAdI is called when a new item is created during delivery creation or when a new item is added to an existing delivery. The following interface is used in this BAdI: IF_LE_SHP_STOR_LOC_DETERMINE
Suppress Delivery Split (LE_SHP_SUPPRESS_DEL_SPLIT)	Shipping: Delivery	You can use this BAdI to suppress a delivery split. When a sales order contains different items with materials that belong to different transportation groups, a subsequent delivery is divided into two or more deliveries. With the help of this BAdI, the split can be suppressed, and the delivery can be created with all sales order items. If delivery split is suppressed using this BAdI, the Transportation Group on the delivery header is empty.

BAdI Description	Business Context	Business Use
		This BAdI is called when a delivery is created from a sales order. The following interface is used in this BAdI: `IF_LE_SHP_SUPPRESS_DEL_SPLIT`
Filter Attachments Before Sending ASN (`LE_SHP_ASN_FILTER_ATTACHMENTS`)	Shipping: Delivery	You can use this BAdI to filter attachments out of outbound delivery documents before they are sent using the SOAP API *Outbound Delivery - Notify (B2B)*. This BAdI is called before advance shipping notifications (ASN) are sent using this SOAP API. This can be used, for example, to prevent the aborting of the sending process by filtering out attachments that are too large or have certain MIME types. The following interface is used in this BAdI: `IF_LE_SHP_ASN_SEND_ATTACHMENTS`
ATP Deactivation for Non-Stock Items (`LE_SHP_ATP_CHECK`)	Shipping: Delivery Item	You can use this BAdI to disable the availability check of an outbound delivery item that refers to a sales document item with special stock type E (Sales Order Stock). This allows overruling the configuration of the availability check in the delivery item category, depending on the current delivery document. The following interface is used in this BAdI: `IF_LE_SHP_ATP_CHECK`
Flexible Grouping in Delivery Notes (`LE_SHP_DELIVERY_NOTE_GROUP`)	Shipping: Delivery Item	You can use this BAdI to group delivery items and handling units into a two-level hierarchy in the Delivery Note form output. You define the grouping criteria. The BAdI implementation decides which fields to use as group keys at each level. The Form Data Provider (FDP) calls this BAdI once per delivery document during payload assembly. This occurs before the XML is passed to Adobe Document Services (ADS) for rendering. If no BAdI implementation is active, or if the implementation leaves all output tables empty, the FDP renders the standard flat delivery item output unchanged. Your standard form templates still work. You don't have to change them. The following interface is used in this BAdI: IF_LE_SHP_DELIVERY_NOTE_GROUP

For more information about using the **Custom Logic** app, see Custom Logic.

To display the complete documentation of the implemented BAdIs, carry out the following steps:

1. Open the **Custom Logic** app.
2. Choose an implementation from the **Implementations** list.
3. Choose the **View Documentation** link.

To search for BAdIs available for implementation, carry out the following steps:

1. Open the **Custom Logic** app.
2. Choose **Create**.
3. Enter the BAdI description from the above table in the search field and press Enter.
4. Choose the **View Documentation** link.

Integration

Delivery Management integrates with other SAP components and features.

For more information, select one of the following integration options:

Integration with Transportation Management

Delivery Management integrates with Transportation Management (TM) as part of the cross-component process Advanced Shipping and Receiving (Adv. SR) to allow you to track inbound, outbound, and customer returns deliveries. For more information about the overall Advanced Shipping and Receiving process and the other components involved, see Warehouse Integration in the Transportation Management documentation and Advanced Shipping and Receiving in the Warehouse Management documentation.

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Deliveries in Advanced Shipping and Receiving

Not all deliveries in a system are relevant for Advanced Shipping and Receiving. The system detects whether a delivery is relevant for Adv. SR by looking at the assigned shipping and receiving points. Any TM-relevant delivery which is assigned to a shipping or receiving point that is relevant for Adv. SR gets an indicator showing that this delivery is relevant for Adv. SR, too. When this indicator is set, the system displays additional fields for Advanced Shipping and Receiving.

Additional Data

The following table explains the fields and their use:

Fields for Advanced Shipping and Receiving

Field	Description

Field	Description
Loading/Unloading Point	Specifies a location that represents a group of doors where you can load or unload goods. There can be one or more loading points or unloading points assigned to a stop as substops in the hierarchy.
Warehouse Execution Status	The Warehouse Execution status specifies if the corresponding delivery document can be processed in the warehouse. It describes whether warehouse execution is allowed to start and how far the warehouse activities have been carried out so far. This applies to deliveries that refer to inventory-managed storage locations, as well as to deliveries that refer to warehouse-managed storage locations. Click Here to See the Available Values for This Field →

Value	Description	Details
BLANK	Not Relevant	The delivery isn't relevant for Warehouse Execution.
1	Delivery is blocked for Warehouse Execution	Warehouse-related activities may not start yet (for example, picking and packing)
2	Delivery is released for Warehouse Execution	Warehouse-related activities may start (for example, picking and packing)
3	Delivery is in Warehouse Processing	The delivery is currently being processed in the warehouse. This status is only available for outbound deliveries.
4	Delivery is ready for Shipping	The delivery is ready to be shipped. This status is only available for outbound deliveries.

Related Information

Transportation Planning - Sample Outbound Process

Transportation Planning - Sample Inbound Process

Order-Based Transportation Consolidation - Sample Outbound Process

Batches in Advanced Shipping and Receiving

The batch management features available in Delivery Management are compatible with Advanced Shipping and Receiving (Adv. SR).

In most cases, you can use batches with Adv. SR without additional configuration steps. However, to use batches in inbound deliveries and customer returns deliveries, there are some additional restrictions.

Inbound Deliveries

If you use handling units with batch management, activate the automatic batch creation feature. For more information, see Automatic Batch Creation for Inbound Deliveries.

Customer Returns Deliveries

If you use handling units for batch management, enter the batch number of the returned items in the sales returns order before creating the customer returns delivery.